复旦数学资料汇编
微分流形

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第一章　微分流形

1.1　习题: Warner GTM 94

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- Manifolds
- Tensors and Differential Forms
- Lie Groups
- Integration on Manifolds
- Sheaves, Cohomology, and the de Rham Theorem
- The Hodge Theorem

1.2　习题: Warner GTM 94/Integration on Manifolds

原网页

1. ($\Leftarrow$) Let $\omega = dr_1 \wedge \ldots \wedge dr_{d+1}$ be the volume form on $\mathbb{R}^{d+1}$ and $\omega'(x) = \delta f_x(i(N(x))\omega_{f(x)})$ for a smooth nowhere-vanishing normal vector field N along (X, f). Then it is a non-vanishing d-form on X, since for $x \in X$ and $X_1, \ldots, X_d \in X_x$ which are linearly independent, $\omega'_x(X_1, \ldots, X_d) \neq 0$. ($\Rightarrow$) Let ω' be a nowhere-vanishing d-form on X. For $x \in X$ choose $X_1, \ldots, X_d \in X_x$ which satisfies $\widetilde{X_1} \wedge \ldots \wedge \widetilde{X_d} = \omega_x$. Then we can pick $N(x) \in \left(\mathbb{R}^{d+1}\right)_{f(x)}$ s.t. $N(x)$ is orthogonal to $df(X_x)$ and $\omega(df(X_1), \ldots, df(X_d), N(x)) = 1$ where $\omega = dr_1 \wedge \ldots \wedge dr_{d+1}$. $N(x)$ is independent of the choice of $X_1, \ldots, X_d \in X_x$, since for $Y_1, \ldots, Y_d$ which also satisfies $\widetilde{Y_1} \wedge \ldots \wedge \widetilde{Y_d} = \omega_x$, one can express $Y_i = \sum_j a_{ij} X_j$, then using Exercise 2.10, we see that $\det a_{ij} = 1$. Thus $\omega(df(Y_1), \ldots, df(Y_d), N(x)) = 1$ also. Then since $df(X_x)^\perp$ is 1-dimensional, $N(x)$ which satisfies the condition above is uniquely determined. Also, it is by definition nowhere-vanishing. It remains to show that N is smooth, but locally on the neighborhood of $x \in X$ we can choose smooth vector field $X_1, \ldots, X_d$ which satisfies $\widetilde{X_1} \wedge \ldots \wedge \widetilde{X_d} = \omega'$ on the neighborhood, and $df(X_1), \ldots, df(X_d)$ is also smooth on the neighborhood of $f(x)$ over $f(X)$. Therefore locally N is also smooth (locally you can use the Gram–Schmidt process, and scaling if necessary, then the result $N(x)$ would be a smooth function of the local coordinates of x).
2. Taking a nowhere-vanishing n-form on $\mathbb{P}^n$ is equivalent to taking a nowhere-vanishing n-form on S^n such that it is preserved by antipodal map. Now regard S^n as imbedded to $\mathbb{R}^{n+1}$, and let N be the outward unit normal vector field on S^n and $\omega = dr_1 \wedge \ldots \wedge dr_{n+1}$. Then $\omega' = i(N)\omega$ is a

nowhere-vanishing n-form on S^n. If $\varphi\colon \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}, x \mapsto -x$, $\varphi|_{S^n}$ is the antipodal map, and

$$\delta\varphi_x\omega'_x = (-1)^n i(N(x))\omega_{-x} = (-1)^{n+1} i(N(-x)\omega_{-x}) = (-1)^{n+1}\omega'_{-x}.$$

Therefore, if n is odd, φ is orientation-preserving, therefore $\mathbb{P}^n$ is orientable. But if n is even, φ is orientation-reversing, so we cannot take a nowhere-vanishing n-form on S^n which is preserved by the antipodal map, which means $\mathbb{P}^n$ is not orientable.

3. At first find local vector fields $X_1, \ldots, X_n$ near $p \in X$ s.t. $X_{1,p}, \ldots, X_{n,p}$ is orthonormal, and let $L(m) = \{g(X_{i,m}, X_{j,m})\}_{i,j}$ where g is the given Riemannian metric. Then near $p \in X$, it is a smooth function to the set of $n \times n$ matrix. Also $L(p) = I$. Therefore locally L is positive definite and also symmetric by definition. Thus there is R s.t. $R^2 = L$, which is also smooth and symmetric. Define $(Y_1, \ldots, Y_n)^t = R^{-1}(X_1, \ldots, X_n)^t$. Then we see that $g(Y_i, Y_j) = \delta_{ij}$, which means that they are local orthonormal frame fields.
4. Use the notation on the problem. Let $V = f_1 e_1 + \ldots + f_n e_n$. Then $\tilde{V} = f_1\omega_1 + \ldots + f_n\omega_n$, so

$$*\tilde{V} = f_1\omega_2 \wedge \ldots \wedge \omega_n - \ldots + (-1)^{n-1} f_n \omega_1 \wedge \ldots \wedge \omega_{n-1}.$$

Therefore on ∂D,

$$*\tilde{V} = f_1\omega_2 \wedge \ldots \wedge \omega_n$$

since $\omega_1|_{\partial D} = 0$. On the other hand, $\langle V, \vec{n}\rangle = f_1$ and $\omega_2 \wedge \ldots \wedge \omega_n$ is the volume form on ∂D. Thus

$$\int_{\partial D} *\tilde{V} = \int_{\partial D} f_1.$$

Since

$$\int_D \operatorname{div} V = \int_D *d*\tilde{V} = \int_D d*\tilde{V} = \int_{\partial D} *\tilde{V}$$

by Stokes' theorem, the result follows.

5.

$$\begin{aligned} -\int_D f\Delta g &= \int_D f(*d*dg) = \int_D f(d*dg) \\ &= \int_D d(f*dg) - \int_D df \wedge *dg \\ &= \int_D d(f*dg) - \int_D \langle df, dg\rangle. \end{aligned}$$

Therefore

$$\begin{aligned} \int_D \langle \operatorname{grad} f, \operatorname{grad} g\rangle - \int_D f\Delta g &= \int_D \langle \widetilde{df}, \widetilde{dg}\rangle - \int_D f\Delta g = \int_D d(f*dg) \\ &= \int_{\partial D} f*dg = \int_{\partial D} *(fdg). \end{aligned}$$

Using the proof of Exercise 4,

$$*(fdg) = \langle \widetilde{fdg}, \vec{n}\rangle = f\frac{\partial g}{\partial n}$$

on ∂D. Therefore Green's 1st identity follows.

For the 2nd identity, interchange f and g and subtract from the former one.

6. If we establish the latter equation, the former one directly follows. For the latter one, since the vector space of 1-forms is 1-dimensional, we only need to show that they have the same nonzero value for some arbitrary n-tuples of vectors. If we put $X_1, \ldots, X_n$, the left side becomes $\omega(X_1, \ldots, X_n)^2$, which is the same as $\tilde{X}_1 \wedge \ldots \wedge \tilde{X}_n(X_1, \ldots, X_n) = \det\{\langle X_i, X_j\rangle\}$ since ω is the volume form.
7. Since $\bar{\lambda}(\sigma\tau) = \bar{\lambda}(\sigma)\bar{\lambda}(\tau)$, we only need to show that it is $\mathcal{C}^\infty$ at e. On the local chart of e with coordinate functions $x_1, \ldots, x_n$ centered at e, ω can be expressed as $f(x_1, \ldots, x_n)dx_1 \wedge \ldots \wedge dx_n$ where f is $\mathcal{C}^\infty$. Since $\bar{\lambda}(\sigma)\omega_e = \delta r_\sigma(\omega_\sigma)$,

$$\begin{aligned}\bar{\lambda}(\sigma)f(0, \ldots, 0)dx_1 \wedge \ldots \wedge dx_n &= f(x_1(\sigma), \ldots, x_n(\sigma))d(x_1 \circ r_\sigma) \wedge \ldots \wedge d(x_n \circ r_\sigma) \\ &= f(x_1(\sigma), \ldots, x_n(\sigma))\frac{\det\{\langle dx_i \circ r_\sigma, dx_j \circ r_\sigma\rangle\}}{\det\{\langle dx_i, dx_j\rangle\}}dx_1 \wedge \ldots \wedge dx_n,\end{aligned}$$

   so

$$\bar{\lambda}(\sigma) = \frac{f(x_1(\sigma), \ldots, x_n(\sigma))}{f(0, \ldots, 0)}\frac{\det\{\langle dx_i \circ r_\sigma, dx_j \circ r_\sigma\rangle\}}{\det\{\langle dx_i, dx_j\rangle\}}$$

   which is $\mathcal{C}^\infty$.
8. (This is one of the characteristics of the Haar measure on a unimodular Lie group. Here is the sketch of the proof.) Since

$$\delta r_\sigma \delta\alpha(\omega) = \delta\alpha\delta l_{\sigma^{-1}}(\omega) = \delta\alpha(\omega),$$

   it is right invariant. Let $\omega' = \delta\alpha(\omega)$ and think of μ, μ' as measures on G corresponding to ω and ω' respectively. Then we can define $f \mapsto \int_G f\mu'$ which is a well–defined right–invariant integral on G. Now consider

$$\int_{G\times G} f(\tau\sigma)\mu(\sigma)\mu'(\tau);$$

   if we first integrate with respect to μ it gives $\left(\int_G f\omega\right)\left(\int_G \omega'\right)$. And if we integrate with respect to μ' first, we get $\int_G f\omega'$. By Fubini's theorem, they must be the same. Also,

$$\int_G \omega' = \int_G \delta\alpha(\omega) = \int_{\alpha(G)} \omega = 1,$$

   so $\int_G f\omega = \int_G f\omega'$. Since $\int_G f\omega' = \int_{\alpha(G)}(f \circ \alpha)\omega$, the result follows.
9. ($\Rightarrow$) Since

$$\langle dl_\sigma X, dl_\sigma Y\rangle = \langle dr_\tau X, dr_\tau Y\rangle = \langle X, Y\rangle,$$

   $\mathrm{Ad}(G)$ in $\mathrm{Aut}(g)$ is actually in $O(g)$. Since $O(g)$ is compact, the closure of $\mathrm{Ad}(G)$ is also compact. ($\Leftarrow$) Give to g a non-degenerate inner product. Let $\ell \in \mathrm{Ad}(G)$. Since $\mathrm{Ad}(G)$ is a subgroup of $\mathrm{Aut}(g)$ and has compact closure, ℓ^n should be bounded for all $n \in \mathbb{Z}$. It is satisfied only when $\|\ell\| = 1$, so all elements of $\mathrm{Ad}(G)$ preserve the inner product. Now for $\sigma \in G$ and $X, Y \in G_\sigma$, define

$$\langle X, Y\rangle_\sigma = \langle dl_{\sigma^{-1}}X, dl_{\sigma^{-1}}Y\rangle_e.$$

   Then by definition it is left–invariant, and since

$$\begin{aligned}\langle dr_\tau X, dr_\tau Y\rangle_{\sigma\tau} &= \langle dl_{\tau^{-1}\sigma^{-1}}dr_\tau X, dl_{\tau^{-1}\sigma^{-1}}dr_\tau Y\rangle_e \\ &= \langle dAd_{\tau^{-1}}dl_{\sigma^{-1}}X, dAd_{\tau^{-1}}dl_{\sigma^{-1}}Y\rangle_e\end{aligned}$$

$$= \langle dl_{\sigma^{-1}} X, dl_{\sigma^{-1}} Y \rangle_e = \langle X, Y \rangle_\sigma,$$

it is also right–invariant.

10. (a) Since $\mathbb{R}^n \cong D(0,1) \subset \mathbb{R}^n$, the singular homology groups of the two must be the same. Then by the de Rham theorem, they are also the same as the de Rham cohomology groups. By Poincaré Lemma, it must be zero for given conditions.

    (b) By the de Rham theorem, it suffices to show that the 0-th singular homology group is $\mathbb{R}$. But since it is isomorphic to the free $\mathbb{R}$-module generated by the path-components, the result follows.

11. By the de Rham theorem, it suffices to calculate the singular homology groups. But it can be easily seen that ${}_\infty H_0(M;\mathbb{R}) = \mathbb{R}, {}_\infty H_1(M;\mathbb{R}) = \mathbb{R}, {}_\infty H_n(M;\mathbb{R}) = 0$ for $n \geq 2$.

12. $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta = 0$. If $\beta = d\gamma$, then $d(\alpha \wedge \gamma) = d\alpha \wedge \gamma + (-1)^k \alpha \wedge \beta$, so $\alpha \wedge \beta$ is exact.

13. By Stokes' theorem,
$$\int_z \alpha = \int_{\text{int}\, z} d\alpha = -8 \int_{\text{int}\, z} dx \wedge dy = -8.$$

14. It is closed by direct calculation. Also $d(\sin xy + x^2) = \alpha$, it is exact. Therefore,
$$\int_z \alpha = \int_{\text{int}\, z} ddf = 0.$$

15. It is closed by direct calculation. Also,
$$\int_{S^1} \alpha = \int_{S^1} \frac{xdy - ydx}{2\pi} = \int_D \frac{dx \wedge dy}{\pi} = 1.$$
    If $\delta i(\alpha)$ is exact, then its integral on S^1 must be zero since S^1 is a compact manifold without boundary. Therefore $\delta i(\alpha)$ is not exact. Also, if α is exact, then $\delta i(\alpha)$ is also exact since δ and d commute. Therefore, α is also not exact.

16. (a) It is because ${}_\infty H_1(S^2;\mathbb{R}) = 0$, and by the de Rham theorem.

    (b) By direct calculation.

    (c)
    $$\int_{S^2} \sigma = 3\,\text{Vol}(S^2),$$
    so it is not zero. If it is exact it must be zero since S^2 is a compact manifold without boundary.

    (d)
    $$*\alpha = \frac{\sum_{i=1}^n (-1)^{i-1} r_i dr_1 \wedge \ldots \wedge \widehat{dr_i} \wedge \ldots \wedge dr_n}{\left(\sum_{i=1}^n r_i^2\right)^{\frac{n}{2}}}.$$
    It is closed by calculation similar to (b).

    (e)
    $$\int_{S^{n-1}} *\alpha = n\,\text{Vol}(S^{n-1}),$$
    so it is not zero, and $*\alpha$ is not exact.

17. $H^1_{\text{de Rham}}(S^2;\mathbb{R}) = 0 \neq \mathbb{R}^2 = H^1_{\text{de Rham}}(T^2;\mathbb{R})$, using the singular homology theory and de Rham theorem.

18. (a) Since $H_1(M;\mathbb{R}) = 0$.

    (b) σ in Exercise 16(b) satisfies the condition.

    (c) Since $H_2(D;\mathbb{R}) = 0$ and $0 \neq [\sigma] \in H_2(M;\mathbb{R})$.

19. Let t be a variable corresponding to the interval $(-\varepsilon, 1+\varepsilon)$ on $M \times (-\varepsilon, 1+\varepsilon)$. For a k-form $\omega = \omega_1 + dt \wedge \omega_2$ on $M \times (-\varepsilon, 1+\varepsilon)$, $i\left(\frac{\partial}{\partial t}\right)\omega = \omega_2$ is a $(k-1)$-form, and its expression does not contain dt term. Therefore, we can define

$$\int_0^1 i\left(\frac{\partial}{\partial t}\right)\omega dt = \int_0^1 \omega_2 dt$$

which is a $(k-1)$-form on M. Denote it $D_k(\alpha)$. For convenience of the notation, denote d_M as a differential operator on M, and $d = d_M + d_t$ as a differential operator on $M \times (-\varepsilon, 1+\varepsilon)$. Then

$$\begin{aligned} D_k d(\omega) &= \int_0^1 i\left(\frac{\partial}{\partial t}\right)(d\omega_1 - dt \wedge d\omega_2)dt \\ &= \int_0^1 L_{\frac{\partial}{\partial t}}(\omega_1) - d_M\omega_2 dt, \end{aligned}$$

and

$$dD_k(\omega) = d_M \int_0^1 \omega_2 dt = \int_0^1 d_M\omega_2 dt.$$

Therefore

$$dD_k(\omega) + D_k d(\omega) = i_1^*\omega_1 - i_0^*\omega_1 = i_1^*\omega - i_0^*\omega$$

by definition of Lie derivative and i_0, i_1 ($i_0^*(dt \wedge \omega_2) = i_1^*(dt \wedge \omega_2) = 0$). It means i_0^* and i_1^* are the same map on the cohomology level. Since $f = F \circ i_0$ and $g = F \circ i_1$, $f^* = i_0^* \circ F^*$ and $g^* = i_1^* \circ F^*$ also induce the same cohomology maps.

20. (a) Since the vector space of n-forms on M is 1-dimensional on each fiber, we only need to show that $\omega = \delta f(i(\vec{n})(dr_1 \wedge \ldots \wedge dr_{n+1}))$ satisfies $\omega(v_1, \ldots, v_n) = 1$ for each point $m \in M$ and $v_1, \ldots, v_n \in M_m$ which is an oriented orthonormal basis of M_m. Since

$$\omega(v_1, \ldots, v_n) = dr_1 \wedge \ldots \wedge dr_{n+1}(\vec{n}, df(v_1), \ldots, df(v_n)) = 1$$

by assumption, the result follows.

(b) From vector calculus we obtain

$$\vec{n} = \frac{1}{\sqrt{f_x^2 + f_y^2 + 1}}\left(-f_x\frac{\partial}{\partial x} - f_y\frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right).$$

Therefore,

$$\begin{aligned} \omega &= \delta\varphi(i(\vec{n})(dx \wedge dy \wedge dz)) \\ &= \delta\varphi\left(\frac{1}{\sqrt{f_x^2 + f_y^2 + 1}}(-f_x dy \wedge dz + f_y dx \wedge dz + dx \wedge dy)\right) \\ &= \frac{1}{\sqrt{f_x^2 + f_y^2 + 1}}(-f_x dy \wedge df + f_y dx \wedge df + dx \wedge dy) \\ &= \frac{1}{\sqrt{f_x^2 + f_y^2 + 1}}\left(f_x^2 + f_y^2 + 1\right)dx \wedge dy = \left(\sqrt{f_x^2 + f_y^2 + 1}\right)dx \wedge dy. \end{aligned}$$

1.3　习题: Warner GTM 94/Lie Groups

原网页

1. Let $e \in U \subset G$ be a locally Euclidean open neighborhood of e. Then by Theorem 3.18, $G = \bigcup_{n=1}^{\infty} U^n$. Now, fix a countable basis of U, say $\{B_i\}$. Then for all open sets $V \subset U^n$ and for all points $g \in V$, g can be expressed as $g = g_1 \dots g_n$ by definition. Also, since multiplication on G is continuous, we can find open neighborhoods of g_k, say V_k, such that $V_1 \dots V_n \subset V$. Then, we can find B_{i_k}'s such that $g_k \in B_{i_k} \subset V_k$, and as a result $g = g_1 \dots g_n \in B_{i_1} \dots B_{i_n} \in V$. Therefore it follows from this that $\{B_i^n\}$ becomes a basis of U^n and U^n is second countable, and so is G.
2. (a) It is definitely a differential manifold. Also, subtraction is continuous under the standard topology.
   (b) It is also almost obvious.
   (c) Since it is a subgroup of $\mathbb{C}^*$, and the zero set of $|z| - 1$, which is nonsingular.
   (d) It is also a manifold, and the operation is continuous by definition of product topology.
   (e) It directly follows from (c) and (d).
   (f) Since matrix multiplication is defined by a polynomial, and inverse by a polynomial divided by determinant which is continuous.
   (g) Since it is the zero set of the elements below the diagonal of $Gl(n, \mathbb{R})$.
   (h) It is surely a manifold. Also, $(s,t)(s_1,t_1)^{-1} = \left(\frac{s}{s_1}, -\frac{st_1}{s_1} + t\right)$ is continuous.
   (i) It is also a manifold, and $(A,v)(A_1,v_1)^{-1} = (AA_1^{-1}, -AA_1^{-1}v_1 + v)$ is continuous.
3. (a) It is a vector space, and Lie bracket operation satisfies the given conditions by Proposition 1.45.
   (b) Trivial Lie bracket trivially satisfies the conditions.
   (c)
   $$[B, A] = BA - AB = -[A, B],$$
   and
   $$\begin{aligned}[[A,B],C] + [[B,C],A] + [[C,A],B] = {} & ABC - BAC - CAB + CBA \\ & + BCA - CBA - ABC + ACB \\ & + CAB - ACB - BCA + BAC = 0.\end{aligned}$$
   (d) $[ax + by, cx + dy] = (ad - bc)y$. It is a Lie algebra by direct calculation.
   (e) Cross product also satisfies the conditions. Note that $(a \times b) \times c = b(a \cdot c) - a(b \cdot c)$.
4. (a) It suffices to check that for all smooth vector fields X_i $(1 \le i \le r)$ on G, $\omega(X_1, \dots, X_r)$ is smooth. But
   $$\omega(X_1, \dots, X_r)(\sigma) = \omega_\sigma(X_{1\sigma}, \dots, X_{r\sigma}) = (\delta l_{\sigma^{-1}} \omega_e)(X_{1\sigma}, \dots, X_{r\sigma}) = \omega_e(dl_{\sigma^{-1}} X_{1\sigma}, \dots, dl_{\sigma^{-1}} X_{r\sigma}),$$
   so we need to check that $\omega_e(dl_{\sigma^{-1}} X_{1\sigma}, \dots, dl_{\sigma^{-1}} X_{r\sigma})$ is smooth. Now let $\tilde{\omega}$ be a smooth form such that $\tilde{\omega}_e = \omega_e$. Then for $\varphi \colon G \times G \to G, (a,b) \mapsto ab$, $\delta\varphi\tilde{\omega}$ is smooth. Also, $[0, X_i]$ is a smooth vector field on $G \times G$, and $\iota \colon G \to G \times G, \tau \mapsto (\tau^{-1}, \tau)$ is smooth. Therefore, $\delta\varphi\tilde{\omega}([0, X_1], \dots, [0, X_r])(\iota(\sigma))$ is smooth with respect to σ. But,
   $$\delta\varphi\tilde{\omega}\big([0, X_1], \dots, [0, X_r]\big)(\iota(\sigma))$$

$$= \delta\varphi\tilde{\omega}_{(\sigma^{-1},\sigma)}\big([0,X_1]_{(\sigma^{-1},\sigma)},\dots,[0,X_r]_{(\sigma^{-1},\sigma)}\big)$$
$$= \tilde{\omega}_e\big(d\varphi[0,X_1]_{(\sigma^{-1},\sigma)},\dots,d\varphi[0,X_r]_{(\sigma^{-1},\sigma)}\big).$$

Furthermore,

$$\begin{aligned} d\varphi[0,X_i]_{(a,b)}(f) &= [0,X_i]_{(a,b)}(f\circ\varphi) \\ &= 0_a(f\circ\varphi\circ\iota_b^1) + X_{ib}(f\circ\varphi\circ\iota_a^2) \\ &= X_{ib}(f\circ l_a) = dl_a X_{ib}(f). \end{aligned}$$

Therefore

$$\tilde{\omega}_e\big(d\varphi[0,X_1]_{(\sigma^{-1},\sigma)},\dots,d\varphi[0,X_r]_{(\sigma^{-1},\sigma)}\big) = \omega_e(dl_{\sigma^{-1}}X_{1\sigma},\dots,dl_{\sigma^{-1}}X_{r\sigma}),$$

and the result follows.

(b) δl_σ is an algebra endomorphism and $E^*_{\text{linv}}(G) = \bigcap_{\sigma\in G}\{\delta l_\sigma\text{-invariant elements}\}$, so $E^*_{\text{linv}}(G)$ is a subalgebra. Also, $\varepsilon\colon E^*_{\text{linv}}(G)\to\Lambda(G_e^*), \omega\mapsto\omega(e)$ is a homomorphism. And since

$$\omega(e)=\omega'(e)\Rightarrow\omega_\sigma=\delta l_{\sigma^{-1}}\omega_e=\delta l_{\sigma^{-1}}\omega'_e=\omega'_\sigma\Rightarrow\omega=\omega',$$

ε is injective. Furthermore, for all $\chi\in\Lambda(G_e^*)$, if we set $\omega(a)=\delta l_{a^{-1}}\chi$, then

$$(\delta l_\tau\omega)(\sigma)=\delta l_\tau(\omega(\tau\sigma))=\delta l_\tau\delta l_{\sigma^{-1}\tau^{-1}}\chi=\delta l_{\sigma^{-1}}\chi=\omega(\sigma),$$

so $\omega\in E^*_{\text{linv}}(G)$. Also, $\varepsilon(\omega)=\chi$, so ε is surjective.

(c)

$$\omega(X)(\sigma)=\omega_\sigma(X_\sigma)=(\delta l_{\sigma^{-1}}\omega)_\sigma(X_\sigma)=\omega_e(dl_{\sigma^{-1}}X_\sigma)=\omega_e(X_e)=\omega(X)(e),$$

so $\omega(X)$ is constant on G. Also, if $\alpha^*\colon G_e^*\to\mathfrak{g}^*$ is the dual map of α defined in Proposition 3.7, then

$$\alpha^*(\varepsilon(\omega))(X)=\alpha(\omega(e))(X)=\omega(e)(X(e))=\omega(X)(e).$$

(d)

$$d\omega(X,Y)=X(\omega(Y))-Y(\omega(X))-\omega([X,Y])=-\omega([X,Y])$$

since $\omega(X),\omega(Y)$ are constants.

(e) Since $\mathfrak{g}$ is closed under the Lie bracket, such c_{ijk}'s exist, and the relation (3) is directly followed by the properties of the Lie bracket, i.e., anti-symmetry and the Jacobi identity. Also, since δl_σ commutes with d, $d\omega_i\in E^2_{\text{linv}}(G)$ and

$$d\omega_i(X_j,X_k)=-\omega_i[X_j,X_k],$$

so we can write

$$d\omega_i=\sum_{j<k}c_{jki}\omega_k\wedge\omega_j+\omega'$$

where $\omega'(X,Y)=0,\forall X,Y\in(g)$. But, since

$$\varepsilon\colon E^2_{\text{linv}}(G)\to\Lambda_2(G_e)\cong\mathfrak{g}\wedge\mathfrak{G}$$

is an isomorphism, ω' must be zero. Hence the Maurer–Cartan equations are derived.

5. Consider id, $\pi_2 \colon \mathbb{Z} \times \mathbb{R} \to \mathbb{Z} \times \mathbb{R}$.
6. ($\Leftarrow$) part is obvious since $d\omega(X, Y) = -\omega[X, Y]$.
   For ($\Rightarrow$) part, $d\omega(X, Y) = -\omega[X, Y] = 0$ by assumption, so by using the Maurer–Cartan equation, $d\omega$ should be in $\mathcal{I}$.
7. (a) See [1], pp. 61–62, or follow the sketch given in the textbook.
   (b) See [1], pp. 64–65, or follow the sketch given in the textbook.
   (c) If $\pi \colon \tilde{X} \to X$ is a covering and X is simply connected, then by (a), there is a unique continuous map $\tilde{\alpha} \colon X \to \tilde{X}$ s.t. $\pi \circ \tilde{\alpha} = \mathrm{id}_X$. Therefore, $\pi \circ (\tilde{\alpha} \circ \pi) = \pi$. Then by uniqueness, $\tilde{\alpha} \circ \pi = \mathrm{id}_{\tilde{X}}$ also. Therefore, $\tilde{\alpha}$ and π are inverses to each other, which means π is a homeomorphism.
8. (Following the proof of [4], Prop. 1.9.) There is a countable open cover $\mathcal{U}$ whose elements are homeomorphic to open ball in Euclidean space. For all $U, U' \in \mathcal{U}, U \cap U'$ has at most countably many components, each of which is path connected. Let $\mathcal{X}$ be a countable set containing one point from each component of $U \cap U'$ (including $U = U'$). For each $U \in \mathcal{U}$ and each $x, x' \in \mathcal{X}$ s.t. $x, x' \in U$, let $p^U_{x,x'}$ be a path joining x and x' and contained in U. Since the fundamental groups based at any two points in the same component of M are isomorphic, and $\mathcal{X}$ contains at least one point in each component of M, we may as well choose a point $q \in \mathcal{X}$ as base point. Define a "special loop" to be a loop based at q that is equal to a finite product of paths of the form $p^U_{x,x'}$. Clearly, the set of special loops is countable, and each special loop determines an element of $\pi_1(M, q)$. To show that $\pi_1(M, q)$ is countable, therefore, it suffices to show that every element of $\pi_1(M, q)$ is represented by a special loop. Suppose $f \colon [0, 1] \to M$ is any loop based at q. The collection of components of sets of the form $f^{-1}(U)$ as U ranges over $\mathcal{U}$ is an open cover of $[0, 1]$, so by compactness it has a finite subcover. Thus there are finitely many numbers $0 = a_0 < a_1 < \ldots < a_k = 1$ such that $[a_{i-1}, a_i] \subset f^{-1}(U)$ for some $U \in \mathcal{U}$. For each i, let f_i be the restriction of f to the interval $[a_{i-1}, a_i]$, reparametrized so that its domain is $[0, 1]$, and let $U_i \in \mathcal{U}$ be a coordinate ball containing the image of f_i. For each i, we have $f(a_i) \in B_i \cap B_{i+1}$, and there is some $x_i \in \mathcal{X}$ that lies in the same component of $B_i \cap B_{i+1}$ as $f(a_i)$. Let g_i be a path in $B_i \cap B_{i+1}$ from x_i to $f(a_i)$, with the understanding that $x_0 = x_k = q$, and g_0 and g_k are both equal to the constant path c_q based at q. Then, because $g_i^{-1} \cdot g_i$ is path homotopic to a constant path,

$$\begin{aligned} f &\sim f_1 \cdot \ldots \cdot f_k \\ &\sim g_0 \cdot f_1 \cdot g_1^{-1} \cdot g_1 \cdot f_2 \cdot g_2^{-1} \cdot \ldots \cdot g_{k-1}^{-1} \cdot g_{k-1} \cdot f_k \cdot g_k^{-1} \\ &\sim \tilde{f}_1 \cdot \tilde{f}_2 \cdot \ldots \cdot \tilde{f}_n \end{aligned}$$

   where $\tilde{f}_i = g_{i-1} \cdot f_i \cdot g_i^{-1}$. For each i, $\tilde{f}_i$ is a path in U_i from x_{i-1} to x_i. Since B_i is simply connected, $\tilde{f}_i$ is path homotopic to $p^{U_i}_{x_{i-1},x_i}$. It follows that f is path homotopic to a special loop, as claimed.
9. ($\Rightarrow$) part is easy since

$$\delta l_\sigma \delta\varphi\omega = \delta(\varphi \circ l_\sigma)\omega = \delta(l_{\varphi(\sigma)} \circ \varphi)\omega = \delta\varphi\omega.$$

   For the converse, let $\mathcal{I}$ be an ideal generated by

$$\delta\pi_1\delta\varphi(\omega_i) - \delta\pi_2(\omega_i)$$

for projections $\pi_1 \colon G \times H \to G, \pi_2 \colon G \times H \to H$. Then by direct calculation similar to 3.15, (6), it is a differential ideal. Therefore, there exists I, maximal connected integral manifold of $\mathcal{I}$ through (e,e). But since $\delta\pi_1, \delta\pi_2$, and $\delta\varphi$ pull left invariant forms back to left invariant forms, by Thm 3.19, Corollary (c), I is a Lie subgroup of $G \times H$. Meanwhile, let (G, g) be a graph of φ where $g(\sigma) = (\sigma, \varphi(\sigma))$. Then also by direct calculation (following 2.33), it is an integral manifold of $\mathcal{I}$. Since G is connected, $g(G) \subset I$ by maximality. Now, let $\pi_1|_I \colon I \to G$ be a restriction map which is a homomorphism. Then, $d\pi_1 \colon I_{(e,e)} \to G_e$ is an isomorphism since for

$$(X,Y) \in I_{(e,e)} \Leftrightarrow \forall i, [\delta\pi_1 \delta\varphi(\omega_i) - \delta\pi_2(\omega_i)](X,Y) = 0 \Leftrightarrow Y = d\varphi(X).$$

Therefore by Theorem 3.26, $\pi_1|_I$ is a covering map. If $(\sigma,\tau) \in I$, then by path–connectedness, there is a path $\ell = (\ell_G, \ell_H)$ from (e,e) to (σ,τ) in I. Then $\left(\ell_G^{-1}, \varphi(\ell_G^{-1})\right) \cdot \ell$ is also a path in I which joins (e,e) and $(e, \varphi(\sigma^{-1}) \cdot \tau)$ since $g(G) \subset I$. However, since the "G coordinate" of this path is constantly e and there is a local isomorphism from a neighborhood of $(e,e) \in I$ to a neighborhood of $e \in G$, it should be a constant path, which means $\varphi(\sigma^{-1}) \cdot \tau = e$. Therefore, since I is a Lie subgroup, $(\sigma,\tau) \in I \Rightarrow (\sigma^{-1}, \tau^{-1}) \in I \Rightarrow \varphi(\sigma) = \tau \Rightarrow I \subset g(G)$. Thus $I = g(G)$ and it means that φ is a homomorphism.

Connectedness is necessary: Consider $G = \mathbb{Z}, H = \mathbb{R}$, $\varphi \colon \mathbb{Z} \to \mathbb{R}, x \mapsto x^2$.

10. If there exists $A \in \mathfrak{gl}(2,\mathbb{R})$ s.t.

$$\begin{pmatrix} -2 & 0 \\ 0 & -1 \end{pmatrix} = e^A.$$

Then regarding A as an element of $\mathfrak{gl}(2,\mathbb{C})$, one can find a matrix $L = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathfrak{gl}(2,\mathbb{C})$ s.t. LAL^{-1} is upper triangular. Then

$$L \begin{pmatrix} -2 & 0 \\ 0 & -1 \end{pmatrix} L^{-1} = Le^A L^{-1} = e^{LAL^{-1}}$$

is also upper triangular.

$$L \begin{pmatrix} -2 & 0 \\ 0 & -1 \end{pmatrix} L^{-1} = \frac{1}{ad-bc} \begin{pmatrix} -2ad+bc & ab \\ -cd & -ad+2bc \end{pmatrix},$$

thus $cd = 0$. If $c = 0$,

$$L \begin{pmatrix} -2 & 0 \\ 0 & -1 \end{pmatrix} L^{-1} = \frac{1}{ad} \begin{pmatrix} -2ad & ab \\ 0 & -ad \end{pmatrix} = \begin{pmatrix} -2 & b \\ 0 & -1 \end{pmatrix}.$$

If $d = 0$,

$$L \begin{pmatrix} -2 & 0 \\ 0 & -1 \end{pmatrix} L^{-1} = \frac{1}{-bc} \begin{pmatrix} bc & ab \\ 0 & 2bc \end{pmatrix} = \begin{pmatrix} -1 & -\frac{a}{c} \\ 0 & -2 \end{pmatrix}.$$

Therefore, the diagonal elements of $e^{LAL^{-1}}$ should be -2 and -1. Meanwhile, if we let λ_1, λ_2 be the eigenvalues of A (and LAL^{-1}), since A is real, two eigenvalues are either both real or complex conjugates. But since LAL^{-1} is upper triangular, the diagonal elements of $e^{LAL^{-1}}$ are e^{λ_1} and e^{λ_2}, which should not be -2 and -1 in either case. Therefore, there is no A satisfying the condition.

11. For η defined as the text, using Exercise 1.24(d),

$$d\eta(X,Y) = d\eta d\iota_e^1(X) + d\eta d\iota_e^2(Y) = X + Y$$

for $\iota_e^1 \colon G \to G \times G, \sigma \mapsto (\sigma, e)$ and $\iota_e^2 \colon G \to G \times G, \sigma \mapsto (e, \sigma)$. Therefore,

$$d\alpha = d\eta(d\sigma, d\beta) = d\sigma + d\beta.$$

Evaluating both sides at $\frac{\partial}{\partial t}\big|_{t=0}$, we obtain the given result.

12. For any $\sigma \in G$, a path ℓ from e to σ, and $\tau \in \ker\varphi$, $\ell \cdot \tau \cdot \ell^{-1}$ is a curve in $\ker\varphi$ also. Since $\ker\varphi$ is discrete, the curve should be constant, which means $\sigma \cdot \tau \cdot \sigma^{-1} = \tau$. Therefore $\ker\varphi \subset Z(G)$. For the second statement, for any Lie group G, let $\pi \colon \tilde{G} \to G$ is the universal covering. Then since π is a local homeomorphism, $\ker\pi$ is discrete, so $\ker\pi \in Z(\tilde{G})$, which says that $\ker\pi$ is abelian. But by the lifting property, the fundamental group of G can be identified with a subgroup of $\ker\pi$, so it is also abelian.

13. Let the Lie algebra of 3.5(d) be $L = (\mathbb{R}^2, [\cdot,\cdot])$. For any non-abelian Lie algebra $\bar{L} = (\mathbb{R}^2, \overline{[\cdot,\cdot]})$ which is defined by $\overline{[e_1, e_2]} = se_1 + te_2$, let

$$f \colon \bar{L} \to L, \;\; ae_1 + be_2 \mapsto (at - bs)e_1 + (as + bt)e_2.$$

It is by definition a linear isomorphism, and

$$\begin{aligned} f(\overline{[ae_1 + be_2, ce_1 + de_2]}) &= (ad - bc)f(se_1 + te_2) = (ad - bc)(s^2 + t^2)e_2, \\ [f(ae_1 + be_2), f(ce_1 + de_2)] &= [(at - bs)e_1 + (as + bt)e_2, (ct - ds)e_1 + (cs + dt)e_2] \\ &= ((at - bs)(cs + dt) - (as + bt)(ct - ds))e_2 \\ &= (ad - bc)(s^2 + t^2)e_2. \end{aligned}$$

Therefore f is a Lie algebra isomorphism. For the statement below, consider $(\mathbb{R}^{>0}, \cdot) \times (\mathbb{R}, +)$ instead of 3.3(h). Then it can be easily checked that $\left\{x\frac{\partial}{\partial x}, x\frac{\partial}{\partial y}\right\}$ is left invariant tangent fields that satisfies

$$\left[x\frac{\partial}{\partial x}, x\frac{\partial}{\partial y}\right] = x\frac{\partial}{\partial y}.$$

Therefore, by Theorem 3.27, Corollary, it is (up to isomorphism) the unique simply connected 2-dimensional non-abelian Lie group.

14. Let $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Then

$$e^A e^B = \begin{pmatrix} 1 & e \\ 0 & e \end{pmatrix}, \quad e^{A+B} = \begin{pmatrix} 1 & e-1 \\ 0 & e \end{pmatrix}.$$

15. If suffices to check that all Jordan canonical forms are in the image, since basis change and the exponential function commute, and the exponent of block diagonal matrices is block diagonal matrices of which blocks consist of the exponent of the original blocks. For A an elementary Jordan matrix, let $A = \lambda I \cdot N$, where the diagonal entries of N are 1's. Since $(I-N)^k = 0$ for $k > n$,

$$\log N = -\sum_{k=1}^{\infty} \frac{(I-N)^K}{k}$$

exists and $e^{\log N} = N$ by calculation dealing with formal series. Also, $\lambda I = e^{\log \lambda I}$ for any $\log \lambda$ regardless of the choice of the branch of log. Since $\log \lambda I$ is in the center, $e^{\log \lambda I + \log N} = e^{\log \lambda I} e^{\log N} = A$.

16. It is a Lie algebra because of Proposition 1.55 and if we denote it by $\mathfrak{h}$, $\mathfrak{h} \to G_e, X \mapsto X(e)$ is an isomorphism. Also, for X a left invariant vector field on G,

$$dr_\sigma d\varphi(X) = d\varphi dl_{\sigma^{-1}}(X) = d\varphi(X),$$

so $d\varphi(X) \in \mathfrak{h}$. Also, for $\eta \colon G \times G \to G, (\sigma, \tau) \to \sigma\tau$, $\eta(\mathrm{id}_G, \varphi)$ is a constant function to e. Thus $0 = d\eta(d\,\mathrm{id}_G, d\varphi) = d\,\mathrm{id}_G + d\varphi$ and $d\varphi(X)(e) = -X(e)$. Also, by the same argument, $d\varphi(\mathfrak{h}) \subset \mathfrak{g}$. Then, since φ is an involution, $d\varphi d\varphi = \mathrm{id}$, so $\mathfrak{g}$ and $\mathfrak{h}$ are isomorphic via $d\varphi$.

17. Consider ($\mathbb{Q}$, discrete topology) $\hookrightarrow$ ($\mathbb{R}$, standard topology).

18. By Theorem 3.50, Corollary (b) and Theorem 3.28, $\mathbb{R}^n$ is the unique simply connected abelian Lie group up to isomorphism. Then, since $\pi \colon \mathbb{R}^n \to G$ is the universal cover of G, $G \simeq \mathbb{R}^n / \ker \pi$. Also, the kernel of a covering homomorphism is discrete. Then following the argument in the text, $\ker \pi = \bigoplus_{i=1}^k \mathbb{Z} v_i$ for some linearly independent elements $v_1, \ldots, v_k$. Therefore, $G \simeq \mathbb{R}^n / \bigoplus_{i=1}^k \mathbb{Z} v_i \simeq \mathbb{R}^{n-k} \times T^k$.

19. Clearly $A_B(V)$ is an abstract subgroup of $\mathrm{Aut}(V)$ and $\mathfrak{d}_B$ is a subalgebra of $\mathrm{End}(V)$. According to Theorem 3.42, $A_B(V)$ is a closed Lie subgroup of $\mathrm{Aut}(V)$ since

$$A_B(V) = \bigcap_{v,w \in V} \ker(\alpha \mapsto (\alpha(v), \alpha(w)) - (v, w))$$

is closed. $\mathfrak{a}$ be the Lie algebra of $A_B(V)$. If $l \in \mathfrak{a}$, $\exp tl \in A_B(V)$. Therefore

$$((\exp tl)(v), (\exp tl)(w)) = (v, w).$$

Taking their derivatives at $t = 0$,

$$(lv, w) + (v, lw) = 0.$$

Therefore $l \in \mathfrak{d}_B$. Conversely, suppose that $l \in \mathfrak{d}_B$. Then $B(l \otimes 1 + 1 \otimes l) = 0$ by definition. Thus, $B(l \otimes 1 + 1 \otimes l)^n = 0$ for $n \geq 1$ and $B \circ e^{t(l \otimes 1 + 1 \otimes l)} = B$. But since

$$e^{t(l \otimes 1 + 1 \otimes l)} = e^{t(l \otimes 1)} e^{t(1 \otimes l)} = (e^{tl} \otimes 1)(1 \otimes e^{tl}) = e^{tl} \otimes e^{tl},$$

$\exp tl \in A_B(V)$, and $l \in \mathfrak{a}$. Therefore, $\mathfrak{a} = \mathfrak{d}_B$.

20. Thanks to the outline, we only need to prove that $A(G)$ is naturally isomorphic with the closed subgroup of $A(\tilde{G})$ consisting of those automorphisms of $\tilde{G}$ which map D onto D. However, if

$g\colon G \to G \in A(G)$, then by the lifting property, there is a unique map $\tilde{g}\colon \tilde{G} \to \tilde{G}$ s.t. $\pi \circ \tilde{g} = g \circ \pi$, and since $\tilde{g} \circ g^{-1} = \mathrm{id}_{\tilde{G}}$ by the uniqueness, $\tilde{g} \in A(\tilde{G})$. By definition $\tilde{g}(D) \subset D$ and $\tilde{g}(D^c) \subset D^c$, thus $\tilde{g}(D) = D$ since it is an automorphism. Conversely, if $h \in A(\tilde{G})$ and $h(D) = D$, there exists a map $\hat{h}\colon G \to G$ s.t. $\hat{h} \circ \pi = \pi \circ h$ and it is also an automorphism since $\ker(\pi \circ h) = \ker \pi$. Then it is easy to see that $\hat{\tilde{g}} = g$ and $\tilde{\hat{h}} = h$.

21. $\varphi\colon U(n) \to S^1 \times SU(n), A \mapsto \left(\det A, \frac{A}{\det A}\right)$ is a diffeomorphism.
22. (a) First, find an eigenvalue λ and a corresponding unit eigenvector v. Since

$$\lambda \|v\| = \lambda \bar{v}^t v = \bar{v}^t A v = \bar{v}^t \bar{A}^t v = \bar{\lambda} \bar{v}^t v = \bar{\lambda} \|v\|,$$

λ is real. Then if $v^t \bar{w} = 0$,

$$\langle v, Aw \rangle = v^t \overline{Aw} = v^t \bar{A} \bar{w} = v^t A^t \bar{w} = \lambda v^t \bar{w} = 0.$$

Therefore $\langle v \rangle$ and $\langle v \rangle^\perp$ are both A-stable. Then we can use induction on the rank of A to find orthonormal basis which consists of $\{v_i\}$ eigenvectors of A. Set $B = (v_1 v_2 \dots v_n)^{-1}$. For the real symmetric case, first find an eigenvalue λ (which is real) and a corresponding eigenvector v which can be complex valued. Since

$$A\bar{v} = \overline{Av} = \overline{\lambda v} = \lambda \bar{v},$$

$\bar{v}$ is also an eigenvector of λ. Therefore, $\mathrm{Re}(v)$ and/or $\mathrm{Im}(v)$ is an real eigenvector of λ and after scaling, we can find an unit real eigenvector. Now use the same induction as above.

(b) For a hermitian matrix A, by (a) we can find $B \in U(n)$ s.t. BAB^{-1} is diagonal. Then $e^A = B^{-1} e^{BA^{-1}} B$. Since eigenvalues of A is real, $e^{BAB^{-1}}$ is a diagonal matrix whose diagonal entries are all positive. Therefore, since $B \in U(n)$, e^A is also positive definite hermitian. Meanwhile, for a positive definite hermitian matrix H, we can find $B \in U(n)$ as above. Then since eigenvalues of H are all positive, we can find a real diagonal matrix A s.t. $e^A = BHB^{-1}$. Then $e^{B^{-1}AB} = H$, and $B^{-1}AB$ is a hermitian matrix. If $e^A = e^{A'}$ for two hermitian matrices A and A', find $B \in U(n)$ s.t. BAB^{-1} is real diagonal, and note that $e^{BAB^{-1}} = e^{BA'B^{-1}}$ is positive definite diagonal. Also, there exists $C \in U(n)$ s.t. $CBA'B^{-1}C^{-1}$ is real diagonal, and $e^{CBAB^{-1}C^{-1}} = e^{CBA'B^{-1}C^{-1}}$ is also positive definite diagonal. But since $CBAB^{-1}C^{-1}$ is also real diagonal by direct calculation, it means $CBAB^{-1}C^{-1} = CBA'B^{-1}C^{-1}$, which says $A = A'$. Therefore exp is well-defined one-to-one onto.

For a real symmetric A, e^A is positive definite hermitian by above and real, so positive definite symmetric. Injectivity is explained above. For a positive definite symmetric matrix H, we can find $B \in O(n)$ as (a), and there exists a real diagonal matrix A s.t. $e^A = BHB^{-1}$. Then note that $e^{B^{-1}AB} = H$ and $B^{-1}AB$ is real symmetric.

23. If $\sigma = PR = P_1 R_1$, then

$$P^2 = PRR^t P^t = P_1 R_1 R_1^t P_1^t = P_1^2.$$

Find A, B real symmetric matrices satisfying $e^A = P$ and $e^B = P_1$, and we see that

$$e^{2A} = (e^A)^2 = P^2 = P_1^2 = (e^B)^2 = e^{2B},$$

so by injectivity, $2A = 2B$ or $A = B$. It follows that $P = P_1$. Therefore, $R = R_1$ also.

24. $\sigma\bar{\sigma}^t$ is positive definite hermitian, so $\exists B \in U(n)$ s.t. $B\sigma\bar{\sigma}^t B^{-1}$ is positive definite diagonal. Set $P = B^{-1}(B\sigma\bar{\sigma}^t B^{-1})^{\frac{1}{2}}B$, which is positive definite hermitian, and $R = P^{-1}\sigma$ a unitary matrix.
25. $U(n)$ is connected, and there exists a path $tI + (1-t)P$ joining I and P a positive definite hermitian matrix. Thus by similar argument of Theorem 3.68 and using Exercise 24, $Gl(n, \mathbb{C})$ is connected.
26. Follow the argument and use actions $SU(n) \times X \to X$ for (b) and $SO(n) \times \mathbb{P}^{n-1} \to \mathbb{P}^{n-1}$ for (c).

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1.4 习题: Warner GTM 94/Manifolds

原网页

1. Let $x = (x_1, \ldots, x_{n+1})$. Then for $x \in S_n - n$, the map $p_n \colon S^d - n \to \mathbb{R}^n$ is defined as $p_n(x) = \frac{1}{1-x_{n+1}}(x_1, \ldots, x_n)$. Similarly, if $x \in S_n - s$, then $p_n(x) = \frac{1}{1+x_{n+1}}(x_1, \ldots, x_n)$. Therefore each function is continuous on its domain, respectively. Also on $p_s(S_n - n - p), p_n \circ p_s^{-1} \colon p_s(S_n - n - p) \to p_n(S_n - n - p) \subset \mathbb{R}^n$ is defined as $p_n \circ p_s^{-1}(y_1, \ldots, y_n) = \frac{1}{k}(y_1, \ldots, y_n)$ $(k = \sum_{i=1}^n y_i^2)$, which is smooth since $(0, \ldots, 0) \notin p_s(S_n - n - p)$. Similarly, $p_s \circ p_n^{-1}$ is also smooth. Therefore, the maximal collection including these coordinate systems is indeed a differentiable structure on S^d.
2. Let $f(t) = t^3$. Then $\mathrm{id}_{\mathbb{R}} \circ f^{-1}(t) = t^{\frac{1}{3}}$, which is not differentiable at $t = 0$, and it means $\mathscr{F} \neq \mathscr{F}_1$. However, if we set $\varphi \colon (\mathbb{R}, \mathscr{F}) \to (\mathbb{R}, \mathscr{F}_1), t \mapsto t^{\frac{1}{3}}$, then $\mathrm{id}_{\mathbb{R}} \circ \varphi \circ f^{-1}(t) = t$, which is a diffeomorphism. Therefore, $(\mathbb{R}, \mathscr{F})$ and $(\mathbb{R}, \mathscr{F}_1)$ are diffeomorphic.
3. By Theorem 1.11, given an open cover $\{U_\alpha\}$, there exists a partition of unity $\{\varphi_\alpha\}$ subordinate to the cover $\{U_\alpha\}$. If we set $V_\alpha = \varphi_\alpha^{-1}(\mathbb{R} \backslash \{0\})$, then it is also an open cover since for every $m \in M$, $\sum_\alpha \varphi_\alpha(m) = 1$, which means $\varphi_\alpha(m) \neq 0$ for some α, and $m \in V_\alpha$. Also, by definition of support, $\overline{V_\alpha} \subset U_\alpha$. Therefore $\{V_\alpha\}$ is the refinement satisfying the condition.
4. Suppose two closed subsets $C, D \subset M$ are given. Then by regularity, $\forall c \in C$, $\exists$ open subset $U_c, V_c \subset M$ s.t. $c \in U_c, D \subset V_c$, and $U_c \cap V_c = \varnothing$. Then $\{U_c\}_{c \in C} \cup M \backslash C$ is an open cover of M. Hence, by paracompactness, there exists a refinement of $\{U_c\}_{c \in C} \cup M \backslash C$ which is locally finite. By discarding the open sets which do not intersect C, one can obtain an open cover of C which is locally finite, say $\{\widetilde{U}_\beta\}$. Then, $\forall d \in D, \exists \widetilde{V_d}$ s.t. $\widetilde{V}_d \cap \widetilde{U}_\beta \neq \varnothing$ for only finite β's, which we can

label as $\beta_1, \ldots, \beta_n$. Since $\{\widetilde{U}_\beta\}$ is a refinement, one can find $c_i \in C$ for each β_i s.t. $\widetilde{U}_{\beta_i} \subset U_{c_i}$. If we set $W_d = \widetilde{V_d} \cap \bigcap_{i=1}^n V_{c_i}$, then one can easily see that $\forall \beta, W_d \cap \widetilde{U}_\beta = \varnothing$ and W_d is an open neighborhood of d. If we set $U = \bigcup \widetilde{U}_\beta, W = \bigcup W_d$, then each open set is a neighborhood of C and D respectively which does not intersect each other. Since C and D are arbitrary, M is normal.

5. (a) If $\vec{a} = (a_1, \ldots, a_d)$ and $\vec{v} = (v_1, \ldots, v_d)$, $\tilde{\psi} \circ \tilde{\varphi}^{-1}(\vec{a}, \vec{v}) = \big(\psi \circ \varphi^{-1}(\vec{a}), \mathcal{J}(\psi \circ \varphi^{-1})(\vec{v})\big)$, which is $\mathcal{C}^\infty$.
   (b) $\tilde{\varphi}\colon \pi^{-1}(U) \to \mathbb{R}^{2d}$ is a bijection between $\pi^{-1}(U)$ and $\varphi(U) \times \mathbb{R}^d$, which is open in $\mathbb{R}^{2d}$. Also, for any (V, ψ), $\tilde{\psi}^{-1}(W') \cap \pi^{-1}(U)$ is of the form $\tilde{\varphi}^{-1}(W)$, which is $\tilde{\varphi}^{-1}\big(\tilde{\varphi} \circ \tilde{\psi}^{-1}\big(\psi(U \cap V) \times \mathbb{R}^d \cap W'\big)\big)$. Therefore, on the defined topology, $\tilde{\varphi}$ is a homeomorphism and that $T(M)$ is a $2d$-dimensional locally Euclidean space directly follows. For second countability, since M is second countable, we can choose countable such (U, φ) that cover $T(M)$. Since $\mathbb{R}^{2d}$ is also second countable, it follows that $T(M)$ is second countable.
   (c) By (a) and (b), it directly follows.
6. By Theorem 1.30, Corollary (a), ψ is a diffeomorphism if and only if $d\psi$ is surjective everywhere. Suppose that $d\psi$ is not surjective at $m \in M$. Since it is nonsingular, $\dim M = p < \dim N = d$. Pick a coordinate system on N, (U, φ), such that $\varphi(U) = \mathbb{R}^d$. Then $\varphi \circ \psi\colon \psi^{-1}(U) \to \mathbb{R}^d$ is surjective. Also, by the second countability of M, one can find countable coordinate systems of $\psi^{-1}(U)$, say $\{(V_\alpha, f_\alpha)\}_{\alpha \in A}$. Then $\mathbb{R}^d = \bigcup_{\alpha \in A} \varphi \circ \psi \circ f_\alpha^{-1}(\mathbb{R}^p)$. Using Proposition 1.35, since $\varphi \circ \psi \circ f_\alpha^{-1}$ is an immersion, one can obtain a countable cover $\{U_\beta\}$ of $\mathbb{R}^p$ such that $\varphi \circ \psi \circ f_\alpha^{-1}(U_\beta)$ is a subset of a slice of a open set in $\mathbb{R}^d$, which is nowhere dense. Therefore, by the Baire category theorem, $\bigcup_{\alpha \in A} \varphi \circ \psi \circ f_\alpha^{-1}(\mathbb{R}^p)$ cannot be all of $\mathbb{R}^d$, contradicting the assumption. Therefore, ψ is a diffeomorphism.
7. (a) For two differentiable structures on A, $\mathscr{F}$ and $\mathscr{F}'$, $\mathrm{id}_A\colon (A, \mathscr{F}) \to (A, \mathscr{F}')$ is continuous, hence $\mathcal{C}^\infty$ by Theorem 1.32. Since its inverse is also $\mathcal{C}^\infty$, it is a diffeomorphism: that is, two structures are the same.
   (b) Let $(\tilde{A}, \iota')$ be A equipped with a manifold structure possibly different from the induced structure form M. Then $\mathrm{id}_A\colon \tilde{A} \to A$ is continuous and hence $\mathcal{C}^\infty$ by Theorem 1.32. Also it is nonsingular since $d\iota' = d(\iota \circ \mathrm{id}_A)$ is. Therefore, by Exercise 6, it is a diffeomorphism, which means that a manifold structure on $\tilde{A}$ is equal to that on A.
8. Use notations of Theorem 1.37. Since $\left\{ \frac{\partial f_i}{\partial s_j}\Big|_{(r_0, s_0)} \right\}$ is nonsingular (i.e., invertible), by continuity, there exists an open set $(r_0, s_0) \in V_0 \times W_0 \subset U$ such that df is surjective. Also, $(r_0, s_0) \in f^{-1}(0)$, hence nonempty. Therefore, by Theorem 1.38, $P = f|^{-1}_{V_0 \times W_0}(0)$ is an imbedded submanifold of $V_0 \times W_0$ of dimension $c - d$. Now, let $\pi_1\colon V_0 \times W_0 \to V_0$ be the first projection. Then, $\pi_1 \circ \iota\colon P \to V_0$ is $\mathcal{C}^\infty$, and $d(\pi_1 \circ \iota)\colon P_{(r_0, s_0)} \to V_{0r_0}$ is an isomorphism. $\big(\because P_{(r_0, s_0)}$ can be identified with $\ker df \subset V_0 \times W_{0(r_0, s_0)}$, and $df|_{W_{0s_0}}$ is an isomorphism, $V_0 \times W_{0(r_0, s_0)} = \ker df \oplus W_{s_0} = V_{0r_0} \oplus W_{0s_0}$. Hence $\ker df \to V_{0r_0}$ is an isomorphism also.) Therefore, by Theorem 1.30, one can find an open set $(r_0, s_0) \in V \times W \subset V_0 \times W_0$ such that $\pi_1 \circ \iota^{-1}_{P \cap V \times W}\colon V \to P \cap V \times W$ is an one-to-one onto $\mathcal{C}^\infty$ map. Therefore, if we set $g = \pi_2 \circ \iota \circ \big(\pi_1 \circ \iota|^{-1}_{P \cap V \times W}\big)\colon V \to W$ (where π_2 is the second projection), then it is the map satisfying the condition.
9. $\frac{\partial f}{\partial x} = 3x^2 + y$, $\frac{\partial f}{\partial y} = x + 3y^2$. Therefore, $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$ if and only if $(x, y) = (0, 0), \big(-\frac{1}{3}, -\frac{1}{3}\big)$.

Therefore, by Theorem 1.38, $f^{-1}\left(f\left(\frac{1}{3},\frac{1}{3}\right)\right)$ is an imbedded submanifold, since $(0,0), \left(-\frac{1}{3},-\frac{1}{3}\right) \notin f^{-1}\left(f\left(\frac{1}{3},\frac{1}{3}\right)\right)$. $f^{-1}(f(0,0))$ is not a manifold since $(0,0)$ is a node. If $p = \left(-\frac{1}{3},-\frac{1}{3}\right)$, then $f(x,y) = f\left(\frac{1}{3},\frac{1}{3}\right)$ iff

$$x^3 + xy + y^3 - \frac{1}{27} = 0$$

iff

$$\left(x + y - \frac{1}{3}\right)\left(x^2 + y^2 + \frac{1}{9} + \frac{x}{3} + \frac{y}{3} - xy\right) = 0$$

iff $x + y = \frac{1}{3}$, since

$$x^2 + y^2 + \frac{1}{9} + \frac{x}{3} + \frac{y}{3} - xy = \frac{1}{2}\left(\left(x + \frac{1}{3}\right)^2 + \left(y + \frac{1}{3}\right)^2 + (x-y)^2\right).$$

Therefore $f^{-1}\left(f\left(-\frac{1}{3},-\frac{1}{3}\right)\right)$ is a line, which is an imbedded submanifold.

10. Since $f(M)$ is compact, $\exists m \in M$ s.t. $|f(m)|$ is the biggest. If $df\colon M_m \to \mathbb{R}^n_{f(m)}$ is nonsingular, then by Theorem 1.30, Corollary (a), $\exists U \subset f(M)$ neighborhood of $f(m)$, which contradicts the maximality of $|f(m)|$.
11. Not closed: $\left(-\frac{\pi}{2},\frac{\pi}{2}\right) \subset \mathbb{R}, g(x) = \tan x$.
    Not imbedded: Second Example of 1.28, $g(x) = x$.
    To prove the converse: if there is a disk $D(x,r) \subset \mathbb{R}^n$, one can construct a $\mathcal{C}^\infty$ function which is nonzero exactly on $D(x,r)$ and zero elsewhere. Since M is locally Euclidean, there exists a basis $\{B_\alpha\}$ of M such that $\forall B_\alpha, \exists g_\alpha$ s.t. $g_\alpha^{-1}(\mathbb{R}\backslash\{0\}) = B_\alpha$. Then by the assumption there is a $\mathcal{C}^\infty$ function f_α on N such that $f_\alpha \circ \psi = g_\alpha$. Since $f_\alpha^{-1}(\mathbb{R}\backslash\{0\}) \cap \psi(M) = \psi(B_\alpha)$, $\psi(B_\alpha)$ is open in $\psi(M)$ with relative topology, which means ψ is an imbedding. Therefore we can say $M \subset N$. If M is not closed in N, then there is $n \in \bar{M}\backslash M$ and (U,φ) a coordinate system of n, and there is a $\mathcal{C}^\infty$ function g on M such that

$$g(m) = \frac{1}{\varphi(m) - \varphi(n)}$$

    on the neighborhood of n on M. However, since n is a limit point of M, it cannot be extended to all of N. Therefore, M should be closed in N.
12. (a) Since $df = \sum 2r_i dr_i$, it is surjective except at the origin. Therefore one can use Theorem 1.38. Also, since it is a submanifold with the relative topology, it should be the same as 1.5(d) by 1.33(b).
    (b) By following the argument, we need only to show that $d\psi_I$ is surjective. Since $\psi(A) = \left(\sum_k a_{ik}a_{jk}\right)_{ij}$, $d\psi_A = \left(\sum_k (a_{jk}da_{ik} + a_{ik}da_{jk})\right)_{ij}$. $\therefore\ d\psi_I = (da_{ij} + da_{ji})_{ij}$. But, since $Gl(n,\mathbb{R})_I \cong M_{n\times n}(\mathbb{R})$, it is surjective onto $\mathrm{Sym}_n(\mathbb{R})_I \cong \mathrm{Sym}_n(\mathbb{R})$.
13. (a) If we set $X = \sum a_i \frac{\partial}{\partial x_i}, Y = \sum b_i \frac{\partial}{\partial x_i}$ locally as Proposition 1.43(b),

$$[X,Y] = \sum_i \sum_j \left(a_i \frac{\partial b_j}{\partial x_i} - b_i \frac{\partial a_j}{\partial x_i}\right) \frac{\partial}{\partial x_j}.$$

    Therefore by 1.43(b), it is a smooth vector field.
    (b) Using the same notation as above,

$$[fX,gY] = \sum_i \sum_j \left(f a_i \frac{\partial g b_j}{\partial x_i} - g b_i \frac{\partial f a_j}{\partial x_i}\right) \frac{\partial}{\partial x_j}$$

$$= \sum_i \sum_j \left(fga_i \frac{\partial b_j}{\partial x_i} - fgb_i \frac{\partial a_j}{\partial x_i} + fa_i b_j \frac{\partial g}{\partial x_i} - ga_j b_i \frac{\partial f}{\partial x_i} \right) \frac{\partial}{\partial x_j}$$
$$= fg[X,Y] + f(Xg)Y - g(Yf)X.$$

(c)

$$[X,Y] = \sum_i \sum_j \left(a_i \frac{\partial b_j}{\partial x_i} - b_i \frac{\partial a_j}{\partial x_i} \right) \frac{\partial}{\partial x_j} = -\sum_i \sum_j \left(b_i \frac{\partial a_j}{\partial x_i} - a_i \frac{\partial b_j}{\partial x_i} \right) \frac{\partial}{\partial x_j} = -[Y,X].$$

(d) Let $Z = \sum c_i \frac{\partial}{\partial x_i}$. Then,

$$[[X,Y],Z] = \sum_i \sum_j \sum_k \left(\left(a_i \frac{\partial b_j}{\partial x_i} - b_i \frac{\partial a_j}{\partial x_i} \right) \frac{\partial c_k}{\partial x_j} - c_j \left(\frac{\partial a_i}{\partial x_j} \frac{\partial b_k}{\partial x_i} + a_i \frac{\partial^2 b_k}{\partial x_j \partial x_i} - \frac{\partial b_i}{\partial x_j} \frac{\partial a_k}{\partial x_i} - b_i \frac{\partial^2 a_k}{\partial x_j \partial x_i} \right) \right) \frac{\partial}{\partial x_k}$$

and by direct calculation, the Jacobi identity gets proved.

14. No. If $X_x = \exp(x)\frac{\partial}{\partial r}$ and $\gamma_0(t)$ is the integral curve at $0 \in \mathbb{R}$, then by definition, $\frac{d\gamma_0}{dt}(t) = \exp(\gamma_0(t))$ and $\gamma_0(0) = 0$. By solving the differential equation, $\gamma_0(t) = -\log(-t+1)$, and it is not defined on $(-\infty, \infty)$.
15. Use the equation of Exercises 13(a) above.
16. Consider the first example of 1.31 and a curve along the horizontal part of the submanifold. It is clear that on $0 \in \mathbb{R}^2$, the tangent vector of the integral curve should be horizontal, whereas N_0 consists of vertical vectors.
17. By 1.48(f), $\bigcup_{t>0} \mathcal{D}_t = \bigcup_{t<0} \mathcal{D}_t = M$, thus by the compactness of M, $\exists t_0 > 0$ s.t. $\mathcal{D}_{t_0} = \mathcal{D}_{-t_0} = M$. Thus by 1.48(g), X_{t_0} is an endomorphism, and $\forall n \in \mathbb{N}$, $X_{t_0}^n$ is defined on all of M. Therefore by 1.48(h), $\forall N > 0, \mathcal{D}_N = M$. Likewise, $\forall N < 0, \mathcal{D}_N = M$. Since it is clear that $D_0 = M$, M is complete.
18. If it is one-to-one, the image should be infinite. Therefore, there exist $a, b, c \in \operatorname{im} f$ s.t. $a < b < c$. By erasing b, a and c can be separated by open sets $(-\infty, b) \cap \operatorname{im} f$ and $(b, \infty) \cap \operatorname{im} f$, hence $\operatorname{im} f$ is disconnected, whereas $\mathbb{R}^2 \backslash f^{-1}(b)$ is connected since $f^{-1}(b)$ consists of one point. This contradicts the continuity of f.
19. (1.60⇐1.61) Follow the remark of 1.61. Then,

$$[Y_a, Y_b] = \sum_{j=1}^{d-c} \left(\frac{\partial f_{jb} \circ \gamma}{\partial y_a} - \frac{\partial f_{ja} \circ \gamma}{\partial y_b} \right) \frac{\partial}{\partial y_{c+j}} + \sum_{j=1}^{d-c} \sum_{k=1}^{d-c} \left(\frac{\partial f_{jb} \circ \gamma}{\partial y_{c+k}} f_{ka} - \frac{\partial f_{ja} \circ \gamma}{\partial y_{c+k}} f_{kb} \right) \frac{\partial}{\partial y_{c+j}}.$$

Since $\mathcal{D}$ is involutive, $[Y_a, Y_b] \in \mathcal{D}$. But the expression above does not include $\frac{\partial}{\partial y_i}$ terms for $1 \leq i \leq c$, it must be zero. Therefore, the assumption 1.61(2) is satisfied. Then we obtain a map α as 1.61(3). Since $a(r_0, s) = s$, $(\mathrm{id}, \alpha)^{-1}$ is $\mathcal{C}^\infty$ local homeomorphism, so shrinking the neighborhood if necessary, we obtain a local chart $\varphi = (\mathrm{id}, \alpha)^{-1} \circ \gamma \colon W \to U \times V$ and $(z_1, \ldots, z_d) = (\mathrm{id}, \alpha)^{-1} \circ (y_1, \ldots, y_d)$ is coordinate functions. Also, if $1 \leq i \leq c$,

$$\frac{\partial}{\partial z_i} = \sum_{j=1}^{d} \frac{\partial y_j}{\partial z_i} \frac{\partial}{\partial y_j} = \frac{\partial}{\partial y_i} + \sum_{j=1}^{d-c} f_{ji} \circ \gamma \frac{\partial}{\partial y_{c+j}} = Y_i$$

by 1.61(4) and (6). Therefore, if we let I be a submanifold defined by $z_j = \text{constant}$ for $c+1 \leq j \leq d$, it is an integral manifold of $\mathcal{D}$.

(1.60⇒1.61) Let $X_i = \frac{\partial}{\partial r_i} + \sum_{j=1}^{n} b_{ji} \frac{\partial}{\partial s_j}$ for $1 \leq i \leq n$ and $\mathcal{D}$ be a distribution generated by X_i's. Then by similar calculation above, $[X_a, X_b] = 0 \in \mathcal{D}$ by assumption 1.61 (2). Thus by 1.60, there exists a cubic coordinate system (W, φ) with coordinate functions $(x_1, \ldots, x_m, y_1, \ldots, y_n)$ defined as the theorem. Let us define α on the neighborhood of $(0,0)$ s.t. $y_j(r, \alpha_s(r)) = y_j(r_0, s), \forall 1 \leq j \leq n$ by implicit function theorem. We can define this because $X_1, \ldots, X_m, \frac{\partial}{\partial s_1}, \ldots, \frac{\partial}{\partial s_1}$ is a basis of tangent space and $X_i(y_j) = 0$ since y_j's are constant on integral manifolds of $\mathcal{D}$, which means $\left\{ \frac{\partial y_j}{\partial s_k} \right\}$ should be nonsingular. Then clearly $\alpha_s(r_0) = s$ and

$$\frac{\partial y_j}{\partial r_i} + \sum_{k=1}^{n} \frac{\partial y_j}{\partial s_k} \frac{\partial \alpha_{s,k}}{\partial r_i} = 0.$$

Also, $X_i(y_j) = 0$ means

$$\frac{\partial y_j}{\partial r_i} + \sum_{k=1}^{n} b_{ki} \frac{\partial y_j}{\partial s_k} = 0.$$

Since $\left\{ \frac{\partial y_j}{\partial s_k} \right\}$ is nonsingular, condition (4) is satisfied.

20. Let $M = \mathbb{R}, N = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and regard N as a submanifold imbedded in M and φ as an inclusion. If $X(t) = \tan(t) \frac{\partial}{\partial r}$, then it satisfies the assumption since φ is one-to-one, but it cannot be extended to any vector field on M.
21. It is clearly one-to-one, $\mathcal{C}^\infty$, and nonsingular. To simplify the calculation, let us think the torus as $T = \mathbb{R}^2/\mathbb{Z} \times \mathbb{Z}$ and $\varphi \colon \mathbb{R} \to T, x \mapsto (x, \alpha x)$. Then $\operatorname{im} \varphi \cap \{(0, y) \in T\}$ is dense in $\{(0, y) \in T\}$, since α is irrational. Also, given x_0, $\operatorname{im} \varphi \cap \{(x_0, y) \in T\}$ is dense in $\{(x_0, y) \in T\}$ since it is the transition of $\operatorname{im} \varphi \cap \{(0, y) \in T\}$. Since x_0 is arbitrary, the image of φ is dense in T.
22. Let t_0 be a point satisfying $\dot{\gamma}(t_0) = 0$. Then, the constant map whose image is $\gamma(t_0)$ is also an integral curve. Therefore, by Theorem 1.48(c), γ should be a constant map.
23. Let $\{U_\alpha\}$ be an open cover of M such that $\forall \alpha, (U_\alpha, \psi_\alpha)$ is a chart. By Theorem 1.11, there exists a partition of unity $\{\varphi_\alpha\}$ subordinate to the cover. Also, on each chart, there exists a positive definite inner product induced by the standard inner product on the Euclidean space, say $\langle\ ,\ \rangle_\alpha$, and it is smooth on each chart. If we set $\langle\ ,\ \rangle_m = \sum_\alpha \langle\ ,\ \rangle_{\alpha,m} \varphi_\alpha(m)$, then it is an inner product which makes M a Riemannian manifold.
24. (a) ($\Rightarrow$) Obvious.

($\Leftarrow$)$\forall (U, \varphi), \forall (V, \psi)$ charts of M and N respectively, $(U \times V, \varphi \times \psi)$ is a chart of $M \times N$. Then $(\varphi \times \psi) \circ \alpha = (\varphi \circ \pi_1 \circ \alpha, \psi \circ \pi_2 \circ \alpha)$ is $\mathcal{C}^\infty$ if $\pi_1 \circ \alpha, \pi_2 \circ \alpha$ are $\mathcal{C}^\infty$.

(b) Let $m \in U, n \in V$, and $(x_1, \ldots, x_s), (y_1, \ldots, y_t)$ be the local coordinates of U, V respectively. Then

$$(M \times N)_{(m,n)} = \bigoplus_i \mathbb{R} \frac{\partial}{\partial x_i}\Big|_{(m,n)} \bigoplus_j \mathbb{R} \frac{\partial}{\partial y_j}\Big|_{(m,n)} \cong M_m \oplus N_n.$$

(c) On $U \times V, \tilde{X} = \sum_i a_i(x_1, \ldots, x_s) \frac{\partial}{\partial x_i}$ and $\tilde{Y} = \sum_j b_j(y_1, \ldots, y_t) \frac{\partial}{\partial y_j}$. Therefore, by direct calculation, $[\tilde{X}, \tilde{Y}] = 0$.

(d) Let

$$v = \sum_i c_i \frac{\partial}{\partial x_i}\bigg|_{(m_0,n_0)} + \sum_j d_j \frac{\partial}{\partial y_j}\bigg|_{(m_0,n_0)} \in (M \times N)_{(m_0,n_0)}.$$

Then

$$d\pi_1(v) = \sum_i c_i \frac{\partial}{\partial x_i}\bigg|_{m_0}, \quad d\pi_2(v) = \sum_j d_j \frac{\partial}{\partial y_j}\bigg|_{n_0}.$$

Also,

$$\frac{\partial f \circ i_{n_0}}{\partial x_i}\bigg|_{m_0} = \frac{\partial f}{\partial x_i}\bigg|_{(m_0,n_0)}, \quad \frac{\partial f \circ i_{m_0}}{\partial y_j}\bigg|_{n_0} = \frac{\partial f}{\partial y_j}\bigg|_{(m_0,n_0)}.$$

Summing up,

$$v(f) = \sum_i c_i \frac{\partial f}{\partial x_i}\bigg|_{(m_0,n_0)} + \sum_j d_j \frac{\partial f}{\partial y_j}\bigg|_{(m_0,n_0)} = v_1(f \circ i_{n_0}) + v_2(f \circ i_{m_0}).$$

1.5 习题: Warner GTM 94/Sheaves, Cohomology, and the de Rham Theorem

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1. (Following Lecture 1, Proposition 1 of [6]) Let f be the 0-section, $x \in X$ and let G be an open set containing $f(x) = 0_x$. Then there is an open set G_1 such that $0_x \in G_1 \subset G$ and $\pi|_{G_1}$ is a homeomorphism of G_1 onto open $\pi(G_1)$. Since $0_x + 0_x = 0_x$, and addition is continuous, there exist open sets H, K with $0_x \in H$, $0_x \in K$ such that $H + K \subset G_1$. Let $L = G_1 \cap H \cap K$, then L is open, $0_x \in L$ and $\pi|_L$ is a homeomorphism of L onto open $\pi(L)$. Clearly $x = \pi(0_x) \in \pi(L)$. If $y \in \pi(L)$ there exists $q \in L$ with $\pi(q) = y$. Then $q \in H \cap K \cap \mathcal{S}_y$, so $q + q \in G_1 \cap \mathcal{S}_y$. However, since $q \in G_1$ and $\pi(G_1)$ is 1-1, $\{q\} = \mathcal{S}_y \cap G_1$. Therefore $q + q = q$, which means $q = 0_y$. Thus if $y \in \pi(L)$, $f(y) = 0_y \in L$. Since $\pi|_L \colon L \to \pi(L)$ is a homeomorphism, and $f|_{\pi(L)}$ is its inverse, it is also continuous. Since $x \in X$ is arbitrary, f is continuous on X.
2. If $f \colon U \to \mathcal{S}$ is a section and $x \in U$, then there exists an open set $f(x) \in V$ s.t. $\pi|_V \colon V \to \pi(V)$ is a homeomorphism and $\pi(V) \subset U$. Since f is continuous, $f^{-1}(V)$ is open, and $x \in f^{-1}(V) \subset \pi(V) \subset U$. Let $W = \pi^{-1} f^{-1}(V) \cap V$. Then it is an open neighborhood of $f(x)$, and $\pi(W) \subset f^{-1}(V)$. Also $f(f^{-1}(V)) \subset W$. But since $\pi \circ f|_{f^{-1}(V)} = \mathrm{id}_{f^{-1}(V)}, \pi(W) = f^{-1}(V)$. Also, $\pi|_W \colon W \to \pi(W) = f^{-1}(V)$ is a homeomorphism, $f(f^{-1}(V)) = W$. Since $f|_{f^{-1}(V)}$ is continuous with a continuous inverse $\pi|_W$, it is a homeomorphism. Therefore f is a local homeomorphism and hence an open map.
3. Let f, g be two sections and $f(x) = g(x)$. Then there exist an open set $x \in U$ and $x \in V$ s.t. $f \colon U \to f(U)$ and $g \colon V \to g(V)$ are local homeomorphisms. Since $f(x) = g(x) \in f(U) \cap g(V)$, $f(U) \cap g(V)$ is nonempty. Also, on $W = \pi(f(U) \cap g(V))$, which is an open neighborhood of x, $f(y) = (f \circ (\pi \circ g))(y) = ((f \circ \pi) \circ g)(y) = g(y)$ if $y \in W$. Therefore $f = g$ on W. Since 0-section is continuous (and indeed a section) by Exercise 1, the second statement follows.
4. It is because $f_x = 0$ does not mean only $f(x) = 0$, but "f is zero on the neighborhood of x". If $f \colon \mathbb{R} \to \mathbb{R}, x \mapsto x$, $f(0) = 0$ but $f_0 \neq 0$.

5. It is because projection of sheaves are local homeomorphisms and a sheaf mapping is locally equal to the composition of a projection and the inverse of another projection.
6. Let $\pi_1 \colon \mathcal{S}_1 \to X$ and $\pi_2 \colon \mathcal{S}_2 \to X$ be two sheaves on X and $\varphi, \psi \colon \mathcal{S}_1 \to \mathcal{S}_2$ be two sheaf mappings. Suppose $\varphi(x) = \psi(x)$. There exists an open set $U \ni x$ such that $\pi_1 \colon U \to \pi_1(U)$ is a local homeomorphism. Also, contracting U if necessary, we can assume that $\varphi \colon U \to \varphi(U), \psi \colon U \to \psi(U), \pi_2 \colon \varphi(U) \to \pi_2(\varphi(U)), \pi_2 \colon \psi(U) \to \pi_2(\varphi(U))$ are local homeomorphisms. Since $\varphi(x) = \psi(x) \in \varphi(U) \cap \psi(U)$, $\varphi(U) \cap \psi(U)$ is nonempty. Also, on $W = \pi_1^{-1}(\pi_2(\varphi(U) \cap \psi(U)))$, which is an open neighborhood of x since $\pi_1(U) \supset \pi_2(\varphi(U) \cap \psi(U))$, $\varphi(y) = \big(\varphi \circ (\pi_1^{-1} \circ \pi_2 \circ \psi)\big)(y) = \big((\varphi \circ \pi_1^{-1} \circ \pi_2) \circ \psi\big)(y) = \psi(y)$ if $y \in W$. Therefore $\varphi = \psi$ on W. By the way, if ψ is a "0-sheaf-mapping", then it is the composition of the 0-section and the projection map, hence continuous. Therefore, it is a sheaf mapping indeed and the second statement follows.
7. First, a lemma:

   Lemma. Let $\pi \colon \mathcal{S} \to X$ be a sheaf and $\mathcal{R} \subset \mathcal{S}$ be a subsheaf. If $f \colon U \to \mathcal{S}$ is a section, then $\mathcal{R} \cap \pi^{-1}(U) + f(U)$ is open in $\mathcal{S}$.

   Proof. Let $p_x \in \mathcal{R} \cap \pi^{-1}(U) + f(U)$ and $\pi(p_x) = x$. By definition, there exist $q_x \in \mathcal{R}$ and $r_x \in f(U)$ s.t. $p_x = q_x + r_x$. Let U be an open neighborhood of p_x which is locally homeomorphic to $\pi(U)$. Since addition is continuous, there exist $V \subset \mathcal{R} \cap \pi^{-1}(U)$ and $W \subset f(U)$ neighborhoods of q_x and r_x respectively, s.t. $\pi(V) = \pi(W), V + W \subset U$, and $\pi|_V \colon V \to \pi(V), \pi|_W \colon W \to \pi(W)$ are homeomorphisms. Let $Y = \pi(V) = \pi(W)$. Then $V + W \subset U \cap \pi^{-1}(Y)$. Also, $y \in U \cap \pi^{-1}(Y)$, then there exists $a_{\pi(y)} \in V \cap \mathcal{S}_{\pi(y)}$ and $b_{\pi(y)} \in W \cap \mathcal{S}_{\pi(y)}$. Then $a_{\pi(y)} + b_{\pi(y)} \in U \cap \mathcal{S}_{\pi(y)}$. But since $\pi|_U \colon U \to \pi(U)$ is 1-1, $y = a_{\pi(y)} + b_{\pi(y)} \in V + W$. Therefore $V + W = U \cap \pi^{-1}(Y)$ is open. But it means $p_x \in V + W \subset \mathcal{R} \cap \pi^{-1}(U) + f(U)$, so $\mathcal{R} \cap \pi^{-1}(U) + f(U)$ is open.

   Now come back to the question. First, we need to show that $\bar{\pi} \colon \mathcal{T} \to M$ is a sheaf.

   1) For $x \in \mathcal{T}$, there exists a $y \in \mathcal{S}$ s.t. $\varphi(y) = x$ and $y \in U$ s.t. $\pi|_U \colon U \to \pi(U)$ is a homeomorphism. Let $f \colon \pi(U) \to U$ be the inverse of $\pi|_U$, which is a section. Since $\varphi^{-1}(\varphi(U)) = U + \mathcal{R} \cap \pi^{-1}(\pi(U))$ and it is open in $\mathcal{S}$ by the Lemma, $\varphi(U)$ is open. Also if $W \subset \varphi(U)$, $\varphi^{-1}(W) = f^{-1}(\bar{\pi}(W)) + \mathcal{R} \cap \pi^{-1}(\bar{\pi}(W))$ is also open in U by Lemma. Therefore $\varphi \colon U \to \varphi(U)$ is continuous. It is clearly onto, and 1-1 since φ preserves fibers. Also, if $V \subset U$ is open, $\varphi^{-1}(\varphi(V)) = V + \mathcal{R} \cap \pi^{-1}(\pi(V))$ is open by the Lemma, so $\varphi(V)$ is open. It means φ^{-1} is also continuous. Therefore, $\varphi|_U$ is a homeomorphism, and $\bar{\pi}|_{\varphi(U)} = \pi \circ \varphi|_U^{-1}$ is a homeomorphism also. Clearly $x \in \varphi(U)$. Therefore $\bar{\pi} \colon \mathcal{R} \to M$ is a local homeomorphism.

   2) $\bar{\pi}^{-1}(m)$ is clearly a K-module.

   3) $\varphi \times \varphi \colon \mathcal{S} \times \mathcal{S} \to \mathcal{T} \times \mathcal{T}$ is also a quotient map, and restricting to $\mathcal{S} \circ \mathcal{S}$, we obtain a quotient map $\varphi \circ \varphi \colon \mathcal{S} \circ \mathcal{S} \to \mathcal{T} \circ \mathcal{T}$. Let $\psi \colon \mathcal{S} \circ \mathcal{S} \to \mathcal{S}$ be a composition map. Then $\bar{\psi} \colon \mathcal{T} \circ \mathcal{T} \to \mathcal{T}$ is a composition map on $\mathcal{T}$. To check that it is continuous, we need to verify that $\bar{\psi}^{-1}(U)$ is open in $\mathcal{T} \circ \mathcal{T}$ for an open set $U \subset \mathcal{T}$. But it is by definition equal to whether $(\varphi \circ \varphi)^{-1}(\bar{\psi}^{-1}(U))$ is open, and it is open since $(\varphi \circ \varphi)^{-1}(\bar{\psi}^{-1}(U)) = \psi^{-1}(\varphi^{-1}(U))$. Therefore the composition map is continuous. Therefore $\bar{\pi} \colon \mathcal{T} \to M$ is a sheaf. It remains to show that φ is a sheaf homomorphism, but it follows from the proof of 1).
8. Let $\varphi \colon \beta(\alpha(\mathcal{S})) \to \mathcal{S}, \rho_{p,U} f \to f(p)$.

1) If $\rho_{p,U}f = \rho'_{p',U'}f', p = p'$, and $\exists V \subset U \cap U'$ and $p \in V$ s.t. $f_V = f'_V$. Therefore $f(p) = f'(p')$ and φ is well-defined.

2) Let $s \in \mathcal{S}$ and $s \in V$ s.t. $\pi|_V \colon V \to \pi(V)$ be a homeomorphism. If $\varphi(\rho_{p,u}f) \in V$, $f(p) \in V$, therefore $f(p) \in f(U) \cap V \neq \varnothing$, which is open. Also $\pi(f(U) \cap V) \subset U \cap \pi(V)$ is open. Then it means $\rho_{p,U}f \in \{\rho_{q,\pi(f(U)\cap V)}f|_{\pi(f(U)\cap V)} | q \in \pi(f(U) \cap V)\} \subset \varphi^{-1}(V)$. Thus φ is continuous.

3) It clearly preserves fibers and composition. On the other hand, let $\psi \colon \mathcal{S} \to \beta(\alpha(\mathcal{S})), f(p) \mapsto \rho_{p,u}f$. (Since π is a local homeomorphism, we can find a section $f = \pi|_U^{-1}$ and $p \in M$ s.t. $f(p) = x$, $\forall x \in \mathcal{S}$.)

1) If $f(p) = f'(p')$ for $f \in \Gamma(\mathcal{S}, U)$ and $f' \in \Gamma(\mathcal{S}, U')$, $p = p' \in U \cap U'$, and $\exists V \subset U \cap U'$ s.t. $f|_V = f'|_V$ by Exercise 3. Therefore $\rho_{p,U}f = \rho_{p',U'}f'$ and ψ is well-defined.

2) For $O_f = \{\rho_{p,U}f \mid p \in U\}$, $\psi^{-1}(O_f) = f(U)$, which is open. Therefore ψ is continuous.

3) It clearly preserves fibers and composition.

Then it is certain that φ and ψ are inverses to each other. Therefore $\beta(\alpha(\mathcal{S}))$ is isomorphic to $\mathcal{S}$.

9. $(\mathcal{S} \otimes \mathcal{T})_m = \varinjlim_{m\in U}(\alpha(\mathcal{S}) \otimes \alpha(\mathcal{T}))_U = \varinjlim_{m\in U}\alpha(\mathcal{S})_U \otimes \alpha(\mathcal{T})_U = \varinjlim_{m\in U}\Gamma(\mathcal{S}, U) \otimes \Gamma(\mathcal{T}, U) = \varinjlim_{m\in U}\Gamma(\mathcal{S}, U) \otimes \varinjlim_{m\in U}\Gamma(\mathcal{T}, U) = \mathcal{S}_m \otimes \mathcal{T}_m$, since direct limit and tensor product commute. The last equality follows from $\beta(\alpha(\mathcal{S})) \simeq \mathcal{S}$ by Exercise 8 .
10. Let $\mathbb{C}_p$ and $\mathbb{C}_q$ be two skyscraper sheaves with $p \neq q$. Then $\mathbb{C}_p \otimes \mathbb{C}_q$ is a zero sheaf, but $\Gamma(\mathbb{C}_p, M) \otimes \Gamma(\mathbb{C}_q, M) \simeq \mathbb{C} \otimes \mathbb{C}$, which is not zero.
11. Let $M = \mathbb{R}$, and $\varphi \colon M \to \mathbb{R}, x \mapsto x$. Denote the given mapping of the exercise as Φ and the 0-section as f. Then $f(M) \subset \mathscr{C}^\infty(M)$ is open, but $\Phi^{-1}(f(M)) = f(M) \cap \mathscr{C}^\infty(M)_0$, which is not open.
12. Let $t \in \Gamma_\Phi(\mathcal{T})$. Since $\operatorname{supp}(t) \in \Phi$, $\exists U_1, U_2, U_3$ s.t. $\operatorname{supp}(t) \subset U_1 \subset \overline{U_1} \subset U_2 \subset \overline{U_2} \subset U_3 \subset \overline{U_3}$ and $\overline{U_1}, \overline{U_2}, \overline{U_3} \in \Phi$. Using the process of Theorem 5.12, we can find $s \in \mathcal{S}_{U_3}$ s.t. $t|_{U_3}$ is the image of s. Then $s|_{U_3\setminus\overline{U_1}}$ has a zero image, so it is in $\mathcal{R}_{U_3\setminus\overline{U_1}}$. Now consider a cover $U_2, \overline{U_1}^c$. Since $\mathcal{R}$ is fine, there exists an endomorphism φ s.t. $\operatorname{supp}(\varphi) \subset \overline{U_1}^c$ and $\varphi = 1$ on U_2^c. Also, $\operatorname{supp}(\varphi \circ s|_{U_3\setminus\overline{U_1}}) \subset \overline{U_1}^c$, so we can extend it to all over U_3 which is valued zero on $\overline{U_1}$. Denote it $s' \in \mathcal{S}_{U_3}$. Then the image of $s - s'$ is also $t|_{U_3}$ and by construction $s - s'|_{U_3\setminus U_2^c} = 0$, so $\operatorname{supp}(s - s') \subset \overline{U_2} \subset U_3$. Thus one can extend it to all over M which is valued zero on U_3^c, and $s - s' \in \Gamma_\Phi(\mathcal{S})$. Also, t is the image of $s - s'$. The statement is proved.
13.

$$\cdots \longrightarrow H^{q-1}(E^*) \xrightarrow{\partial} H^q(C^*) \xrightarrow{f} H^q(D^*) \xrightarrow{g} H^q(E^*) \xrightarrow{\partial} H^{q+1}(C^*) \longrightarrow \cdots$$

$$\begin{array}{ccccccccc}
0 & \longrightarrow & C^{q-1} & \longrightarrow & D^{q-1} & \longrightarrow & E^{q-1} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & C^{q} & \longrightarrow & D^{q} & \longrightarrow & E^{q} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & C^{q+1} & \longrightarrow & D^{q+1} & \longrightarrow & E^{q+1} & \longrightarrow & 0
\end{array}$$

1) $f\partial = 0$. Let $[e] \in E^{q-1}$ for $e \in Z(E^{q-1})$ and $d \in D^{q-1}, d' \in D^q, c \in C^q$ s.t. $d \mapsto e, d \mapsto d', c \mapsto d'$. Then by definition of ∂, $f\partial[e] = f[c] = [d'] = 0 \in H^q(D^*)$.

2) $\operatorname{im}\partial \supset \ker f$. Let $c \in Z(C^q)$ and $d' \in D^q$ s.t. $c \mapsto d'$. If $f[c] = [d'] = 0$, $\exists d \in D^{q-1}$ s.t. $d \mapsto d'$.

Let $e \in E^{q-1}$ s.t. $d \mapsto e$. Then $e \mapsto 0 \in E^q$ since $c \mapsto d' \mapsto 0 \in E^q$. Therefore $e \in Z(E^{q-1})$, and $\partial[e] = [c]$ by definition of ∂.

3) $gf = 0$. Since the original sequence is exact.

4) $\operatorname{im} f \supset \ker g$. Let $d \in Z(D^q)$ and $d \mapsto e \in E^q$. If $g[d] = [e] = 0, \exists e' \in E^{q-1}$ s.t. $e' \mapsto e$. Let $d' \in D^{q-1}$ s.t. $d' \mapsto e'$ and $\tilde{d} \in D^q$ s.t. $d' \mapsto \tilde{d}$. Then $d - \tilde{d} \mapsto 0 \in E^q$, so $\exists c \in C^q$ s.t. $c \mapsto d - \tilde{d}$. Now let $c' \in C^{q+1}$ s.t. $c \mapsto c'$. Then $c' \mapsto 0 \in D^{q+1}$ since $d - \tilde{d} \mapsto 0 \in D^{q+1}$, so $c' = 0$ since $C^{q+1} \to D^{q+1}$ is injective. Therefore $c \in Z(C^q)$, and $f[c] = [d - \tilde{d}] = [d]$.

5) $\partial g = 0$. Let $d \in Z(D^q), e \in E^q$ s.t. $d \mapsto e$. Then $d \mapsto 0 \in D^{q+1}$, so $e \mapsto 0 \in E^{q+1}$, which means $e \in Z(E^q)$. Also, $\partial g[d] = \partial[e] = 0$ by definition of ∂, since $0 \in C^{q+1} \mapsto 0 \in D^{q+1}$.

6) $\operatorname{im} g \supset \ker \partial$. Let $e \in Z(E^q), d \in D^q, d' \in D^{q+1}, c' \in C^{q+1}$ s.t. $d \mapsto e, d \mapsto d', c' \mapsto d'$. Then $\partial[e] = [c'] = 0$ means $\exists c \in C^q$ s.t. $c \mapsto c'$. Let $\tilde{d} \in D^q$ s.t. $c \mapsto \tilde{d}$. Then $d - \tilde{d} \mapsto 0 \in D^{q+1}$, so$d - \tilde{d} \in Z(D^q)$. Also, $g[d - \tilde{d}] = [e]$.

14. (a) Since $Z(\Gamma(\mathscr{C}_0 \otimes \mathscr{S})) = \Gamma(\mathscr{K} \otimes \mathscr{S}) = Z(\Gamma(\tilde{\mathscr{C}_0} \otimes \mathscr{S})), \varphi|_{Z(\Gamma(\mathscr{C}_0 \otimes \mathscr{S}))} = \mathrm{id}_{\Gamma(\mathscr{K} \otimes \mathscr{S})}$. Hence condition (a) is satisfied.

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \Gamma(\mathscr{K} \otimes \mathscr{S}) & \longrightarrow & \Gamma(\mathscr{C}_0 \otimes \mathscr{S}) & \longrightarrow & \Gamma(\mathscr{C}_1 \otimes \mathscr{S}) & \longrightarrow & \Gamma(\mathscr{C}_2 \otimes \mathscr{S}) \longrightarrow \cdots \\
 & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \varphi_0} & & \downarrow{\scriptstyle \varphi_1} & & \downarrow{\scriptstyle \varphi_2} \\
0 & \longrightarrow & \Gamma(\mathscr{K} \otimes \mathscr{S}) & \longrightarrow & \Gamma(\tilde{\mathscr{C}_0} \otimes \mathscr{S}) & \longrightarrow & \Gamma(\tilde{\mathscr{C}_1} \otimes \mathscr{S}) & \longrightarrow & \Gamma(\tilde{\mathscr{C}_2} \otimes \mathscr{S}) \longrightarrow \cdots
\end{array}$$

(b) Since the following diagram commutes:

$$\begin{array}{ccc}
\mathscr{C}_q \otimes \mathscr{S} & \longrightarrow & \mathscr{C}_q \otimes \mathscr{T} \\
\downarrow & & \downarrow \\
\tilde{\mathscr{C}_q} \otimes \mathscr{S} & \longrightarrow & \tilde{\mathscr{C}_q} \otimes \mathscr{T}
\end{array}$$

(c) By using 5.17.

15. If $U = \bigcup_i V_i, f_i \in A^p(V_i)$ s.t. $f_i = f_j$ on $V_i \cap V_j$, then set $f \in A^p(U)$ s.t. $f(x_1, \ldots, x_{p+1}) = f_i(x_1, \ldots, x_{p+1})$ if $x_1, \ldots, x_{p+1} \in V_i$ and otherwise $f = 0$. Then f is well-defined, and $f|_{V_i} = f_i$. However, if $U = \mathbb{R}, V_1 = (-\infty, 1), V_2 = (-1, \infty)$, and $f \in A^1(U)$ s.t. $f(2, -2) = 1$ and otherwise $f = 0$, then $f|_{V_1} = 0$ and $f|_{V_2} = 0$, but $f \neq 0$.

16.

$$0 \longrightarrow \mathbb{Z}/2\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}/4\mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

is exact, but

$$0 \longrightarrow \mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z} \xrightarrow{\times 2 \otimes \mathrm{id} = 0} \mathbb{Z}/4\mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z} \xrightarrow{\pi \otimes \mathrm{id}} \mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

is not exact.

17. Choose any sheaf $\mathscr{K}$ which is not fine and consider the sheaf of germs of discrete sections of $\mathscr{K}$ which is fine.

18. Since exactness is a local property, we only need to consider the module cases. However, since stalks of torsionless sheaves are flat, the result follows. Or a long exact sequence can be divided into short exact sequences, and use Proposition 5.15.

19. Naturally $f\colon M \to N$ induces $f_*\colon H^p_\Delta(N; G) \to H^p_\Delta(M; G)$ s.t. $f_*([\varphi])(\sigma) = \varphi(f(\sigma))$ which is well-defined. Also that it satisfies functorial properties is straightforward. Therefore the last statement follows from a basic property of a functor.

20. Denote the given map $\varphi\colon H^p_{\Delta\infty}(M; \mathbb{R}) \to {}_\infty H_p(M; \mathbb{R})^*$. Let $[f] \in \ker \varphi$. Then $f(\sigma) = 0$ for any cycle σ. Now set $g \in S^{p-1}(M, \mathbb{R})$ s.t. for $(p-1)$-boundary $\partial\tau$, $g(\partial\tau) = f(\tau)$ and otherwise 0. It

is well-defined since if $\partial\tau = \partial\tau'$ then $\tau - \tau'$ is a cycle, so $f(\tau - \tau') = 0$. Then $dg = f$, so $[f] = 0$. Therefore φ is injective. By the way, if $g \in {}_\infty H_p(M;\mathbb{R})^*$, set $f \in S^p(M,\mathbb{R})$ s.t. $f(\sigma) = g([\sigma])$. Then it is well-defined, and $df(\tau) = f(\partial\tau) = g([\partial\tau]) = 0$, so f is a cocycle. Also, $\varphi(f) = g$ by the setting. Hence φ is surjective.

21. If the periods of σ and τ are all integer-valued, they must be zero since $\{\sigma\}(\{az\}) = a\{\sigma\}(\{z\})$ for all $a \in \mathbb{R}$ and p-cycle z, and similarly for τ. Therefore, $\sigma \wedge \tau$ is exact and has integer periods, i.e. zero. (Maybe the purpose of this exercise is to consider the cohomology theory over $\mathbb{Z}$, but I am not sure.)

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1.6 习题: Warner GTM 94/Tensors and Differential Forms

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1. (a) Let $l\colon V\times W\to U$ be a bilinear map. Then by the universal property of free module, $\bar{l}\colon \mathcal{F}(V\times W)\to U$ is well defined, and by the bilinearity, generators of 2.1(1) are in the kernel of $\bar{l}$. Therefore, the map $\tilde{l}\colon V\otimes W\to U$ is well defined. Also, it satisfies $\tilde{l}\circ\varphi=l$ by the definition of the map. Since $\tilde{l}(v\otimes w)$ is determined by the value of l and elements of the form $v\otimes w$ generate $V\otimes W$, such $\tilde{l}$ must be unique. The rest of the argument is directly followed by the properties of universal objects.

   (b) Since $V\times W\cong W\times V$, it is true because of the universal property.

   (c) For $v\in V$, let $f_v\colon W\times U\to (V\times W)\times U\to (V\otimes W)\times U\to (V\otimes W)\otimes U$ be the map $(w,u)\mapsto((v,w),u)\mapsto(v\otimes w,u)\mapsto(v\otimes w)\otimes u$. Since it is bilinear, one can define $\tilde{f}_v\colon W\otimes U\to (V\otimes W)\otimes U$. Then, if we set $g\colon V\times(W\otimes U)\to(V\otimes W)\otimes U, (v,x)\mapsto\tilde{f}_v(x)$ for all $x\in W\otimes U$, it is also bilinear, hence one can define $\tilde{g}\colon V\otimes(W\otimes U)\to(V\otimes W)\otimes U$, and $\tilde{g}(v\otimes(w\otimes u))=(v\otimes w)\otimes u)$. By the same method, one can also define $\tilde{h}\colon (V\otimes W)\otimes U\to V\otimes(W\otimes U)$, and it is easy to see that $\tilde{g}$ and $\tilde{h}$ are the inverse to each other. Therefore, $V\otimes(W\otimes U)\cong(V\otimes W)\otimes U$.

   (d) Let $f\colon V^*\times W\to \mathrm{Hom}(V,W)$ be a function such that $f(\varphi,w)(v)=\varphi(v)\cdot w$. Since it is bilinear, one can define a map $\alpha\colon V^*\otimes W\to\mathrm{Hom}(V,W)$.

   (1) Let $\{v_i\},\{w_j\}$ be a basis of V,W, respectively, and $\{v_i^*\}$ be the dual basis corresponding to $\{v_i\}$, i.e., $v_a^*(v_b)=\delta_{ab}$. Then elements of $V^*\otimes W$ can be represented as $\sum_{i,j}a_{ij}v_i^*\otimes w_j$. If

$\alpha\big(\sum_{i,j} a_{ij} v_i^* \otimes w_j\big) = 0$, then $\forall v \in V, \sum_{i,j} a_{ij} v_i^*(v) \cdot w_j = 0$. In particular, if one substitutes v_i for v, then $\sum_j a_{ij} w_j = 0, \forall i$. Therefore, by the linear independence of the basis, $a_{ij} = 0, \forall i, j$, and it means α is injective.

(2) Let $\varphi \in \mathrm{Hom}(V, W)$. Then if $\varphi(v_i) = \sum_j a_{ij} w_j$, set $\psi = \alpha\big(\sum_{i,j} a_{ij} \otimes w_j\big)$. Then ψ and φ have the same value at v_i, so they must be the same linear map. Therefore, α is surjective. Therefore, α is an isomorphism. Hence, $\dim V^* \otimes W = \dim \mathrm{Hom}(V, W) = (\dim V)(\dim W)$, and $\dim V \otimes W = \dim V^{**} \otimes W = (\dim V^*)\,(\dim W) = (\dim V)(\dim W)$.

(e) By arguments similar to the above, $\{e_i \otimes f_j\}$ span $V \otimes W$. Since the number of $\{e_i \otimes f_j\}$ is the same as the dimension of $V \otimes W$ by (d), it must be a basis.

2. (a) Let $V = \mathbb{R}^2$, and consider $e_1 \otimes e_2 - e_2 \otimes e_1 \in V \otimes V$. If it is decomposable, say $(ae_1 + be_2) \otimes (ce_1 + de_2)$, then $ac = bd = 0, ad = 1, bc = -1$, which is impossible.

(b) If the dimension is 1 or 2, then it is trivially true. Let $\dim V = 3, V = \langle v_1, v_2, v_3 \rangle$. Then we only need to check that all elements of $\Lambda_2(V)$ are decomposable. However, for any element

$$x = av_1 \wedge v_2 + bv_2 \wedge v_3 + cv_3 \wedge v_1 \in \Lambda_2(V),$$

if $a = 0$, then $x = (bv_2 - cv_1) \wedge v_3$, and if $a \neq 0$, then $x = \big(v_1 - \frac{b}{a} v_3\big) \wedge (av_2 - cv_3)$. Therefore x is always decomposable.

(c) If $V = \mathbb{R}^4$, then $e_1 \wedge e_2 + e_3 \wedge e_4$ is indecomposable. If it is decomposable, say

$$(a_1e_1 + a_2e_2 + a_3e_3 + a_4e_4) \wedge (b_1e_1 + b_2e_2 + b_3e_3 + b_4e_4),$$

then

$$a_1b_2 - b_1a_2 = a_3b_4 - b_3a_4 = 1,$$

$$a_1b_3 - b_1a_3 = a_1b_4 - b_1a_4 = a_2b_3 - b_2a_3 = a_2b_4 - b_2a_4 = 0.$$

If $a_1 = 0, b_1a_2 = -1, b_1a_3 = b_1a_4 = 0$, so $a_3 = a_4 = 0$, which contradicts $a_3b_4 - b_3a_4 = 1$.

If $a_1 \neq 0, b_3 = \frac{b_1a_3}{a_1}, b_4 = \frac{b_1a_4}{a_1}$, which also contradicts $a_3b_4 - b_3a_4 = 1$.

(d) No. Let $\alpha = e_1 \wedge e_2 + e_3 \wedge e_4 \in \Lambda_2(\mathbb{R}^4)$. Then $\alpha \wedge \alpha = 2e_1 \wedge e_2 \wedge e_3 \wedge e_4 \in \Lambda_4(\mathbb{R}^4)$, which is not zero.

3. (a) That $u \wedge v \in \lambda_{k+l}(V)$ is a direct consequence of the associativity of tensor products. Since the last equation is bilinear, we only need to show if u and v are decomposable. However, if $u = u_1 \wedge \ldots \wedge u_k$ and $v = v_1 \wedge \ldots \wedge v_l$,

$$\begin{aligned} v \wedge u &= v_1 \wedge \ldots \wedge v_l \wedge u_1 \wedge \ldots \wedge u_k \\ &= (-1)^l u_1 \wedge v_1 \wedge \ldots \wedge v_l \wedge u_2 \wedge \ldots \wedge u_k \\ &= \ldots \\ &= (-1)^{kl} u_1 \wedge \ldots \wedge u_k \wedge v_1 \wedge \ldots \wedge v_l = (-1)^{kl} u \wedge v. \end{aligned}$$

(b) Let us follow the argument in the text. Since an arbitrary element of $I(V)$ has zero determinant, $e_1 \otimes \ldots \otimes e_n$ cannot be in $I(V)$, i.e., $e_1 \wedge \ldots \wedge e_n \neq 0$. That $\{e_\Phi\}$ spans $\Lambda(V)$ is obvious. If $\sum_\Phi a_\Phi e_\Phi = 0$, since homogeneous parts should be zero, $\forall 0 \leq k \leq n$, $\sum_{|\Phi|=k} a_\Phi e_\Phi = 0$. However, for some Φ_0, if we wedge $e_{\Phi_0^c}$ on the equation, then we get $a_{\Phi_0} e_1 \wedge \ldots \wedge e_n = 0$, hence

$a_{\Phi_0} = 0$. Since Φ_0 is arbitrary, it means $\{e_\Phi\}$ is linearly independent. Therefore, $\{e_\Phi\}$ is a basis. The rest of the argument trivially follows.

(c) $\tilde{h}$ can be defined because of the universal property of tensor products and $I(V)$ should be in the kernel of the induced map since h is alternating. Its uniqueness is also followed by the universal property of tensor products. If $W = \mathbb{R}$, since $\tilde{h} \in \Lambda_k(V)^*$ and $h \in A_k(V)$, $\varphi^* \colon \Lambda_k(V)^* \to A_k(V)$ gives an isomorphism.

4. Since (2), (3), and (4) are bilinear forms, we only need to consider when f and g are both induced by decomposable elements. Therefore if $\varphi \colon \Lambda(V)^* \to A(V)$ is the isomorphism, and $f = \varphi \circ \alpha(w_1^* \wedge \ldots \wedge w_p^*), g = \varphi \circ \alpha(w_{p+1}^* \wedge \ldots \wedge w_{p+q}^*)$. Then

$$\begin{aligned}
f \wedge_\alpha g(v_1, \ldots, v_{p+q}) &= w_1^* \wedge \ldots \wedge w_p^* \wedge w_{p+1}^* \wedge \ldots \wedge w_{p+q}^*(v_1 \wedge \ldots \wedge v_{p+q}) \\
&= \det(w_i^* v_j)_{1 \le i,j \le p+q} \\
&= \sum_{\sigma \in S_{p+q}} \operatorname{sgn}(\sigma) \prod_{i=1}^{p+q} w_i^* v_{\sigma(i)} \\
&= \sum_{\pi : p,q \text{ shuffles}} \sum_{\sigma_1 \in S_p} \sum_{\sigma_2 \in S_q} \operatorname{sgn}(\pi) \operatorname{sgn}(\sigma_1) \operatorname{sgn}(\sigma_2) \prod_{i_1=1}^{p} w_{i_1}^* v_{\pi(\sigma_1(i_1))} \prod_{i_2=1}^{q} w_{p+i_2}^* v_{\pi(p+\sigma_2(i_2))} \operatorname{sgn} \\
&= \sum_{\pi : p,q \text{ shuffles}} \operatorname{sgn}(\pi) f(v_{\pi(1)}, \ldots, v_{\pi(p)}) g(v_{\pi(p+1)}, \ldots, v_{\pi(p+q)}).
\end{aligned}$$

On the other hand, let $f = \varphi \circ \beta(w_1^* \wedge \ldots \wedge w_p^*), g = \varphi \circ \beta(w_{p+1}^* \wedge \ldots \wedge w_{p+q}^*)$. Then

$$\begin{aligned}
f \wedge_\beta g(v_1, \ldots, v_{p+q}) &= w_1^* \wedge \ldots \wedge w_p^* \wedge w_{p+1}^* \wedge \ldots \wedge w_{p+q}^*(v_1 \wedge \ldots \wedge v_{p+q}) \\
&= \frac{1}{(p+q)!} \det(w_i^* v_j)_{1 \le i,j \le p+q} \\
&= \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} \operatorname{sgn}(\sigma) \prod_{i=1}^{p+q} w_i^* v_{\sigma(i)} \\
&= \frac{1}{(p+q)!} \sum_{\pi : p,q \text{ shuffles}} \sum_{\sigma_1 \in S_p} \sum_{\sigma_2 \in S_q} \operatorname{sgn}(\pi) \operatorname{sgn}(\sigma_1) \operatorname{sgn}(\sigma_2) \prod_{i_1=1}^{p} w_{i_1}^* v_{\pi(\sigma_1(i_1))} \prod_{i_2=1}^{q} w_{p+i_2}^* v_{\pi(p+} \\
&= \frac{1}{(p+q)!} \sum_{\pi : p,q \text{ shuffles}} \operatorname{sgn}(\pi) \det(w_i^* v_{\pi(i)})_{1 \le i \le p} \det(w_j^* v_{\pi(j)})_{p+1 \le j \le p+q} \\
&= \frac{p!q!}{(p+q)!} \sum_{\pi \in S_{p+q}} \operatorname{sgn}(\pi) \det(w_i^* v_{\pi(i)})_{1 \le i \le p} \det(w_j^* v_{\pi(j)})_{p+1 \le j \le p+q} \\
&= \frac{1}{(p+q)!} \sum_{\pi \in S_{p+q}} \operatorname{sgn}(\pi) f(v_{\pi(1)}, \ldots, v_{\pi(p)}) g(v_{\pi(p+1)}, \ldots, v_{\pi(p+q)}).
\end{aligned}$$

Therefore,

$$\begin{aligned}
f \wedge_\alpha g(v_1, \ldots, v_{p+q}) &= \sum_{\pi : p,q \text{ shuffles}} \operatorname{sgn}(\pi) f(v_{\pi(1)}, \ldots, v_{\pi(p)}) g(v_{\pi(p+1)}, \ldots, v_{\pi(p+q)}) \\
&= \frac{1}{p!q!} \sum_{\pi \in S_{p+q}} \operatorname{sgn}(\pi) f(v_{\pi(1)}, \ldots, v_{\pi(p)}) g(v_{\pi(p+1)}, \ldots, v_{\pi(p+q)}) \\
&= \frac{(p+q)!}{p!q!} f \wedge_\beta g.
\end{aligned}$$

5. (where γ is an integral curve of X)

$$\begin{aligned} L_X(f)|_m &= \lim_{t\to 0}\frac{\delta X_t(f_{X_t(m)}) - f_m}{t} = \lim_{t\to 0}\frac{f_{X_t(m)} - f(m)}{t} \\ &= \lim_{t\to 0}\frac{f(\gamma_m(t)) - f(m)}{t} = d\gamma\left(\left.\frac{d}{dt}\right|_0\right)(f) \\ &= X_m(f). \end{aligned}$$

6. (Since $\sqrt{t}$ is defined on $[0,\varepsilon)$, the limit should be calculated when $t\to 0^+$.) We will use L'Hôpital's rule.

   Let $g(x,y,z,w) = f(Y_{-x}X_{-y}Y_zX_w)(m)$. Then,

$$\begin{aligned} \frac{\partial g}{\partial x}(x,y,z,w) &= -Y_{Y_{-x}X_{-y}Y_zX_w(m)}(f) \\ \frac{\partial g}{\partial y}(x,y,z,w) &= -X_{X_{-y}Y_zX_w(m)}(f\circ Y_{-x}) \\ \frac{\partial g}{\partial z}(x,y,z,w) &= Y_{Y_zX_w(m)}(f\circ Y_{-x}\circ X_{-y}) \\ \frac{\partial g}{\partial w}(x,y,z,w) &= X_{X_w(m)}(f\circ Y_{-x}\circ X_{-y}\circ Y_z) \\ \frac{\partial^2 g}{\partial x^2}(x,y,z,w) &= Y_{Y_{-x}X_{-y}Y_zX_w(m)}(Y(f)) \\ \frac{\partial^2 g}{\partial y^2}(x,y,z,w) &= X_{X_{-y}Y_zX_w(m)}(X(f\circ Y_{-x})) \\ \frac{\partial^2 g}{\partial z^2}(x,y,z,w) &= Y_{Y_zX_w(m)}(Y(f\circ Y_{-x}\circ X_{-y})) \\ \frac{\partial^2 g}{\partial w^2}(x,y,z,w) &= X_{X_w(m)}(X(f\circ Y_{-x}\circ X_{-y}\circ Y_z)) \\ \frac{\partial^2 g}{\partial x\partial y}(x,y,z,w) &= X_{X_{-y}Y_zX_w(m)}(Y(f)\circ Y_{-x}) \\ \frac{\partial^2 g}{\partial x\partial z}(x,y,z,w) &= -Y_{Y_zX_w(m)}(Y(f)\circ Y_{-x}\circ X_{-y}) \\ \frac{\partial^2 g}{\partial x\partial w}(x,y,z,w) &= -X_{X_w(m)}(Y(f)\circ Y_{-x}\circ X_{-y}\circ Y_z) \\ \frac{\partial^2 g}{\partial y\partial z}(x,y,z,w) &= -Y_{Y_zX_w(m)}(X(f\circ Y_{-x})\circ X_{-y}) \\ \frac{\partial^2 g}{\partial y\partial w}(x,y,z,w) &= -X_{X_w(m)}(X(f\circ Y_{-x})\circ X_{-y}\circ Y_z) \\ \frac{\partial^2 g}{\partial z\partial w}(x,y,z,w) &= X_{X_w(m)}(Y(f\circ Y_{-x}\circ X_{-y})\circ Y_z) \end{aligned}$$

   Therefore,

$$\begin{aligned} \lim_{t\to 0^+}\frac{f(\beta(t)) - f(\beta(0))}{t} &= \lim_{t\to 0^+}\frac{f(\beta(t^2)) - f(\beta(0))}{t^2} \\ &= \lim_{t\to 0^+}\frac{g(t,t,t,t) - g(0,0,0,0)}{t^2} \\ &= \lim_{t\to 0^+}\frac{dg(t,t,t,t)/dt}{2t} \\ &= \lim_{t\to 0^+}\frac{d^2g(t,t,t,t)/dt^2}{2}. \end{aligned}$$

Since

$$dg(t,t,t,t)/dt|_{t=0} = -Y_m(f) - X_m(f) + Y_m(f) + X_m(f) = 0,$$

we have

$$\begin{aligned}\lim_{t\to 0^+} \frac{d^2 g(t,t,t,t)/dt^2}{2} &= \frac{1}{2}\left(Y_m(Y(f)) + X_m(X(f)) + Y_m(Y(f)) + X_m(X(f))\right)\\ &+ X_m(Y(f)) - Y_m(Y(f)) - X_m(Y(f)) - Y_m(X(f))\\ &- X_m(X(f)) + X_m(Y(f))\\ &= X_m(Y(f)) - Y_m(X(f)) = [X,Y]|_m(f).\end{aligned}$$

7. For $m \in M$, let $x_1, \ldots, x_n$ be local coordinates around m. Since the equation is linear with respect to ω, it is enough to consider the case when ω is decomposable, i.e., $\omega = g dx_{m_1} \wedge \ldots \wedge dx_{m_p}$. Then,

$$\begin{aligned}L_{Y_0}(\omega(Y_1,\ldots,Y_p)) &= Y_0(g \det(Y_1(x_{m_i}),\ldots,Y_p(x_{m_i}))_{1\le i\le p})\\ &= Y_0(g)\det(Y_1(x_{m_i}),\ldots,Y_p(x_{m_i}))_{1\le i\le p}\\ &+ \sum_{j=1}^{p} g \det(Y_1(x_{m_i}),\ldots,Y_0(Y_j(x_{m_i})),\ldots,Y_p(x_{m_i}))_{1\le i\le p},\end{aligned}$$

$$\begin{aligned}L_{Y_0}(\omega)(Y_1,\ldots,Y_p) &= Y_0(g)\det(Y_1(x_{m_i}),\ldots,Y_p(x_{m_i}))_{1\le i\le p}\\ &+ \sum_{j=1}^{p} g \det(Y_1(x_{m_i}),\ldots,Y_j(Y_0(x_{m_i})),\ldots,Y_p(x_{m_i}))_{1\le i\le p},\end{aligned}$$

$$\begin{aligned}\omega(Y_1,\ldots,L_{Y_0}Y_j,\ldots,Y_p) &= \det(Y_1,\ldots,Y_0(Y_j(x_{m_i})),\ldots,Y_p)_{1\le i\le p}\\ &- \det(Y_1,\ldots,Y_j(Y_0(x_{m_i})),\ldots,Y_p)_{1\le i\le p}.\end{aligned}$$

(e) follows by combining these equations.

8. $(\Rightarrow)\forall(m,n)\in M\times N, \forall Y_2,\ldots,Y_p \in (M\times N)_{(m,n)}$,

$$\begin{aligned}i(X)\omega_{(m,n)}(Y_2,\ldots,Y_p) &= i(X)\delta\pi(\alpha)_{(m,n)}(Y_2,\ldots,Y_p)\\ &= \delta\pi(\alpha)_{(m,n)}(X_{(m,n)},Y_2,\ldots,Y_p)\\ &= \alpha_m(d\pi(X_{(m,n)}),d\pi(Y_2),\ldots,d\pi(Y_p)) = 0.\end{aligned}$$

Also,

$$\begin{aligned}i(X)d\omega_{(m,n)}(Y_2,\ldots,Y_p) &= i(X)d(\delta\pi(\alpha))_{(m,n)}(Y_2,\ldots,Y_p)\\ &= d(\delta\pi(\alpha))_{(m,n)}(X_{(m,n)},Y_2,\ldots,Y_p)\\ &= \delta\pi(d\alpha)_{(m,n)}(X_{(m,n)},Y_2,\ldots,Y_p)\\ &= d\alpha_m(d\pi(X_{(m,n)}),d\pi(Y_2),\ldots,d\pi(Y_p)) = 0.\end{aligned}$$

Since $L_X = i(X)\circ d + d\circ i(X)$, $L_X\omega = 0$.

($\Leftarrow$) Let $(m,n) \in M \times N$, and choose local coordinates of $m \in M, n \in N$, say $\{x_1,\ldots,x_s\},\{y_1,\ldots,y_t\}$, such that local coordinates of $(m,n) \in M \times N$ consist of $\{x_1,\ldots,x_s,y_1,\ldots,y_t\}$. Then, ω can be described as

$$\sum_{|\Phi|+|\Psi|=p} g_{\Phi,\Psi} dx_\Phi \wedge dy_\Psi.$$

And we can take a vector field X on $M \times N$ as $\frac{\partial}{\partial x_i} \times (\mathcal{C}^\infty$ function which is zero outside the local chart and 1 at $(m,n))$. Then $X_{(m,n)} = \frac{\partial}{\partial x_i}\big|_{(m,n)}$ and $d\pi(X) = 0$. Therefore, by assumption, $L_X\omega = 0$. But,

$$L_X\omega = \sum_{|\Phi|+|\Psi|=p} \frac{\partial g_{\Phi,\Psi}}{\partial x_i} dx_\Phi \wedge dy_\Psi.$$

Therefore $\forall x_i, \frac{\partial g_{\Phi,\Psi}}{\partial x_i} = 0$. If we let $\iota_{m'}\colon N \to M \times N, n' \mapsto (m',n')$,

$$(\delta\iota_m\omega)_n = \sum_{|\Psi|=p} g_{\Phi,\Psi}(m,n) dy_\Psi.$$

Therefore, by the result above, on the small neighborhood of $m \in M$ and for all m' in the neighborhood, $(\delta\iota'_m\omega)_n$ is the same as $(\delta l_m\omega)_n$. Since (m,n) is arbitrary, we can say that $\delta\iota_m\omega$ is locally the same regardless of the choice of m. Also, since M is connected, we conclude that $\delta\iota_m\omega$ is always the same regardless of the choice of $m \in M$. Let α be this p-form on N. Then, $\omega = \delta\pi(\alpha)$. Indeed, for all $(m,n) \in M \times N$, and $X_1,\ldots,X_p \in (M \times N)_{(m,n)}$,

$$\begin{aligned}(\omega - \delta\pi(\alpha))_{(m,n)}(X_1,\ldots,X_p) &= \omega_{(m,n)}(X_1,\ldots,X_p) - \delta\pi(\delta\iota_m\omega)_{(m,n)}(X_1,\ldots,X_p)\\ &= \omega_{(m,n)}(X_1,\ldots,X_p) - \omega_{(m,n)}(d(\iota_m\circ\pi)(X_1),\ldots,d(\iota_m\circ\pi)(X_p))\\ &= \omega_{(m,n)}(X_1 - d(\iota_m\circ\pi)(X_1),\ldots,X_p - d(\iota_m\circ\pi)(X_p)).\end{aligned}$$

The last term is zero since if we set X as a vector field of $M \times N$ such that

$$X_{(m,n)} = X_1 - d(\iota_m\circ\pi)(X_1)$$

and $d\pi X = 0$, (it is possible since

$$d\pi(X_1 - d(\iota_m\circ\pi)(X_1)) = 0,$$

the last term is

$$(i(X)\omega)_{(m,n)}(X_2 - d(\iota_m\circ\pi)(X_2),\ldots,X_p - d(\iota_m\circ\pi)(X_p))$$

which should be zero by the assumption.

9. ($\Leftarrow$) If $a_1v_1 + \ldots + a_rv_r = 0$ and some of a_i's are nonzero, then without loss of generality, one can assume $a_1 \neq 0$. Dividing the equation by a_1, one can also assume $a_1 = 1$. Then

$$v_1 \wedge \ldots \wedge v_n = (-a_2v_2 - \ldots - a_rv_r) \wedge v_2 \wedge \ldots \wedge v_r = 0.$$

($\Rightarrow$) Suppose $v_1,\ldots,v_p$ are linearly independent, then there is a linear map $\varphi\colon V \to k^r, v_i \mapsto e_i$. By the universal property, it induces a map $\Lambda_r(\varphi)\colon \Lambda_r(V) \to \Lambda_r(k^r)$. Since $\varphi(v_1 \wedge \ldots \wedge v_p) = e_1 \wedge \ldots \wedge e_p \neq 0$ by 2.6, $v_1 \wedge \ldots \wedge v_p \neq 0$.

10. ($\Rightarrow$) Let $v_i = \sum_j a_{ij} w_j$. Then by direct calculation, $v_1 \wedge \ldots \wedge v_r = (\det A) w_1 \wedge \ldots \wedge w_r$.
($\Leftarrow$) If $w \in \langle w_1, \ldots, w_r \rangle \setminus \langle v_1, \ldots v_r \rangle$, $v_1 \wedge \ldots \wedge v_r \wedge w \neq 0$ by Exercise 9, which contradicts $w_1 \wedge \ldots \wedge w_r \wedge w = 0$. Therefore $\langle w_1, \ldots, w_r \rangle \subset \langle v_1, \ldots v_r \rangle$ and vice versa.
11. (Condition$\Rightarrow$(a)) Since $d_{\omega_i} \in \mathfrak{I}$ by definition of differential ideals.
((a)$\Rightarrow$(b))
$$d\omega = d\omega_1 \wedge \ldots \wedge \omega_r + \ldots + (-1)^{r+1} \omega_1 \wedge \ldots \wedge d\omega_r = \alpha \wedge \omega$$
by using (a).
((b)$\Rightarrow$(a))
$$d\omega = d\omega_1 \wedge \ldots \wedge \omega_r + \ldots + (-1)^{r+1} \omega_1 \wedge \ldots \wedge d\omega_r = \alpha \wedge \omega.$$
Since $\omega \wedge \omega_i = 0$, by wedging ω_i on the equation, we get $d\omega_i \wedge \omega = 0$. Therefore, $d\omega_i \in \langle \omega_1, \ldots, \omega_r \rangle$.
((a)$\Rightarrow$condition)
$$d\langle \omega_1, \ldots, \omega_r \rangle \subset \langle d\omega_1, \ldots, d\omega_r, \omega_1, \ldots, \omega_r \rangle$$
by the property of derivations, and the result directly follows.
12. For another basis $w_1, \ldots, w_n$, let $w_i = \sum_j a_{ij} v_j$. Then
$$\begin{aligned} Aw_1 \wedge \ldots \wedge Aw_n &= \sum_j a_{1j} Av_j \wedge \ldots \wedge \sum_j a_{nj} Av_j \\ &= \det(a_{ij}) Av_1 \wedge \ldots \wedge Av_n. \end{aligned}$$
Therefore by Exercise 10, $\det A$ does not depend on the choice of basis. If we choose the standard basis $e_1, \ldots, e_n$, the equation of $\det A$ follows directly. Also, for two matrices A, B,
$$Bv_1 \wedge \ldots \wedge Bv_n = (\det B) v_1 \wedge \ldots \wedge v_n,$$
so
$$\begin{aligned} ABv_1 \wedge \ldots \wedge ABv_n &= (\det B) Av_1 \wedge \ldots \wedge Av_n \\ &= (\det B)(\det A) v_1 \wedge \ldots \wedge v_n, \end{aligned}$$
which proves $\det A \det B = \det AB$.
13. The orthonormality comes from direct calculations.
(5) Since $**$ is linear, we only need to show this property for the basis elements. If $\Phi = \{i_1, \ldots, i_p\} \subset \{1, \ldots, n\}$ and let f be the element which satisfies $e_\Phi \wedge f = e_1 \wedge \ldots \wedge e_n$, then
$$f \wedge e_\Phi = (-1)^{p(n-p)} e_1 \wedge \ldots \wedge e_n.$$
Therefore $** e_\Phi = \pm * f = (-1)^{p(n-p)} e_\Phi$.
(6) Since the equation is bilinear in v and w, we only need to show this for the basis elements. Therefore we can assume that $v = e_{\Phi_1}, w = e_{\Phi_2}$. If $v \neq w$,
$$\langle v, w \rangle = *(w \wedge *v) = *(v \wedge *w) = 0$$
by definition of star and inner product. If $v = w$,
$$\langle v, w \rangle = *(w \wedge *v) = *(v \wedge *w) = 1$$
also by definition. Hence (6) is proved.

14. It suffices to show that for all $v \in \Lambda_{p+1}(V)$ and $w \in \Lambda_p(V)$,

$$\langle \gamma(v), w \rangle = \langle v, \xi \wedge w \rangle.$$

But it is equivalent to (using Exercise 13):

$$\begin{aligned}
&(-1)^{np}\langle w, *(\xi \wedge *v)\rangle = \langle \xi \wedge w, v\rangle \\
\Leftrightarrow &(-1)^{np} * \left(w \wedge (-1)^{p(n-p)}\xi \wedge *v\right) = \langle \xi \wedge w, v\rangle \\
\Leftrightarrow &(-1)^{p} * (w \wedge \xi \wedge *v) = \langle \xi \wedge w, v\rangle \\
\Leftrightarrow &* (\xi \wedge w \wedge *v) = \langle \xi \wedge w, v\rangle
\end{aligned}$$

which is true by Exercise 13.

15. (It is true only if $\xi \neq 0$) Obviously $(\xi\wedge)^2 = 0$. Conversely, if $\xi \wedge w = 0$, then

$$w = \xi \wedge \gamma\left(\frac{w}{\langle \xi, \xi\rangle}\right).$$

To prove this, it is enough to show that for all $v \in \Lambda_{p+1}(V)$,

$$\langle \xi \wedge \gamma(w), v\rangle = \langle \xi, \xi\rangle\langle w, v\rangle.$$

But

$$\begin{aligned}
\langle \xi \wedge \gamma(w), v\rangle &= \langle \gamma(w), \gamma(v)\rangle \\
&= \langle (-1)^{np} * (\xi \wedge *w), (-1)^{np} * (\xi \wedge *v)\rangle \\
&= *(*(\xi \wedge *w) \wedge (-1)^{p(n-p)}\xi \wedge *v) \\
&= *(\xi \wedge *v \wedge *(\xi \wedge *w)) \\
&= \langle \xi \wedge *v, \xi \wedge *w\rangle.
\end{aligned}$$

Let us consider a basis $\{v_i\}$ of V which contains ξ and an expression of w in terms of the basis derived by $\{v_i\}$, say $\sum_{|I|=p+1} a_I v_I$ where $I = \{v_{i_1}, \ldots, v_{i_{p+1}}\} \subset \{v_i\}$ and $v_I = v_{i_1} \wedge \ldots \wedge v_{i_{p+1}}$ (to clarify the definition, it is assumed that orders of elements of all finite subset of $\{v_i\}$ are determined, possibly by using the Axiom of Choice if necessary). Since $\xi \wedge w = 0$, it follows that all I which $a_I \neq 0$ should contain ξ. Therefore, considering an expression of $*w$ derived by w, we see that there is no ξ in the expression of $*w$. It means that if calculating $\langle \xi \wedge *v, \xi \wedge *w\rangle$ by splitting matrices and getting determinants, the first column of matrices should be $(\langle \xi, \xi\rangle, 0, \ldots, 0)$. It means that $\langle \xi \wedge *v, \xi \wedge *w\rangle$ should be $\langle \xi, \xi\rangle\langle *v, *w\rangle = \langle \xi, \xi\rangle\langle v, w\rangle$.

16. $\sum_i \theta_i \wedge \omega_i = 0$. If we wedge $\omega_1 \wedge \ldots \wedge \widehat{\omega_i} \wedge \ldots \wedge \omega_p$, we get

$$\theta_i \wedge \omega_1 \wedge \ldots \wedge \omega_p = 0.$$

Therefore $\theta_i \in \langle \omega_1, \ldots, \omega_p\rangle$. So it is possible to write $\theta_i = \sum_j A_{ij}\omega_j$, where A_{ij} are $\mathcal{C}^\infty$ functions. One can derive $A_{ij} = A_{ji}$ by just putting these expressions into the given equation and noting that $\omega_i \wedge \omega_j$'s are linearly independent.

1.7 习题: Warner GTM 94/The Hodge Theorem

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1. If Δ operates on $E^p(M)$, then

$$\Delta = \delta d + d\delta = (-1)^{n(p+2)+1} * d * d + (-1)^{n(p+1)+1} d * d * .$$

(Be careful with the signs.) Therefore

$$*\Delta = (-1)^{n(p+2)+1+(n-p)p} d * d + (-1)^{n(p+1)+1} * d * d*$$

and

$$\Delta* = (-1)^{n(n-p+2)+1} * d * d * + (-1)^{n(n-p+1)+1+p(n-p)} d * d.$$

Since

$$n(p+2) + 1 + (n-p)p \equiv n(n-p+1) + 1 + p(n-p) \bmod 2$$

and

$$n(p+1) + 1 \equiv n(n-p+2) + 1 \bmod 2,$$

$*\Delta = \Delta*$ as desired.

2. (a) Obviously G is linear. To show that it is bounded, note that there exists a $c > 0$ such that $\|\beta\| \leq c\|\Delta\beta\|$ for all $\beta \in (H^p)^\perp$ by 6.8(4). Hence if we let $G(\alpha) = \omega$, then

$$\|G(\alpha)\| = \|\omega\| \leq c\|\Delta\omega\| = c\|\alpha - H(\alpha)\| \leq c\|\alpha\|.$$

Thus G is bounded.

(b) For all $\alpha, \beta \in (H^p)^\perp$, $\Delta G\alpha = \alpha, \Delta G\beta = \beta$ by definition. Thus,

$$\langle G\alpha, \beta\rangle = \langle G\alpha, \Delta G\beta\rangle = \langle \Delta G\alpha, G\beta\rangle = \langle \alpha, G\beta\rangle.$$

(c) Suppose $\{\alpha_n\}$ is a bounded sequence. Then $\{G\alpha_n\}$ is also bounded by (a), and $\{\Delta G\alpha_n\}$ is also bounded because

$$\|\Delta G\alpha_n\| = \|\alpha_n - H(\alpha_n)\| \leq \|\alpha_n\|.$$

Then use 6.6 to obtain the result.

3. It suffices to show that

$$\int_{\mathbb{R}^n} \left(1 + x_1^2 + \ldots + x_n^2\right)^{-k} dx_1 \ldots dx_n$$

converges for $k \geq \left[\frac{n}{2}\right] + 1$. By a change of variables $y_i = \frac{x_i}{\sqrt{1+x_n^2}}$ for $1 \leq i \leq n-1$ and $y_n = x_n$, it is equal to

$$\begin{aligned}
&\int_{\mathbb{R}^n} \left(1 + y_n^2\right)^{-k} \left(1 + y_1^2 + \ldots + y_{n-1}^2\right)^{-k} \left(1 + y_n^2\right)^{\frac{n-1}{2}} dy_1 \ldots dy_n \\
&= \int_{\mathbb{R}^n} \left(1 + y_1^2 + \ldots + y_{n-1}^2\right)^{-k} \left(1 + y_n^2\right)^{\frac{n-1}{2}-k} dy_1 \ldots dy_n \\
&= \int_{\mathbb{R}^{n-1}} \left(1 + y_1^2 + \ldots + y_{n-1}^2\right)^{-k} dy_1 \ldots dy_{n-1} \int_{\mathbb{R}} \left(1 + y_n^2\right)^{\frac{n-1}{2}-k} dy_n
\end{aligned}$$

(because $\dfrac{\partial(x_1,\ldots,x_n)}{\partial(y_1\ldots,y_n)} = (1+y_n^2)^{\frac{n-1}{2}}$). The former part converges by induction on n, and the latter one converges when $2k-(n-1)>1$, or $k>\frac{n}{2}$. It is equivalent to $k \geq \left[\frac{n}{2}\right]+1$ because k is an integer.

4.
$$\sum_{[\alpha]=0}^{t} \|D^\alpha \varphi\|^2 = \sum_{[\alpha]=0}^{t} \sum_{\xi} \xi^{2\alpha} |\varphi_\xi|^2.$$
Thus we only need to show that there exists $c>0$ which depends on t and n such that
$$c(1+|\xi|^2)^t \leq \sum_{[\alpha]=0}^{t} \xi^{2\alpha} \leq (1+|\xi|^2)^t$$
for all ξ. The second inequality is clear because each term in $\sum\limits_{[\alpha]=0}^{t} \xi^{2\alpha}$ is in the expansion of $(1+|\xi|^2)^t$ with coefficients ≥ 1. Also, if we set $c=n^{-t}$, then it is easy to see that the first inequality holds for similar reasons.

5. Since both are defined by pointwise inner products, there exists a positive-definite Hermitian matrix M_x for each $x\in\mathbb{R}^n$ such that $\varphi \circ' \bar{\psi} = \varphi \circ \overline{A\psi}$. Also it is smooth because the volume form on M is smooth.

6. I will deal only with the case of $\alpha = f dx_1 \wedge \ldots \wedge dx_p$. The general case is similar. By definition,
$$d\alpha = (-1)^p \sum_{i=p+1}^{n} \frac{\partial f}{\partial x_i} dx_1 \wedge \ldots \wedge dx_p \wedge dx_i,$$
and
$$*\alpha = f dx_{p+1} \wedge \ldots \wedge dx_n$$
(assuming $*1 = dx_1 \wedge \ldots \wedge dx_n$). Also
$$\delta\alpha = (-1)^{n(p+1)+1} * d * \alpha = (-1)^{n(p+1)+1} \sum_{i=1}^{p} (-1)^{p(i-1)+(p-1)(p-i)} \frac{\partial f}{\partial x_i} dx_1 \wedge \ldots \wedge \widehat{dx_i} \wedge \ldots \wedge dx_p.$$
Furthermore, by tedious calculation, we get
$$\Delta\alpha = -\sum_{i=1}^{n} \frac{\partial^2 f}{\partial x_i^2} dx_1 \wedge \ldots \wedge dx_p.$$

7. By 6.16(14),
$$-\frac{\partial^2 \varphi}{\partial x \partial y} = \sum_{(a,b)\in\mathbb{Z}^2} ab\varphi_{(a,b)} e^{i(ax+by)}.$$
Thus
$$\left\| \frac{\partial^2 \varphi}{\partial x \partial y} \right\|^2 = \sum_{(a,b)\in\mathbb{Z}^2} a^2 b^2 |\varphi_{(a,b)}|^2$$
by the Parseval identity. Similarly, we get
$$\|\Delta\varphi\|^2 = \sum_{(a,b)\in\mathbb{Z}^2} (a^2+b^2)^2 |\varphi_{(a,b)}|^2.$$
Thus the given inequality follows from $a^2b^2 \leq \frac{(a^2+b^2)^2}{4}$.

8. If $\{f_n\}$ is bounded in C^1, then it is uniformly bounded and equicontinuous by the assumption, thus by the Arzelà–Ascoli theorem it has a convergent subsequence.
9. The differential operators given in the following 4 problems are all elliptic. Therefore we can deduce that a solution exists if and only if the given function is in the orthogonal complement of the kernel of the formal adjoint.
   (a) Adjoint: $D^*(u) = -u'$
   (b) Adjoint: $D^*(u) = -u' - u$
   (c) Adjoint: $D^*(u) = u''$
   (d) Adjoint: $D^*(u) = u'' + \pi u$
10. By the Hodge decomposition theorem, we may represent any form ω as

$$\omega = \delta\alpha + d\beta + \gamma$$

with γ harmonic. If it is closed, then $d\omega = d\delta\alpha = 0$. Since $\langle \delta\alpha, \delta\alpha\rangle = \langle d\delta\alpha, \alpha\rangle = 0$, we see that $\delta\alpha = 0$. Hence $\omega = d\beta + \gamma$ and the result follows.
11. ($\Rightarrow$) Since $\mathcal{P}$ is a (infinite dimensional) vector space, the element $u = \sum_\xi u_\xi e^{ix\cdot\xi}$ as a formal sum is uniquely determined. It remains to show that in fact $u \in H_{-\infty}$. By 6.22(2), we know that

$$|l(\varphi)| \le \text{const} \sum_{[\alpha]\le k} \|D^\alpha \varphi\|_\infty \le \text{const}\, \|\varphi\|_N$$

for some $N \gg 0$. In other words,

$$\left|\sum_\xi u_\xi \varphi_\xi\right| \le \text{const}\, \|\varphi\|_N.$$

By squaring both sides, it becomes

$$\left|\sum_\xi u_\xi \varphi_\xi\right|^2 \le \text{const}\, \|\varphi\|_N^2.$$

Since multiplying $e^{i\theta}$ ($\theta \in \mathbb{R}$) does not change the absolute values of φ_ξ, we may assume that $u_\xi \| \varphi_\xi$. Then

$$\sum_\xi |u_\xi|^2|\varphi_\xi|^2 \le \left(\sum_\xi |u_\xi||\varphi_\xi|\right)^2 = \left|\sum_\zeta u_\xi \varphi_\xi\right|^2 \le \text{const}\, \|\varphi\|_N^2.$$

Since $\mathcal{P}$ is dense in H_{-N}, the inequality above is also true when $\varphi \in H_{-N}$. Now set

$$|\varphi_\xi| = (1 + |\xi|^2)^{-\frac{N+M}{2}}$$

for some $M > \left[\frac{n}{2}\right] + 1$. Then $\varphi \in H_{-N}$ by 6.16(7), and the inequality says that

$$\sum_\xi |u_\xi|^2|\varphi_\xi|^2 = \sum_\xi |u_\xi|^2 (1 + |\xi|^2)^{-N-M} < \infty,$$

that is to say $u \in H_{-N-M}$.

($\Leftarrow$) If $u \in H_{-\infty}$, there exists $s < 0$ such that $u \in H_s$. Then

$$\begin{aligned} |l(\varphi)| = |\langle u, \varphi\rangle_0| &\le \|u\|_s \|\varphi\|_{-s} \\ &\le \text{const} \sqrt{\sum_{[\alpha]\le -s} \|D^\alpha \varphi\|^2} \\ &\le \text{const} \sqrt{\sum_{[\alpha]\le -s} \|D^\alpha \varphi\|_\infty^2} \\ &\le \text{const} \sum_{[\alpha]\le -s} \|D^\alpha \varphi\|_\infty \end{aligned}$$

as desired.

12. It suffices to show that if $\int_M \alpha = 0$, then it is an exact form. $E^n = \operatorname{im} d \oplus H_n$ because $\operatorname{im} \delta = 0$. Let $\alpha = d\gamma + \varepsilon$ with ε harmonic. Since $[\alpha] \mapsto \int_M \alpha$ is the natural isomorphism, $[\alpha] = [\varepsilon] = 0$. Hence by 6.11, $\varepsilon = 0$. In other words, α is exact.
13. Let $\varphi_n = \sum_{a\in\mathbb{Z}} u_a^n e^{iax}$, where

$$u_a^n = \begin{cases} \frac{1}{a^2} & \text{if } |a| < n \\ 0 & \text{otherwise} \end{cases}.$$

Then

$$\|\varphi_n\|^2 = \sum_{a\in\mathbb{Z}} |u_a^n|^2 < \sum_{a\in\mathbb{Z}} \frac{1}{a^4}$$

and

$$\|\Delta\varphi_n\|^2 = \sum_{a\in\mathbb{Z}} a^2 |u_a^n|^2 < \sum_{a\in\mathbb{Z}} \frac{1}{a^2}$$

are both bounded. Also, φ_n is a finite sum, so $\varphi_n \in \mathcal{P}$. However, its limit

$$\varphi = \sum_{a\in\mathbb{Z}} \frac{1}{a^2} e^{iax} \notin H_3.$$

Thus it is not in $\mathcal{P}$.

14. The statement can be proved by following the proof of 6.32. The only property of Δ used here is that Δ is an elliptic operator. Also, by a "Reduction to the Periodic Case" argument, we only need to show this locally.
15. It is elliptic because $x + iy \neq 0$ for $(x, y) \neq (0, 0)$. Hence every holomorphic function is $\mathcal{C}^\infty$. Also every holomorphic function is harmonic because

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right).$$

If a holomorphic function f has compact support, then by Green's 1st identity,

$$\int_{\partial D} f \frac{\partial f}{\partial n} = \int_D \langle \operatorname{grad} f, \operatorname{grad} f\rangle - \int_D f\Delta f$$

for every disc D. Since $\Delta f = 0$, if we take a sufficiently large disc D, then it becomes

$$\int_D \langle \operatorname{grad} f, \operatorname{grad} f\rangle = 0.$$

By letting the radius of D go to infinity, we see that

$$\int_{\mathbb{C}} \langle \text{grad } f, \text{grad } f \rangle = 0,$$

and it means grad $f = 0$ everywhere, or f is constant. But f has compact support, so f has to be zero everywhere.

16. (a) If $\Delta u = \lambda u$ and $u \neq 0$,

$$\lambda \langle u, u \rangle = \langle \lambda u, u \rangle = \langle \Delta u, u \rangle = \langle (d\delta + \delta d)u, u \rangle = \langle du, du \rangle + \langle \delta u, \delta u \rangle \geq 0.$$

Thus $\lambda \geq 0$.

(b) If not, it contradicts 6.6 if we have an orthonormal sequence.

(c) If not, it also contradicts 6.6 if we have an orthonormal sequence.

(d) If $\Delta u = \lambda u$ and $\Delta v = \lambda' v$, then

$$\lambda \langle u, v \rangle = \langle \Delta u, v \rangle = \langle u, \Delta v \rangle = \lambda' \langle u, v \rangle.$$

Thus $\langle u, v \rangle = 0$.

(e)

$$\langle \psi_j, G\psi_j \rangle = \langle \Delta G\psi_j, G\psi_j \rangle = \langle dG\psi_j, dG\psi_j \rangle + \langle \delta G\psi_j, \delta G\psi_j \rangle \geq 0.$$

For the latter part, we need to show that $l\big((\Delta - 1/\eta)^* \alpha\big) = 0$. It is equivalent to

$$\lim_{j\to\infty} \eta \langle G\varphi_j, (\Delta - 1/\eta)^* \alpha \rangle = 0 \Leftrightarrow \lim_{j\to\infty} \eta \langle (\Delta - 1/\eta) G\varphi_j, \alpha \rangle = 0 \Leftrightarrow \lim_{j\to\infty} \langle \eta\varphi_j - G\varphi_j, \alpha \rangle = 0,$$

which is true. Also it is not trivial; take N such that $\|G\varphi_N - \eta\varphi_N\| < \varepsilon$ and $\|G\varphi_i - G\varphi_N\| < \varepsilon$ for $i > N$. Since $\|\varphi_N\| = 1$ and by the inequality

$$|\langle v, w \rangle| \geq \|v\|^2 - |\langle v, v - w \rangle| \geq \|v\|(\|v\| - \|v - w\|),$$

we see that $|\langle G\varphi_i, \eta\varphi_N \rangle| \geq \eta(\eta - 2\varepsilon) > 0$ for sufficiently small $\varepsilon > 0$. Thus $l(\varphi_N) \neq 0$.

(f) It is the same as (e).

(g) $\alpha - \sum_{i=1}^{k} \langle \alpha, u_i \rangle u_i \in (H^p)^\perp$, so there exists $\beta \in (H^p)^\perp$. If $k + 1 \leq j$,

$$\langle \alpha, u_j \rangle = \left\langle \alpha - \sum_{i=1}^{k} \langle \alpha, u_i \rangle u_i, u_j \right\rangle = \langle G\beta, u_j \rangle = \langle \beta, Gu_j \rangle = \frac{1}{\lambda_j} \langle \beta, u_j \rangle.$$

Thus

$$\langle \alpha, u_j \rangle u_j = \frac{1}{\lambda_j} \langle \beta, u_j \rangle u_j = G(\langle \beta, u_j \rangle u_j).$$

Then it follows from it and the definition of λ. $\lambda_n \to \infty$ because of (c).

(h) Locally it is true, and by the compactness of M, we can find c and k such that it is still true globally. Then it follows from (g).

17. By the Hodge decomposition, $E^p(M) = \text{im } \Delta \oplus H^p = \text{im } \Delta^2 \oplus V \oplus H^p$ with $V = \ker \Delta^2 \cap \text{im } \Delta$. If $\Delta\alpha \in V$, $\langle \Delta\alpha, \Delta\alpha \rangle = \langle \Delta^2\alpha, \alpha \rangle = 0$, so $\Delta\alpha = 0$. Thus $V = 0$ and $E^p(M) = \text{im } \Delta^2 \oplus H^p$. It means $\Delta^2\alpha = \beta$ is solvable if and only if $\beta \perp H^p$.

18. We can assume the matrix consisting of the coefficients of the highest order terms is symmetric because $\frac{\partial}{\partial x_i}$ and $\frac{\partial}{\partial x_j}$ commutes. Also, L is elliptic at x if for all $v \in \mathbb{R}^n \backslash \{0\}$, $v^t(a_{ij}(x))_{ij}v > 0$, which means that $(a_{ij}(x))_{ij}$ is positive definite. But for $\Box$, the corresponding matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is not positive definite, so it is not elliptic. For example, if we let

$$u(x,y) = \begin{cases} x^3 + 3xy^2 & \text{if} \quad x \geq |y| \\ 3x^2y + y^3 & \text{if} \quad -t \leq x \leq t \\ -3x^2y - y^3 & \text{if} \quad t \leq x \leq -t \\ -x^3 - 3xy^2 & \text{if} \quad x \leq |t| \end{cases}$$

it is easy to see that $\Box u = 0$, but $u \notin \mathcal{C}^3$.

19. Since $\Delta + c$ is elliptic, if $\beta \perp \ker(\Delta + c)$ then the equation can be solved. But $(\Delta + c)\alpha = 0 \Leftrightarrow \Delta\alpha = -c\alpha \Leftrightarrow \alpha = 0$ by assumption, so it can be solved for all $\beta \in E^p(M)$. (In fact, the minimal eigenvalue of Δ is 0.)
20. The first assertion is true because l_σ are isometries, thus they commute with d and δ. Then the second assertion directly follows from it. The third assertion is that $\varphi_i \circ l_\sigma$ is continuous as a function of σ, which is also true by the properties of Lie group.
21. The formal adjoint exists by argument similar to 6.24 (integration by parts) and the fact that the integral of exact forms over M is zero. Also, the proof of 6.5 and 6.6 is still valid because it is valid locally, and we can glue the solution with respect to the transition maps of vector bundles. Then (a) follows from 6.6, and (b) follows from arguments similar to 6.8.
22. For n sufficiently large, or $n > 10$, then

$$\|\varphi_n\|_\infty = \varphi(0,0) = \log\log n.$$

On the other hand, by 6.16(15),

$$\text{const}\,\|\varphi_n\|_1^2 \leq \|\varphi_n\|^2 + \left\|\frac{\partial\varphi_n}{\partial x}\right\|^2 + \left\|\frac{\partial\varphi_n}{\partial y}\right\|^2.$$

Since $\{\varphi_n\}$ is monotonically increasing,

$$\|\varphi_n\|^2 \to \int_{r\leq\frac{1}{2}} \left(\log\log\frac{1}{r}\right)^2 dxdy < \infty$$

by the monotone convergence theorem. Also,

$$\frac{\partial\varphi_n}{\partial x} = \frac{x}{r\left(\frac{1}{n}+r\right)\log\left(\frac{1}{n}+r\right)}$$

and we can use a similar argument to see that

$$\left\|\frac{\partial\varphi_n}{\partial x}\right\|^2 \to \int_{r\leq\frac{1}{2}} \left(\frac{x}{r^2\log r}\right)^2 dxdy = \frac{\pi}{\log 2}.$$

Similarly,

$$\left\|\frac{\partial\varphi_n}{\partial y}\right\|^2 \to \frac{\pi}{\log 2}.$$

Thus $\|\varphi_n\|_1$ is bounded and there is no constant c satisfying the given condition.

23. By the assumption, there exists a solution u of $Lu = f$. Also, since $\mathcal{C}^\infty(M) = \operatorname{im} L^* \oplus \ker L$, we may take $u = L^*v \in \operatorname{im} L^*$. Since $f = f\circ\gamma = Lu\circ\gamma = L(u\circ\gamma)$, $u\circ\gamma$ is also a solution. Meanwhile, $L(\beta\circ\gamma) = 0 \Leftrightarrow L\beta\circ\gamma = 0 \Leftrightarrow L\beta = 0$, $\varphi_\gamma\colon \mathcal{C}^\infty(M) \to \mathcal{C}^\infty(M), \beta \mapsto \beta\circ\gamma$ preserves $\ker L$. Also, φ_γ is an isomorphism (with the inverse $\varphi_{\gamma^{-1}}$) and preserves the inner product (because it preserves the volume form), we conclude that φ_γ also preserves $\operatorname{im} L^*$. Thus $u\circ\gamma \in \operatorname{im} L^*$. Write $u\circ\gamma = L^*v'$. Then

$$LL^*v = LL^*v' \Rightarrow LL^*(v-v') = 0 \Rightarrow 0 = \langle LL^*(v-v'), v-v'\rangle = \langle L^*(v-v'), L^*(v-v')\rangle \Rightarrow L^*v = L^*v',$$

we conclude that $u = u\circ\gamma$, or u is invariant under γ.

1.8 习题: 微分流形上课习题

原网页

1. 验证 $\mathbf{C}^n$ 上所有多元全纯函数所构成的集合 $\mathcal{O}$ 是一个环。

Pf: 对于 $f, g \in \mathcal{O}$，即 f, g 满足 Cauchy-Riemann 方程组，这等价于 f, g 作为每个分量的函数（如 $f(\cdot, z_2, \cdots, z_n)$，$z_2, \cdots, z_n \in \mathbf{C}$ 固定）满足 Cauchy-Riemann 方程，则 f, g 作为每个分量的函数全纯，于是 $f+g, fg$ 作为每个分量的函数也全纯，那么 $f+g, fg$ 也满足 Cauchy-Riemann 方程组，得到 $f+g, fg \in \mathcal{O}$.

或使用 $f \in \mathcal{O}$ 等价于 f 复可微（即 $f(z+\Delta z) = f(z) + f'(z)\cdot\Delta z + o(\Delta z)$ 当 $\Delta z \to 0$），复可微函数相加、相乘所得函数复可微.

2. 设 $f\colon M^m \to N^n$ 在 p 点是淹没，则存在 p 点附近的局部坐标卡 (U, φ) 和 $q = f(p)$ 点附近的局部坐标卡 (V, ψ) 使 $\psi\circ f\circ\varphi^{-1}\colon \varphi(U) \to \psi(V)$ 的表达式为 $(x_1, \cdots, x_m) \mapsto (x_1, \cdots, x_n)$.

Pf: 设 p 点附近有局部坐标卡 (U', φ')，q 点附近有局部坐标卡 (V', ψ')，$\psi'\circ f\circ\varphi^{-1} : \varphi(U') \to \psi'(V), (x_1, \cdots, x_m) \mapsto (y_1, \cdots, y_n)$ 的 Jacobi 矩阵 J 在 p 点满秩.

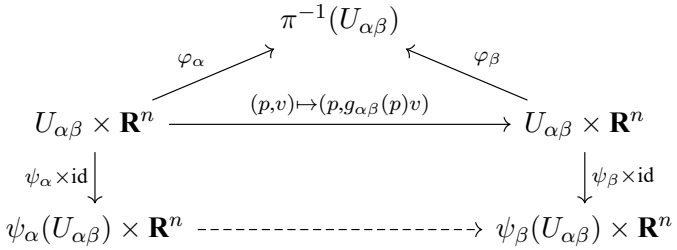
通过调整 φ' 坐标分量的次序（记为 φ），不妨设 J 左边的 $n\times n$ 子方阵非奇异，由隐函数定理，存在 $f(p)$ 的邻域 $V \subset V'$ 与 p 的邻域 $U \subset U'$ 使得有微分同胚 $F\colon \psi(V) \to \varphi(U), (y_1, \cdots, y_n) \mapsto (x_1, \cdots, x_n)$. 令 $\psi = F\circ\psi'$ 即可，$\psi\circ f\circ\varphi^{-1} = F\circ(\psi'\circ f\circ\varphi^{-1})$.

注： 更一般的结论称为常秩定理：设 f 在 p 点的秩为 r，则 f 在合适的局部坐标系下的表达式为 $(x_1, \cdots, x_r, x_{r+1}, \cdots, x_m) \mapsto (x_1, \cdots, x_r, 0, \cdots, 0)$.

3. 证明光滑流形 (复流形) 上的光滑向量丛 (全纯向量丛)E 是光滑流形 (复流形).

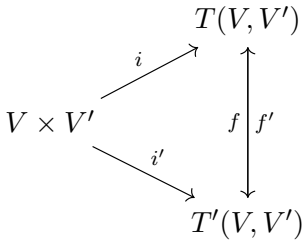
Pf: 设此向量丛为 (E, π, M)，$\{(U_\alpha, \psi_\alpha)\}$ 是覆盖流形 M 的一族（较细的）坐标卡，使得 $U_\alpha \times \mathbf{R}^n (U_\alpha \times \mathbf{C}^n)$ 与 $\pi^{-1}(U_\alpha)$ 的同胚，记为 φ_α.

那么 $\{(\pi^{-1}(U_\alpha),(\psi_\alpha\times\mathrm{id})\circ\varphi_\alpha^{-1}\}$ 就是覆盖 $E=\pi^{-1}(M)$ 的一族坐标卡，且对于 $U_{\alpha\beta}:=U_\alpha\cap U_\beta\neq\varnothing$，转移函数 $\big((\psi_\beta\times\mathrm{id})\circ\varphi_\beta^{-1}\circ\varphi_\alpha\circ(\psi_\alpha^{-1}\times\mathrm{id})\big)(x,v)=((\psi_\beta\circ\psi_\alpha^{-1})(x),g_{\alpha\beta}(\psi_\alpha^{-1}(x))v)$ 是 $\psi_\alpha(U_{\alpha\beta})\times\mathbf{R^n}$ 到 $\psi_\beta(U_{\alpha\beta})\times\mathbf{R^n}$ 的光滑函数 (全纯函数)，其中 $g_{\alpha\beta}(p)=\varphi_\beta^{-1}\circ\varphi_\alpha\big|_{p\times\mathbf{R}^n}$ 是光滑函数 (全纯函数).



4. 验证：由“万有性质”定义的张量积是唯一的（同构意义下）.

Pf: 设 $(T(V,V'),i)$ 和 $(T'(V,V'),i')$ 是 V 和 V' 的两个张量积，由万有性质，存在 $T(V,V')$ 到 $T'(V,V')$ 的线性映射 f 使得 $i\circ f=i'$ 以及 $T'(V,V')$ 到 $T(V,V')$ 的线性映射 f' 使得 $i'\circ f'=i$. 那么，$i=i'\circ f=i\circ f\circ f'$，再根据万有性质，使得 $i=i\circ g$ 的线性映射 $g\colon T(V,V')\to T(V,V')$ 是唯一的，因此 $f\circ f'=\mathrm{id}_{T(V,V')}$. 同理有 $f'\circ f=\mathrm{id}_{T'(V,V')}$，所以 $T(V,V')$ 与 $T'(V,V')$ 同构.



5. 证明：若同一底空间 B 上的三个向量丛满足 $0\to E_1\xrightarrow{f_1}E_2\xrightarrow{f_2}E_3\to 0$ 正合，则 E_1 可看作 E_2 的子丛（即 $f_1(E_1)$ 是 E_2 的子丛），E_3 丛同构于 E_2/E_1 这个商丛.

Pf: 对任何 $b\in B$，$f_1((E_1)_b)\subset(E_2)_b$，所以 $f_1(E_1)\subset E_2$. 由于线性映射 $f_1\colon(E_1)_b\to f_1((E_1)_b)$ 既单又满，即是同构，所以可以把 E_1 看作 $f(E_1)$.

对任何 $b\in B$，由 $\mathrm{im}\,f_1|_b=\ker f_2|_b$ 及同态基本定理，$(E_2)_b/(E_1)_b\simeq(E_2)_b/f_1((E_1)_b)\simeq(E_2)_b/\ker f_2|_b\simeq\mathrm{im}\,f_2|_b=(E_3)_b$，不难验证这些纤维的同构拼在一起就是 $E_3=\bigcup\limits_{b\in B}E_b$ 与 E_2/E_1 之间的丛同构.

6. 设 M 是光滑流形，正向集 $I=\{U\text{ 为}M\text{ 的开集}\}$，$U\le V$ 当且仅当 $V\subseteq U$。环 $A_U:=C^\infty(U)$.

(1) 对 $U\le V$，定义 $i_{UV}\colon A_U\to A_V, f\mapsto f|_V$，验证 $\{(A_U,i_{UV})\}$ 是正向系统.

(2) 设 $\{\mathcal{F}(U)\}_{U\in I}$ 是 U 上的导算子模，对 $U\le V$，定义 $i_{UV}\colon\mathcal{F}(U)\to\mathcal{F}(V), D_U\mapsto D_U|_V$，验证 $\{(\mathcal{F}(U),i_{UV})\}$ 是正向系统.

Pf: (1) $i_{UU}=\mathrm{id}_U$.

对 $U\le V\le W$（即 $W\subset V\subset U$），$(i_{UV}\circ i_{VW})(f)=(f|_V)|_W=f|_W=i_{UW}f$.

(2) $i_{UU}(D_U)=D_U|_U=D_U$.

要说明对 $U\le V\le W$ 有 $(i_{UV}\circ i_{VW})D_U=i_{UW}D_U$，也就是验证对任意 $p\in W$ 与 $f\in$

$C^\infty(U)$，$((D_U|_V)|_W f)(p) = (D_U|_W f)(p)$.
取 p 的邻域 X 使 $\overline{X} \subset W$，存在 $\eta \in C^\infty(M)$ 使得 $\eta|_X \equiv 1$，$\eta|_{M\setminus W} \equiv 0$，那么也有 $\overline{X} \subset V$ 且 $\eta|_{M\setminus V} \equiv 0$，于是 $((D_U|_V)|_W f)(p) = (\eta(D_U|_V)f)(p) = (\eta^2 D_U f)(p)$. 又由 $\eta^2 \in C^\infty(M)$ 也使得 $\eta^2|_X \equiv 1$，$\eta^2|_{M\setminus W} \equiv 0$，得到 $((D_U|_W)f)(p) = (\eta^2 D_U f)(p)$.

7. 根据外微分的延拓的定义，证明

$$\langle dw, X_0, \cdots, X_k\rangle = \sum_{i=0}^{k}(-1)^i X_i\langle w, X_0, \cdots, \hat{X}_i, \cdots, X_k\rangle + \sum_{i<j}(-1)^{i+j}\langle w, [X_i, X_j], X_0, \cdots, \hat{X}_i, \cdots, \hat{X}_j, \cdots,$$

Pf: 见陈维桓《微分流形初步》第 4 章定理 2.2（P188）.

8. 对于外微分 d 作用在 k 次微分式形式 ω 的表达式

$$\langle dw, X_0, \cdots, X_k\rangle = \sum_{i=0}^{k}(-1)^i X_i\langle w, X_0, \cdots, \hat{X}_i, \cdots, X_k\rangle + \sum_{i<j}(-1)^{i+j}\langle w, [X_i, X_j], X_0, \cdots, \hat{X}_i, \cdots, \hat{X}_j, \cdots,$$

验证 $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$.

Pf: 在上一题中我们已经证明了表达式不依赖于局部坐标的选取，因此可以用局部坐标的表达式完成这一问的证明.

9. 可微流形之间的光滑映射 $f, g\colon M \to N$ 称为同伦，如果有一个光滑映射

$$H\colon I \times M \to N,$$

使得：$H(0, x) = f(x), H(1, x) = g(x)$，其中 $I = [0, 1]$. 对于 k- 形式 ω，定义对于 $(k-1)$- 形式 $\mathrm{I}\,\omega$ 为

$$\langle \mathrm{I}\,\omega, v_1, \cdots, v_{k-1}\rangle(x) = \int_0^1 \langle H^*\omega, \frac{\partial}{\partial t}, v_1, \cdots, v_{k-1}\rangle(t, x)dt.$$

证明：

$$g^*\omega - f^*\omega = d\,\mathrm{I}\,\omega + \mathrm{I}\,d\omega.$$

Pf: 设 $v_1, \cdots, v_k \in \Gamma(N)$，有

$$\begin{aligned}
\langle d\,\mathrm{I}\,\omega, v_1, \cdots, v_k\rangle(x) &= \sum_{i=1}^{k}(-1)^{i-1} v_i\langle \mathrm{I}\,\omega, v_1, \cdots, \hat{v_i}, \cdots, v_k\rangle(x) + \sum_{i<j}(-1)^{i+j}\langle \mathrm{I}\,\omega, [v_i, v_j], \cdots, \hat{v_i}, \cdots, \hat{v_j}, \cdots\rangle \\
&= \sum_{i=1}^{k}(-1)^{i-1} v_i \int_0^1 \langle H^*\omega, \frac{\partial}{\partial t}, v_1, \cdots, \hat{v_i}, \cdots, v_k\rangle(t, x)\,dt \\
&\quad + \sum_{i<j}(-1)^{i+j} \int_0^1 \langle H^*\omega, \frac{\partial}{\partial t}, [v_i, v_j], \cdots, \hat{v_i}, \cdots, \hat{v_j}, \cdots\rangle(t, x)\,dt, \\
\langle \mathrm{I}\,d\omega, v_1, \cdots, v_k\rangle(x) &= \int_0^1 \langle H^* d\omega, \frac{\partial}{\partial t}, v_1, \cdots, v_k\rangle(t, x)\,dt
\end{aligned}$$

$$
\begin{aligned}
&= \int_0^1 \langle dH^*\omega, \frac{\partial}{\partial t}, v_1, \cdots, v_k\rangle(t,x)\,dt \\
&= \int_0^1 \frac{\partial}{\partial t}\langle H^*\omega, v_1, \cdots, v_k\rangle(t,x)\,dt + \sum_{i=1}^{k}(-1)^i v_i\langle H^*\omega, \frac{\partial}{\partial t}, v_1, \cdots, \hat{v_i}, \cdots, v_k\rangle(t,x)\,dt \\
&+ \sum_{i<j}(-1)^{i+j}\int_0^1 \langle H^*\omega, [v_i, v_j], \frac{\partial}{\partial t}, \cdots, \hat{v_i}, \cdots, \hat{v_j}, \cdots\rangle(t,x)\,dt \\
&+ \sum_{i<j}(-1)^{i}\int_0^1 \langle H^*\omega, [\frac{\partial}{\partial t}, v_j], \frac{\hat{\partial}}{\partial t}, \cdots, \hat{v_i}, \cdots\rangle(t,x)\,dt.
\end{aligned}
$$

注意到 $[\frac{\partial}{\partial t}, v_j] = 0$，于是

$$
\begin{aligned}
\langle d\,\mathrm{I}\,\omega + \mathrm{I}\,d\omega, v_1, \cdots, v_k\rangle(x) &= \int_0^1 \frac{\partial}{\partial t}\langle H^*\omega, v_1, \cdots, v_k\rangle(t,x)\,dt \\
&= \langle H^*\omega, v_1, \cdots, v_k\rangle(1,x) - \langle H^*\omega, v_1, \cdots, v_k\rangle(0,x) \\
&= \langle g^*\omega, v_1, \cdots, v_k\rangle(x) - \langle f^*\omega, v_1, \cdots, v_k\rangle(x).
\end{aligned}
$$

10. (1) 证明：对于 k- 单形 σ，重心重分完成后，σ 是一些单形的并；
(2) 计算重心重分后 k- 单形的个数；
(3) 证明：当重心重分的次数趋于无穷时，单形的直径趋于 0.

Pf: (1) 设 $\sigma = \langle x_0, \cdots, x_k\rangle$.
用归纳容易说明，当 σ 重心重分完成后得到的是一组形如 $\langle b(\tau_0), b(\tau_1), \cdots, b(\tau_k)\rangle$ 的单形，其中 $\tau_0 = x_{i_0}, \tau_1 = \langle x_{i_0}, x_{i_1}\rangle, \cdots, \tau_k = \langle x_{i_0}, x_{i_1}, \cdots, x_{i_k}\rangle = \sigma$，即该单形为

$$
\left\{ \sum_{i=0}^{k}\lambda_i x_i \;\middle|\; \sum_{i=0}^{k}\lambda_i = 1, \lambda_{i_0} \geq \cdots \geq \lambda_{i_k} \geq 0 \right\},
$$

所以 σ 是这样一些单形的并.
(2) 即 $0, \cdots, k$ 的不同排列 $i_0, \cdots, i_k$ 的个数，为 $(k+1)!$ 个.
(3) 设 σ 重心重分一次后得到单形 $\sigma_1 = \langle b(\tau_0), b(\tau_1), \cdots, b(\tau_k)\rangle$，下证其直径不超过 σ 直径的 $\frac{k}{k+1}$ 倍，即可得到结论.
紧集 σ_1 上必存在两点使得其距离是 σ_1 的直径，这两点是 $b(\tau_l), b(\tau_m)$，这是因为对任何 σ_1 的不是顶点的点，沿两个相反的方向移动它，至少沿其中一个方向时它与另一点的距离增加.
不妨设

$$
b(\tau_l) = \frac{1}{l+1}\sum_{j=0}^{l} x_j,\ b(\tau_m) = \frac{1}{m+1}\sum_{j=0}^{m} x_j,\ 0 \leq l < m \leq k,
$$

则

$$
\begin{aligned}
\operatorname{diam}\sigma_1 &= |b(\tau_l) - b(\tau_m)| \\
&\leq \max_{0\leq i\leq l}|x_i - b(\tau_m)| \\
&= \max_{0\leq i\leq l}|x_i - \frac{1}{m+1}\sum_{j=0}^{m} x_j|
\end{aligned}
$$

$$
\begin{aligned}
&\leq \max_{0\leq i\leq l}\sum_{j=0}^{m}\frac{1}{m+1}|x_i-x_j| \\
&\leq \frac{m}{m+1}\operatorname{diam}\sigma \quad \text{因为}l<m \\
&\leq \frac{k}{k+1}\operatorname{diam}\sigma.
\end{aligned}
$$

1.9 习题: 陈维桓微分流形

原网页

- 第一章预备知识
- 第二章微分流形
- 第三章切向量场
- 第四章外微分式

1.10 习题: 陈维桓微分流形/第一章预备知识

原网页

9. (1) 代入 $\mathrm{D}_v f=\langle\nabla f(P),v\rangle$ 即可。
   (2) $\mathrm{D}_u(g\cdot h)=\langle\nabla(g\cdot h)(P),u\rangle=\langle g(P)\cdot\nabla h(P),u\rangle+\langle h(P)\cdot\nabla g(P),u\rangle=g(P)\cdot\mathrm{D}_u h+h(P)\cdot\mathrm{D}_u g$。
11. (1) $J_{\tilde f}=\left(\frac{\partial\tilde f_i}{\partial\tilde x_j}\right)_{n\times m}=\left(\frac{\partial\tilde f_i}{\partial x_k}\frac{\partial x_k}{\partial\tilde x_j}\right)_{n\times m}=\left(\frac{\partial(a_l^i f_l)}{\partial x_k}\frac{\partial x_k}{\partial(b_s^j x_s)}\right)_{n\times m}=\left(a_j^i\frac{\partial f_i}{\partial x_j}(b_k^j)^{-1}\right)_{n\times m}=AJ_fB^{-1}$。
    (2) 由 A,B 均可逆得 $\operatorname{rk}J_f'=\operatorname{rk}J_f$。
    (3) 完全同理。将曲纹坐标系看成坐标变换即可，此时的坐标变换函数为曲纹坐标变换所对应的 Jacobi 矩阵。
12. 只需证明在所给区域 $\frac{\partial(x^1,\cdots,x^n)}{\partial(u^1,\cdots,u^n)}\neq 0$ 即可。
    写出行列式并消去第一行并使用数学归纳法能够证明（具体过程可参考大多数数学分析教材）

$$
\frac{\partial(x^1,\cdots,x^n)}{\partial(u^1,\cdots,u^n)}=r_1r_2\cdots r_n(u^n)^{n-1}(\cos u^1)^{n-2}(\cos u^2)^{n-3}\cdots\cos u^{n-2},
$$

    由 $x_1\neq 0$ 得 $u^n,\cos u^1,\cdots,\cos u^{n-2}\neq 0$，所以 $\frac{\partial(x^1,\cdots,x^n)}{\partial(u^1,\cdots,u^n)}\neq 0$。
17. f 是线性变换是显然的。
    要证明与基底选取无关，只需取另一组基底 $\tilde\delta_i$，验证这两组基底下的计算结果一致即可。设另一组基底为 $\tilde\delta_i=a^{ij}\delta_j$，则 $\tilde\delta^i=\tilde a_{ji}\delta^j$，其中 $\tilde a_{ij}a^{jk}=\delta_{ik}$，那么（已省略对 i 求和）

$$
\begin{aligned}
\tilde f(u,\tilde\delta^i)\tilde\delta_i&=\tilde f(u,\tilde a_{ji}\delta^j)a^{ik}\delta_k \\
&=\tilde a_{ji}a^{ik}\tilde f(u,\delta^j)\delta_k \\
&=\delta_{jk}\tilde f(u,\delta^j)\delta_k \\
&=\tilde f(u,\delta^i)\delta_i.
\end{aligned}
$$

18. 一方面，对于每一个 r 重线性映射 t，可以定义 $F(\alpha,v_1,\cdots,v_n):=\alpha t(v_1,\cdots,v_n)$;
    另一方面，对于每一个 $(1,r)$ 型张量 $F(\alpha,v_1,\cdots,v_n)$，可以定义 $t(v_1,\cdots,v_n)=\sum_{i=1}^{m}F(\delta^i,v_1,\cdots,v_n)\delta_i$。类似于上一题，由 F 的线性型可知 t 的定义与基底选取无关。

20. 取 V 的两个线性无关的向量 v_1, v_2，W 的两个线性无关的向量 w_1, w_2，$v_1 \otimes w_1, v_2 \otimes w_2$ 均在此集合中，但 $v_1 \otimes w_1 + v_2 \otimes w_2$ 并不在其中。这是因为对于 $v = \sum_{i=1}^{m} a_i v_i$ 和 $w = \sum_{j=1}^{n} b_j w_j$，$v \otimes w = \sum_{i=1}^{m}\sum_{j=1}^{n} a_i b_j v_i \otimes w_j$ 的系数构成的矩阵 $(a_i b_j)_{\substack{1\le i\le m,\\ 1\le j\le n}}$ 的秩为 1，而 $v_1 \otimes w_1 + v_2 \otimes w_2$ 的系数的矩阵 $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ 的秩为 2.
21. 若 $e_i = a_i^k \delta_k$ 为另一组基底，且在 e_i 基底下，f 的表示矩阵为 $c_m^j a_i^l b_l^m$，其中 (c_j^i) 是 (a_j^i) 的逆矩阵，则
$$\tilde{b_i^j}\tilde{b_k^i}\tilde{b_j^k} = c_m^j a_i^l c_j^i a_k^h c_n^k a_j^m b_l^m b_h^l b_m^n. = b_l^m b_h^l b_m^h = B_3$$
只需对 $(3,3)$ 型张量 $F(\alpha^1, \alpha^2, \alpha^3, v_1, v_2, v_3) := \alpha^1(f(v_2))\alpha^2(f(v_3))\alpha^3(f(v_1))$ 缩并三次即可表示出 B_3。
22. 是 $F(\alpha, v) = \alpha(v)$，因为 $F(\delta^j, \delta_i) = \delta^j(\delta_i) = \delta_i^j$。
23. 不存在。因为该张量无法缩并。
    事实上，任取一个基变换，容易发现该张量一定与基选取有关。
24.
$$\begin{aligned}\sigma(f)(v_1, \cdots, v_q) &= \alpha^1 \otimes \cdots \otimes \alpha^q(v_{\sigma(1)}, \cdots, v_{\sigma(q)}) \\ &= \alpha^1(v_{\sigma(1)}) \cdots \alpha^q(v_{\sigma(q)}) \\ &= \alpha^{\tau(1)}(v_1) \cdots \alpha^{\tau(q)}(v_q) \\ &= \alpha^{\tau(1)} \otimes \cdots \otimes \alpha^{\tau(q)}(v_1, \cdots, v_q).\end{aligned}$$
25. (1) 由 $\xi = S_q(\xi)$ 容易得到 ξ 对称；另一方面，当 ξ 对称时，有 $S_q(\xi) = \frac{\sum_{\sigma\in\mathscr{S}} \sigma(\xi)}{q!} = \frac{\sum_{\sigma\in\mathscr{S}} \xi}{q!} = \frac{|\mathscr{S}(q)|\xi}{q!} = \xi$.

    (2) 由 $\xi = A_q(\xi)$ 容易得到 ξ 反对称；另一方面，当 ξ 对称时，有 $A_q(\xi) = \frac{\sum_{\sigma\in\mathscr{S}} \mathrm{sign}(\sigma)\sigma(\xi)}{q!} = \frac{\sum_{\sigma\in\mathscr{S}} \mathrm{sign}(\sigma)^2\xi}{q!} = \frac{|\mathscr{S}(q)|\xi}{q!} = \xi$.
26. 设 σ 为二阶张量上的对换，取 $\xi_1 = \frac{\xi+\sigma(\xi)}{2}$ 与 $\xi_2 = \frac{\xi-\sigma(\xi)}{2}$. 则 $\xi = \xi_1 + \xi_2$ 且 ξ_1 与 ξ_2 分别为对称与反对称张量.
27. 通过列举 $\mathscr{S}$ 中元素并利用 φ 给定性质，可计算得 $A_3(\varphi) = \frac{1}{6}\sum_{\sigma\in\mathscr{S}} \mathrm{sign}(\sigma)\sigma(\varphi) = \frac{1}{6}(2-4)\varphi = -\frac{1}{3}\varphi$. 由 $A_3(\varphi)$ 的反对称性可推知 φ 反对称.
    同时，取任意 $v_1, v_2, v_3 \in V$，由 $\varphi(v_2, v_1, v_3) = \varphi(v_1, v_2, v_3) = -\varphi(v_2, v_1, v_3)$ 得到 $\varphi = 0$.
28. 任取 $u, v \in V$，由条件知 $a(u+v, u+v) - a(u, u) - a(v, v) = 0$，化简即得 $a(u, v) + a(v, u) = 0$，即 $S_2(a) = 0$.
29. 取定一组基 $\{\delta^i\}$. 根据题目条件，取 $a = E^{ij} + E^{ji}$ 可得 $b_{ij} + b_{ji} = 0$，从而可得 b 反对称.
30. 令 $g := f - \tilde{f}$，则对任意 $x \in V$ 有 $g(x, x) = 0$，从第 28 题可得 g 反对称，但 g 又对称，因而 $g = 0$，从而 $f = \tilde{f}$.
31. 仅需依据行列式的逆序数定义对于 $i_1 \ldots i_r$ 是不是 $j_1 \ldots j_r$ 的置换分别进行讨论即可，此不赘述.

32. (1)

$$\begin{aligned}\frac{1}{r!}\delta^{k_1\dots k_r}_{j_1\dots j_r}\delta^{i_1\dots i_{r+s}}_{k_1\dots k_r j_{r+1}\dots j_{r+s}} &= \frac{1}{r!}\delta^{k_1\dots k_r}_{j_1\dots j_r}\delta^{i_1\dots i_r}_{k_1\dots k_r}\delta^{i_{r+1}\dots i_{r+s}}_{j_{r+1}\dots j_{r+s}}\\ &= \frac{|\mathscr{S}(r)|}{r!}\delta^{i_1\dots i_r}_{j_1\dots j_r}\delta^{i_{r+1}\dots i_{r+s}}_{j_{r+1}\dots j_{r+s}}\\ &= \delta^{i_1\dots i_{r+s}}_{j_1\dots j_{r+s}}.\end{aligned}$$

(2) 所求即为 n 个数中任取 r 个的排列，从而 $\delta^{i_1\dots i_r}_{i_1\dots i_r} = P^n_r = \frac{n!}{(n-r)!}$.

33. 对于任意的 $\sigma \in \mathscr{S}$ 有

$$\det a = \operatorname{sign}(\sigma) a^{\sigma(i_1)}_{i_1}\cdots a^{\sigma(i_n)}_{i_n},$$

因此

$$\det a = \frac{1}{n!}\sum_{\sigma\in\mathscr{S}}\operatorname{sign}(\sigma)a^{\sigma(i_1)}_{i_1}\cdots a^{\sigma(i_n)}_{i_n} = \frac{1}{n!}\delta^{i_1\cdots i_n}_{j_1\cdots j_n}a^{j_1}_{i_1}\cdots a^{j_n}_{i_n}.$$

34. 构造 (r,r) 型张量 $f(\alpha^1,\cdots,\alpha^r,v_1,\cdots,v_r) := \sum_{\sigma\in\mathscr{S}(r)}\operatorname{sign}(\sigma)\alpha^1(v_{\sigma(1)})\cdots\alpha^r(v_{\sigma(r)})$ 取定 V 的一组基 $\{e_i\}$ 和对偶基 $\{e^i\}$，则 f 在这组基下的分量

$$\begin{aligned}f^{i_1\cdots i_r}_{j_1\cdots j_r} &= f(e^{i_1},\cdots,e^{i_r},e_{j_1},\cdots,e_{j_r})\\ &= \sum_{\sigma\in\mathscr{S}(r)}\operatorname{sign}(\sigma)e^{i_1}(e_{j_{\sigma(1)}})\cdots e^{i_r}(e_{j_{\sigma(r)}})\\ &= \sum_{\sigma\in\mathscr{S}(r)}\operatorname{sign}(\sigma)\delta^{i_1}_{j_{\sigma(1)}}\cdots\delta^{i_r}_{j_{\sigma(r)}} = \delta^{i_1\cdots i_r}_{j_1\cdots j_r}.\end{aligned}$$

35. 不能，因为该分量不是协变的. 事实上，任取一个基变换, 容易发现该张量一定与基选取有关.

36. 设 $\{e_i\}$ 是另一组基，满足 $e_i = a^j_i\delta_j$. 设 $(\bar{b}^j_i)$ 是 f 在基 $\{e_i\}$ 下的表示矩阵，则 $\bar{b}^j_i = a^m_i c^j_n b^n_m$. 其中 (c^j_i) 是 (a^j_i) 的逆矩阵. 在基底 $\{e_i\}$ 下计算得

$$\begin{aligned}\tilde{B}'_r &= \delta^{i_1\cdots i_r}_{j_1\cdots j_r}\bar{b}^{j_1}_{i_1}\cdots\bar{b}^{j_r}_{i_r} = \delta^{i_1\cdots i_r}_{j_1\cdots j_r}(a^{m_1}_{i_1}c^{j_1}_{n_1}b^{n_1}_{m_1})\cdots(a^{m_r}_{i_r}c^{j_r}_{n_r}b^{n_r}_{m_r})\\ &= \sum_{\sigma\in\mathscr{S}(r)}\delta^{i_1\cdots i_r}_{i_{\sigma(1)}\cdots i_{\sigma(r)}}(a^{m_1}_{i_1}c^{i_{\sigma(1)}}_{n_1}b^{n_1}_{m_1})\cdots(a^{m_r}_{i_r}c^{i_{\sigma(r)}}_{n_r}b^{n_r}_{m_r})\\ &= \sum_{\sigma\in\mathscr{S}(r)}\operatorname{sign}(\sigma)(a^{m_1}_{i_1}c^{i_1}_{n_{\sigma^{-1}(1)}})\cdots(a^{m_r}_{i_r}c^{i_r}_{n_{\sigma^{-1}(r)}})b^{n_1}_{m_1}\cdots b^{n_r}_{m_r}\\ &= \sum_{\sigma\in\mathscr{S}(r)}\operatorname{sign}(\sigma)\delta^{m_1}_{n_{\sigma^{-1}(1)}}\cdots\delta^{m_r}_{n_{\sigma^{-1}(r)}}b^{n_1}_{m_1}\cdots b^{n_r}_{m_r}\\ &= \det(\delta^{m_i}_{n_j})b^{n_1}_{m_1}\cdots b^{n_r}_{m_r} = \delta^{m_1\cdots m_r}_{n_1\cdots n_r}b^{n_1}_{m_1}\cdots b^{n_r}_{m_r} = \tilde{B}_r.\end{aligned}$$

37. 见 P40,41.

38. 设 $\{e_i\},\{e^i\}$ 是另一组基和对偶基. 设 $e_i = a^j_i\delta_j$，则 $e^i = c^j_i\delta^j$，其中 (c^j_i) 是 (a^j_i) 的逆矩阵. 那么

$$\begin{aligned}&\sum_{1\le i_1<\cdots<i_r\le n}(e_{i_1}\wedge\cdots\wedge e_{i_r})\otimes\left(e^{i_1}\wedge\cdots\wedge e^{i_r}\right)\\ &= \frac{1}{r!}(e_{i_1}\wedge\cdots\wedge e_{i_r})\otimes\left(e^{i_1}\wedge\cdots\wedge e^{i_r}\right)\\ &= \frac{1}{r!}a^{j_1}_{i_1}\cdots a^{j_r}_{i_r}c^{i_1}_{k_1}\cdots c^{i_r}_{k_r}(\delta_{j_1}\wedge\cdots\wedge\delta_{j_r})\otimes\left(\delta^{k_1}\wedge\cdots\wedge\delta^{k_r}\right)\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{r!}\delta_{k_1}^{j_1}\cdots\delta_{k_r}^{j_r}\left(\delta_{j_1}\wedge\cdots\wedge\delta_{j_r}\right)\otimes\left(\delta^{k_1}\wedge\cdots\wedge\delta^{k_r}\right)\\
&= \frac{1}{r!}\left(\delta_{j_1}\wedge\cdots\wedge\delta_{j_r}\right)\otimes\left(\delta^{j_1}\wedge\cdots\wedge\delta^{j_r}\right)\\
&= \sum_{1\le j_1<\cdots<j_r\le n}\left(\delta_{j_1}\wedge\cdots\wedge\delta_{j_r}\right)\otimes\left(\delta^{j_1}\wedge\cdots\wedge\delta^{j_r}\right).
\end{aligned}
$$

39. 这是第 43 题的特殊情况.

41. 由第 37 题, $\delta^{i_1}\wedge\cdots\delta^{i_r}(u_1,\cdots,u_r)=\det(u_l^{i_k})$.
设 $\alpha^k=\alpha_i^k\delta^i$, 则

$$
\begin{aligned}
\alpha^1\wedge\cdots\alpha^r(u_1,\cdots,u_r) &= \sum_{i_1,\cdots,i_r}\alpha_{i_1}^1\cdots\alpha_{i_r}^r\delta^{i_1}\wedge\delta^{i_r}(u_1,\cdots,u_r)\\
&= \sum_{i_1,\cdots,i_r}\alpha_{i_1}^1\cdots\alpha_{i_r}^r\det(u_l^{i_k})\\
&= \sum_{i_1,\cdots,i_r}\det(\alpha_{i_k}^k u_l^{i_k})\\
&= \det((\sum_{i=1}^n\alpha_i^k\delta^i)u_l)\\
&= \det(\alpha^k(u_l)).
\end{aligned}
$$

42.

$$
\begin{aligned}
x_1\wedge\cdots\wedge x_r &= \sum_{i_1,\cdots,i_r}a_1^{i_1}\cdots a_r^{i_r}e_{i_1}\wedge\cdots\wedge e_{i_r}\\
&= \sum_{i_1<\cdots<i_r}\sum_{\sigma}a_1^{i_{\sigma(1)}}\cdots a_r^{i_{\sigma(r)}}e_{i_{\sigma(1)}}\wedge\cdots\wedge e_{i_{\sigma(r)}}\\
&= \sum_{i_1<\cdots<i_r}\sum_{\sigma}\mathrm{sign}(\sigma)a_1^{i_{\sigma(1)}}\cdots a_r^{i_{\sigma(r)}}e_{i_1}\wedge\cdots\wedge e_{i_r}\\
&= \sum_{i_1<\cdots<i_r}\det(a_l^{i_k})e_{i_1}\wedge\cdots\wedge e_{i_r}.
\end{aligned}
$$

43.

$$
\begin{aligned}
\varphi\wedge\psi(v_1,\cdots,v_{r+s}) &= \frac{1}{r!s!}\sum_{\sigma}\mathrm{sign}(\sigma)\varphi\otimes\psi(v_{\sigma(1)},\cdots,v_{\sigma(r+s)})\\
&= \frac{1}{r!s!}\sum_{i_1,\cdots,i_r}\varphi(v_{i_1},\cdots,v_{i_r})\sum_{\sigma(k)=i_k}\mathrm{sign}(\sigma)\psi(v_{\sigma(r+1)},\cdots,v_{\sigma(r+s)})\\
&= \sum_{\substack{i_1<\cdots<i_r\\ i_{r+1}<\cdots<i_{r+s}}}\mathrm{sign}(\sigma_{i_1,\cdots,i_r})\varphi(v_{i_1},\cdots,v_{i_r})\psi(v_{i_{r+1}},\cdots,v_{i_{r+s}}),
\end{aligned}
$$

其中 $\sigma_{i_1,\cdots,i_r}(1,2,\cdots,r+s)=(i_1,\cdots,i_r,i_{r+1},\cdots,i_{r+s})$, 故 $\mathrm{sign}(\sigma_{i_1,\cdots,i_r})=\delta_{1,\cdots,r+s}^{i_1,\cdots,i_{r+s}}$, 代入即证.

44. (1) 易证.
(2)

$$
i(u_1)(\varphi\wedge\psi)(u_2,\cdots,u_{r+s})=\sum_{\substack{i_1<\cdots<i_r\\ i_{r+1}<\cdots<i_{r+s}}}\delta_{1,\cdots,r+s}^{i_1,\cdots,i_{r+s}}\varphi(u_{i_1},\cdots,u_{i_r})\psi(u_{i_{r+1}},\cdots,u_{i_{r+s}})
$$

$$
\begin{aligned}
&= \sum_{\substack{i_1<\cdots<i_r\\ i_{r+1}<\cdots<i_{r+s}}} \Big[\sum_{1\in\{i_1,\cdots,i_r\}} \delta^{1,i_2,\cdots,i_{r+s}}_{1,2,\cdots,r+s}\varphi(u_1,u_{i_2},\cdots,u_{i_r})\psi(u_{i_{r+1}},\cdots,u_{i_{r+s}}) \\
&\quad + \sum_{1\notin\{i_1,\cdots,i_r\}} \delta^{i_1,\cdots,i_r,1,i_{r+2},\cdots,i_{r+s}}_{1,\cdots,r,r+1,r+2,\cdots,r+s}\varphi(u_{i_1},\cdots,u_{i_r})\psi(u_1,u_{i_{r+2}},\cdots,u_{i_{r+s}})\Big] \\
&= \sum_{\substack{i_2<\cdots<i_r\\ i_{r+1}<\cdots<i_{r+s}}} \delta^{i_2,\cdots,i_{r+s}}_{2,\cdots,r+s}(i(u_1)\varphi)(u_{i_2},\cdots,u_{i_r})\psi(u_{i_{r+1}},\cdots,u_{i_{r+s}}) \\
&\quad + (-1)^r \sum_{\substack{i_1<\cdots<i_r\\ i_{r+2}<\cdots<i_{r+s}}} \delta^{i_1,\cdots,i_r,i_{r+2},\cdots,i_{r+s}}_{2,\cdots,r+1,r+2,\cdots,r+s}\varphi(u_{i_1},\cdots,u_{i_r})(i(u_1)\psi)(u_{i_{r+2}},\cdots,u_{i_{r+s}}) \\
&= ((i(u_1)\varphi)\wedge\psi + (-1)^r\varphi\wedge(i(u_1)\psi))(u_2,\cdots,u_{r+s}).
\end{aligned}
$$

45. (1) 易证.

(2)

$$
\begin{aligned}
e(\alpha)(\varphi\wedge\psi) &= \alpha\wedge(\varphi\wedge\psi) \\
&= (\alpha\wedge\varphi)\wedge\psi \\
&= (e(\alpha)\varphi)\wedge\psi,
\end{aligned}
$$

$$
\begin{aligned}
(\alpha\wedge\varphi\wedge\psi)(u_1,\cdots,u_{r+s+1})s &= (\varphi\wedge\alpha\wedge\psi)(u_2,\cdots,u_{r+1},u_1,u_{r+2},\cdots,u_{r+s+1}) \\
&= (-1)^r(\varphi\wedge\alpha\wedge\psi)(u_1,\cdots,u_{r+s+1}),
\end{aligned}
$$

故 $e(\alpha)(\varphi\wedge\psi) = (-1)^r\varphi\wedge(\alpha\wedge\psi) = (-1)^r\varphi\wedge(e(\alpha)(\psi))$.

46. 由第 44、45 题，

$$
\begin{aligned}
i(u)\circ e(\alpha)\varphi + e(\alpha)\circ i(u)\varphi &= (i(u)\alpha)\wedge\varphi - \alpha\wedge(i(u)\varphi) + \alpha\wedge(i(u)\varphi) \\
&= \alpha(u)\varphi.
\end{aligned}
$$

即证.

47. (1)(2) 我们直接证明 (2) 的 $n\geq 1$ 的情形.

$n=1$ 显然成立. 对 n 归纳：假设 n 已成立，对于 $\dim V = n+1$ 中的 n 次外形式

$$
\Omega = \sum_{i=1}^{n+1} a_i\delta^1\wedge\cdots\wedge\hat{\delta^i}\wedge\cdots\wedge\delta^{n+1},
$$

a_i 不全为 0，$\{\delta_i\}$ 为 V 的一组基.

考虑 $\psi = \sum b_i\delta^i$, 则 $\psi\wedge\Omega = (\sum a_ib_i(-1)^{i+1})\delta^1\wedge\cdots\wedge\delta^{n+1}$,

易见存在 $\{b_i\}$ 使得 $\sum a_ib_i(-1)^{i+1} = 0$.

即存在 ψ 使得 $\psi\wedge\Omega = 0$.

由定理 5.8（P43），存在 φ 使得 $\Omega = \psi\wedge\varphi$.

注意到存在 $W, \dim W = n$ 使得 $V^* = \mathbf{R}\psi\oplus W^*$,

考虑 φ 在 W^* 的投影 $\tilde{\varphi}$, 可见 $\Omega = \psi\wedge\tilde{\varphi}$.

由归纳假设，$\tilde{\varphi}$ 可分解，于是 Ω 可分解.

(3) 当 $\dim V=4$ 时 $\delta^1\wedge\delta^2+\delta^3\wedge\delta^4$ 不可分解,
否则存在 $\sum\limits_{i=1}^{4}a_i\delta^i\neq 0$ 使得 $\Big(\sum\limits_{i=1}^{4}a_i\delta^i\Big)\wedge(\delta^1\wedge\delta^2+\delta^3\wedge\delta^4)=0$,
展开得 $a_1=a_2=a_3=a_4=0$，矛盾.

48. 设 V^* 的基底为 $\{\delta^i\}$，2 次外形式 α 可表为

$$\alpha=\sum_{i<j}\alpha_{ij}\delta^i\wedge\delta^j.$$

假定 α_{ij} 不全为 0 , 不妨设 $\alpha_{12}\neq 0$, 于是

$$\begin{aligned}\alpha&=(\alpha_{12}\delta^1+\alpha_{32}\delta^3+\cdots+\alpha_{n2}\delta^n)\wedge(\delta^2+\frac{\alpha_{13}}{\alpha_{12}}\delta^3+\cdots+\frac{\alpha_{1n}}{\alpha_{12}}\delta^n)+\sum_{3\le i<j\le n}\alpha'_{ij}\delta^i\wedge\delta^j\\&=\sigma^1\wedge\sigma^2+\sum_{3\le i<j\le n}\alpha'_{ij}\delta^i\wedge\delta^j,\end{aligned}$$

其中 $\sigma^1=\alpha_{12}\delta^1+\alpha_{32}\delta^3+\cdots+\alpha_{n2}\delta^n,\sigma^2=\delta^2+\dfrac{\alpha_{13}}{\alpha_{12}}\delta^3+\cdots+\dfrac{\alpha_{1n}}{\alpha_{12}}\delta^n$.
若 α'_{ij} 不全为零，不妨设 $\alpha'_{34}\neq 0$，继续上面的过程。因 n 是有限数, 故上述步骤必在有限步之后终止, 即 α 可表成

$$\alpha=\sigma^1\wedge\sigma^2+\cdots+\sigma^{2r-1}\wedge\sigma^{2r},$$

易见 $\sigma^1,\cdots,\sigma^{2r}$ 是线性无关的 1 次形式.
由 $\sigma^{2i-1}\wedge\sigma^{2i}\wedge\sigma^{2j-1}\wedge\sigma^{2j}=\sigma^{2j-1}\wedge\sigma^{2j}\wedge\sigma^{2i-1}\wedge\sigma^{2i}$ 可得 $\alpha^r=r!\sigma^1\wedge\sigma^2\wedge\cdots\wedge\sigma^{2r-1}\wedge\sigma^{2r}\neq 0$，
由 α^{r+1} 为高于 $2r$ 次的外形式得 $\alpha^{r+1}=0$，所以 $2r$ 只与 α 有关.
Remark: 可以直接通过反对称二次型的合同标准型说明.

49. (1) 对称性和双线性是显然的.
由于 $(\langle v_\alpha,v_\beta\rangle)_{\alpha,\beta}$ 为内积的 Gram 矩阵，于是正定性成立.
(2)$\langle\delta_{i_1}\wedge\cdots\wedge\delta_{i_p},\delta_{j_1}\wedge\cdots\wedge\delta_{j_p}\rangle=\det(\delta^{j_\beta}_{i_\alpha})=\delta^{i_1\cdots i_p}_{j_1\cdots j_p}$.

50. (1) 注意到映射 $\wedge^pA:\prod\limits_{i=1}^{p}V\to W$ 为交错多线性映射，
由 $\wedge^pV$ 的泛性质，$\wedge^pA:\prod\limits_{i=1}^{p}V\to W$ 诱导了 $\wedge^pA:\wedge^pV\to W$.

(2)$\wedge^pA(\delta_{i_1}\wedge\cdots\wedge\delta_{i_p})=\delta^{i_1\cdots i_p}_{\alpha_1\cdots\alpha_p}a^{\alpha_1}_{i_1}\cdots a^{\alpha_p}_{i_p}$.

(3)$\wedge^p(AB)(z_1,\cdots,z_p)=(ABz_1)\wedge\cdots\wedge(ABz_p)=\wedge^pA(Bz_1,\cdots,Bz_p)=\wedge^pA\circ\wedge^pB(z_1,\cdots,z_p)$.

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2. $\forall p \in M$，$\forall h \in C_p^\infty$，我们有：

$$\begin{aligned}
(1)\& \quad & [aX+bY,Z](h) \\
\& &= (aX+bY)\circ Z(h) - Z\circ(aX+bY)(h) \\
\& &= aX\circ Z(h) - aZ\circ X(h) + bY\circ Z(h) - bZ\circ Y(h) \\
\& &= a[X,Z](h) + b[Y,Z](h) \\
\& &= (a[X,Z]+b[Y,Z])(h); \\
(2)\& \quad & [X,Y](h) \\
\& &= X\circ Y(h) - Y\circ X(h) \\
\& &= -(Y\circ X(h) - X\circ Y(h)) \\
\& &= -[Y,X](h); \\
(3)\& \quad & ([X,[Y,Z]]+[Y,[Z,X]]+[Z,[X,Y]])(h) \\
\& &= X\circ Y\circ Z(h) - X\circ Y\circ(h) - Y\circ Z\circ X(h) + Z\circ Y\circ X(h) \\
\& \quad & + Y\circ Z\circ X(h) - Y\circ X\circ Z(h) - Z\circ X\circ Y(h) + X\circ Z\circ Y(h) \\
\& \quad & + Z\circ X\circ Y(h) - Z\circ Y\circ X(h) - X\circ Y\circ Z(h) + Y\circ X\circ Z(h) \\
\& &= 0.
\end{aligned}$$

4. 我们只需要在局部上证明。

   必要性：设 M 为 m 维光滑流形，在 p 处取局部坐标系 (U,φ)，则切向量场 X 可以局部表示为

$$X|_U = X^i \frac{\partial}{\partial x^i}, \quad X^i \in C^\infty(U) \qquad (\forall i = 1,\cdots,m).$$

   从而 $X(f)|_U = X^i \frac{\partial f}{\partial x^i} \in C^\infty(U)$, $\forall f \in C^\infty(U)$.

   充分性：设 M 为 m 维光滑流形，在 p 处取局部坐标系 (U,φ), 则切向量场 X 可以局部的表示为

$$X|_U = X^i \frac{\partial}{\partial x^i},$$

   则只要证

$$X^i \in C^\infty(U), \quad \forall i = 1,\cdots,m.$$

   而 $\forall x \in U$，

$$(X(f))(x) = X^i(x) \cdot \frac{\partial f}{\partial x^i}(x),$$

   其中

$$\frac{\partial f}{\partial x^i}(x) = \left.\frac{\partial (f\circ\varphi^{-1})}{\partial x^i}\right|_{\phi(x)},$$

   则只需要依次取 $f = x^i$ $(i = 1,\cdots,m)$ 即可得到命题成立。

5. 复习：给定微分流形 M 以及光滑函数 $f \in C^\infty(M)$，切向量场 $X, \tilde{X} \in \mathscr{X}(M)$ 是 f- 相关的当且仅当 $f_* X_p = \tilde{X}_{f(p)}$, $\forall p \in M$.

(1) 由于 $\forall h,g\in C^\infty(M)$，$\alpha\in\mathbf{R}$，

$$\begin{aligned}
\text{(a)}\qquad &f_*X(h+g)\\
&=X((h+g)\circ f)\circ f^{-1}\\
&=X(h\circ f)\circ f^{-1}+X(g\circ f)\circ f^{-1}\\
&=f_*X(h)+f_*X(g),\\
\text{(b)}\qquad &f_*X(\alpha h)\\
&=X(\alpha h\circ f)\circ f^{-1}\\
&=\alpha X(h\circ f)\circ f^{-1}\\
&=\alpha f_*X(h),\\
\text{(c)}\qquad &f_*X(h\cdot g)\\
&=X((h\circ f)\cdot(g\circ f))\circ f^{-1}\\
&=(g\circ f)X((h\circ f)\circ f^{-1}+(h\circ f)X((g\circ f)\circ f^{-1}\\
&=h\cdot f_*X(g)+g\cdot f_*X(h),\\
\text{(d)}\qquad &f_*X(g)=X(g\circ f)\circ f^{-1}\in C^\infty(M).
\end{aligned}$$

则由定理 2.3（P129），f_*X 是 N 上的光滑切向量场.

又 $\forall p\in N$，$\forall g\in C^\infty(N)$，

$$\begin{aligned}
&(f_*X)_{f(p)}(g)\\
&=X(g\circ f)\circ f^{-1}(f(p))\\
&=X(g\circ f)(p),\\
&=f_*X_p(g),
\end{aligned}$$

可见 f_*X 与 X 是 f- 相关的。

(2) 结合（1）与定理 2.6（P133）结论告诉我们，f_* 保持 Poisson 括号积，我们只需要证明其是线性的双射。线性在 (1)(b) 中已经证明，下证其为双射：

由 (1)(c) 知 $f_*(0)=0$ 故 f_* 是单射，又对 N 上任意的光滑切向量场 Y 有

$$\begin{aligned}
&(f_*((f^{-1})_*Y))(g)\\
&=((f^{-1})_*Y)(g\circ f)\circ f^{-1}\\
&=Y(g\circ f\circ f^{-1})\circ f^{-1}\circ f\\
&=Y(g),
\end{aligned}$$

因此 f_* 是满射。

7. 假设存在这样的光滑切向量场 $Y=y\dfrac{d}{dt}$，其中 $y\in C^\infty(\mathbf{R})$，则 $\forall t_0\in\mathbf{R}$，$f\in C^\infty(\mathbf{R})$，

$$\begin{aligned}
&\Phi_*(\frac{d}{dt}\Big|_{t=t_0})(f)=Y\Big|_{t=t_0^2}(f)\\
&\Rightarrow \frac{df(t^2)}{dt}\Big|_{t=t_0}=y\frac{df(t)}{dt}\Big|_{t=t_0^2}\\
&\Rightarrow y(t_0^2)=2t_0,
\end{aligned}$$

从而 $y(t)=2\sqrt{t},\forall t\geq 0$, 那么 y 在 $t=0$ 处不光滑，假设不成立。

10. 未必存在，下一题就是一个反例。

错误示例：如果 φ 是满射的话就存在这样的光滑切向量场 Y.

$\forall p\in N$，令

$$Y_p=\varphi_* X_{\varphi^{-1}(p)}.$$

因为 $\varphi(p)=\varphi(q)$ 时

$$\varphi_{*p}X(p)=\varphi_{*q}X(q),$$

这样的定义是合理的，利用定理 2.3 容易验证这样定义出的 Y 是光滑切向量场，则根据定义易见 Y 和 X 是 φ- 相关的。

(原作者注：确实，利用定理 2.3 是无法验证向量场的光滑性的，草率了，感谢同学指正)

11. 为使记号方便，我们用 f 来表示题目中的 φ，再沿用第二章 §1 例 2 中的记号（P58），则

$$\psi\circ f\circ\varphi^{-1}:\mathbf{R}\to\mathbf{R},$$

$$\psi\circ f\circ\varphi^{-1}(x)=x^3,$$

于是 f 是 C^∞- 映射。

类似于第 10 题，假设存在这样的光滑切向量场 $Y=y\dfrac{d}{dt}$，其中 $y\in C^\infty(\mathbf{R})$，则 $\forall t_0\in\mathbf{R}$，$g\in C^\infty(\mathbf{R})$，

$$\begin{aligned}
&f_*(\frac{d}{dt}\Big|_{t=t_0})(g)=Y\Big|_{t=t_0}(g)\\
\&\Rightarrow\ &\frac{dg(t^3)}{dt}\Big|_{t=t_0}=y\frac{dg}{dt}\Big|_{t=t_0^2}\\
\&\Rightarrow\ &y(t_0^3)=3t_0^2,
\end{aligned}$$

于是 y 在 $t_0=0$ 处不光滑，假设不成立。

12. 计算得

$$\begin{aligned}
[X,Y]&=x\frac{\partial}{\partial z}-z\frac{\partial}{\partial x},\\
[Y,Z]&=-\frac{\partial}{\partial y}+\frac{\partial}{\partial z},\\
[X,Z]&=-\frac{\partial}{\partial y}+\frac{\partial}{\partial x}.
\end{aligned}$$

13. 先计算向量场经过点 $p=(x_0,y_0)$ 的积分曲线 $\gamma_p(t)$。设 $\gamma_p(t)=(x(t),y(t))$，则

$$\frac{d}{dt}\gamma_p(t)=(y(t),-x(t)),$$

得到方程组

$$\begin{cases}\dfrac{dx}{dt}=y,\\ \dfrac{dy}{dt}=-x,\end{cases}$$

代入初始点 $\gamma_p(0)=(x_0,y_0)$ 得到积分曲线

$$\gamma_p(t)=\sqrt{x_0^2+y_0^2}(\cos(t_0-t),\sin(t_0-t)),$$

其中 t_0 满足

$$\sqrt{x_0^2+y_0^2}(\cos(t_0),\sin(t_0))=p=(x_0,y_0).$$

因此 X 所生成的单参数变换群为

$$\varphi(t,p)=\sqrt{x_0^2+y_0^2}(\cos(t_0-t),\sin(t_0-t)).$$

14. 设向量场 Y 经过点 p 的积分曲线为 $\gamma(t)$ 满足 $\gamma(t_0)=p$。因为

$$f_*(\frac{d}{dt}\gamma(t)\Big|_{t=t_0})=X\big|_{f(p)}=\frac{d}{dt}f\circ\gamma(t)\Big|_{t=t_0},$$

所以 $f\circ\gamma$ 是 X 的积分曲线。

15. 验证其是单参数变换群是容易的。

$$\frac{\partial\varphi}{\partial t}=(\lambda xe^{\lambda t},\mu ye^{\mu t}),$$

由此可知单参数变换群诱导的切向量场为 $\lambda x\dfrac{\partial}{\partial x}+\mu y\dfrac{\partial}{\partial y}$。

16. (i)

$$\theta(0,x,y)=(x,y),$$

$$\begin{aligned}\theta(s,\theta(t,x,y))&=\theta(s,x\cos\lambda t+y\sin\lambda t,-x\sin\lambda t+y\cos\lambda t)\\&=(x\cos\lambda(s+t)+y\sin\lambda(s+t),-x\sin\lambda(s+t)+y\cos\lambda(s+t))\\&=\theta(s+t,x,y),\end{aligned}$$

故 θ 是单参数变换群。

(ii) 计算可知，

$$\frac{\partial}{\partial t}\theta(t,x,y)|_{t=0}=(\lambda y,-\lambda x),$$

故诱导切向量场为 $X_{(x,y)}=\lambda y\frac{\partial}{\partial x}-\lambda x\frac{\partial}{\partial y}$。

17. (i)

$$\theta(0,A)=I\cdot A=A,$$

$$\begin{aligned}\theta(s,\theta(t,A))&=\begin{bmatrix}1&s\\0&1\end{bmatrix}\cdot\theta(t,A)\\&=\begin{bmatrix}1&s\\0&1\end{bmatrix}\cdot\begin{bmatrix}1&t\\0&1\end{bmatrix}\cdot A\\&=\begin{bmatrix}1&s+t\\0&1\end{bmatrix}\cdot A\\&=\theta(s+t,A),\end{aligned}$$

故 θ 是单参数变换群。

(ii) 计算可知，

$$\frac{\partial}{\partial t}\theta(t,A)|_{t=0}=\begin{bmatrix}1&1\\0&1\end{bmatrix}\cdot A,$$

故诱导切向量场为 $X=(x_{11}+x_{21})\frac{\partial}{\partial x_{11}}+(x_{12}+x_{22})\frac{\partial}{\partial x_{12}}+x_{21}\frac{\partial}{\partial x_{21}}+x_{22}\frac{\partial}{\partial x_{22}}$。

18. 我们有微分方程

$$\begin{cases} \frac{\partial}{\partial t}\varphi(t,x)|_{t=0} = \varphi(t,x)^2, \\ \varphi(0,x) = x, \end{cases}$$

解得 $\varphi(t,x) = \frac{x}{1-xt}$。通过初始点选取恰当的定义域即可。

19. 本问题仅为计算题，以下直接给出答案：

(i) $X = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}$，$\varphi(t,x,y) = (xe^t, ye^t)$.

(ii) $X = -x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y}$，$\varphi(t,x,y) = (xe^{-t}, ye^{-t})$.

(iii) $X = -y\frac{\partial}{\partial x} + x\frac{\partial}{\partial y}$，$\varphi(t,x,y) = (x\cos t - y\sin t, x\sin t + y\cos t)$.

(iv) $X = (x^2-y^2)\frac{\partial}{\partial x} + 2xy\frac{\partial}{\partial y}$，$\varphi(t,x,y) = (\frac{x-tx^2-ty^2}{(1-tx)^2+(ty)^2}, \frac{y}{(1-tx)^2+(ty)^2})$.

(v) $X = -x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}$，$\varphi(t,x,y) = (xe^{-t}, ye^t)$.

(vi) $X = (y^2-x^2)\frac{\partial}{\partial x} + 2xy\frac{\partial}{\partial y}$.

20. 充分性：对等式两边关于 t 求偏导，得到

$$\begin{aligned} F_*(X_p) &= F_* \circ \frac{\partial \varphi}{\partial t}(0,p) \\ &= \frac{\partial}{\partial t} F \circ \varphi(t,x)|_{t=0,x=p} \\ &= \frac{\partial}{\partial t}\psi(t,F(x))|_{t=0,x=p} \\ &= Y_{F(p)}, \end{aligned}$$

故 Y 和 X 是 F- 相关的。

必要性：由第 14 题可直接推出。Y 和 X 是 F- 相关的，且 $\varphi(t,p)$ 在固定 p 时为 X 的积分曲线，故由第 14 题可知 $F\circ\varphi(t,p)$ 为 Y 的积分曲线。注意到 $\psi(t,F(p))$ 也为 Y 的积分曲线，由积分曲线的唯一性可知 $F\circ\varphi_t = \psi_t \circ F$。

21. 由定理 3.5（P145），$\mathscr{L}_{\widetilde{X}}\widetilde{Y} = [\widetilde{X},\widetilde{Y}]$，$\mathscr{L}_X Y = [X,Y]$。

由定理 2.6（P133），$[\widetilde{X},\widetilde{Y}]$ 和 $[X,Y]$ 是 f- 相关的。

故 $\mathscr{L}_{\widetilde{X}}\widetilde{Y}$ 和 $\mathscr{L}_X Y$ 是 f- 相关的。

22. $\forall Z \in \mathscr{X}(M)$，有

$$\begin{aligned} \mathscr{L}_{[X,Y]}Z &= [[X,Y],Z] \\ &= [X,Y]\circ Z - Z\circ[X,Y] \\ &= X\circ Y\circ Z - Y\circ X\circ Z - Z\circ X\circ Y + Z\circ Y\circ X. \end{aligned}$$

另一方面，

$$\begin{aligned} (\mathscr{L}_X\circ\mathscr{L}_Y - \mathscr{L}_Y\circ\mathscr{L}_X)Z &= \mathscr{L}_X[Y,Z] - \mathscr{L}_Y[X,Z] \\ &= [X,[Y,Z]] - [Y,[X,Z]] \\ &= X\circ(Y\circ Z - Z\circ Y) - (Y\circ Z - Z\circ Y)\circ X \\ &\quad - Y\circ(X\circ Z - Z\circ X) + (X\circ Z - Z\circ X)\circ Y \\ &= X\circ Y\circ Z - Y\circ X\circ Z - Z\circ X\circ Y + Z\circ Y\circ X, \end{aligned}$$

故 $\mathscr{L}_{[X,Y]} = \mathscr{L}_X\circ\mathscr{L}_Y - \mathscr{L}_Y\circ\mathscr{L}_X$。

23.

$$\begin{aligned}\mathscr{L}_X Y &= [X,Y] = [X, Y^j e_j] \\ &= X(Y^j)e_j + Y^j[X,e_j] \\ &= X(Y^j)e_j + Y^j[X^i e_i, e_j] \\ &= X(Y^j)e_j + X^i Y^j [e_i,e_j] - Y^j e_j(X^i)e_i \\ &= X(Y^j)e_j + X^i Y^j c_{ij}^k e_k - Y^j e_j(X^i)e_i \\ &= (X(Y^i) + (X^k c_{kj}^i - e_j(X^i))Y^j)e_i \\ &= (X(Y^i) - a_j^i Y^j)e_i.\end{aligned}$$

24. (i)

$$\mathscr{L}_{X+Y}Z = [X+Y,Z] = [X,Z] + [Y,Z] = (\mathscr{L}_X + \mathscr{L}_Y)Z,$$

故有 $\mathscr{L}_{X+Y} = \mathscr{L}_X + \mathscr{L}_Y$。

(ii)

$$\mathscr{L}_{f\cdot X}Y = [f\cdot X, Y] = f[X,Y] - Y(f)X = f\mathscr{L}_X Y - Y(f)X.$$

25. 首先, 通过计算得到

$$\begin{aligned}\mathscr{L}_X Y &= [X,Y] = X\circ Y - Y\circ X \\ &= x^2\frac{\partial^2}{\partial x^2} - \Big(2x\frac{\partial}{\partial x} + x^2\frac{\partial^2}{\partial x^2}\Big) \\ &= -2x\frac{\partial}{\partial x}.\end{aligned}$$

代入 $u = x$, $v = x + y$, 得到

$$\begin{aligned}\mathscr{L}_X Y &= -2x\frac{\partial}{\partial x} \\ &= -2u\Big(\frac{\partial u}{\partial x}\frac{\partial}{\partial u} + \frac{\partial v}{\partial x}\frac{\partial}{\partial v}\Big) \\ &= -2u\Big(\frac{\partial}{\partial u} + \frac{\partial}{\partial v}\Big).\end{aligned}$$

26. 在 **R** 上, 取 $p = 0$, 以及

$$X = 0,\ \tilde{X} = x\frac{\partial}{\partial x},\ Y = \frac{\partial}{\partial x},$$

则在 p 点成立 $X(p) = 0 = \tilde{X}(p)$. 但经过计算, 我们分别得到

$$\mathscr{L}_X Y(p) = 0, \quad \mathscr{L}_{\tilde{X}} Y(p) = \frac{\partial}{\partial x},$$

两者并不相等.

27. 根据定理 2.7（P134）, f_* 保 Poisson 括号积, 故成立

$$f_*\mathscr{L}_X Y = f_*[X,Y] = [f_*X, f_*Y] = \mathscr{L}_{f_*X} f_* Y.$$

28. 由于 L^h 是 U 上的 h 维光滑分布, 存在 p 的某个邻域 $V \subset U$, 以及定义在 V 上的 h 个处处线性无关的向量场, 记为 $X_1, \cdots, X_h$, 满足 $L^h|_V = \text{span}\{X_1, \cdots, X_h\}$. 那么 v 可以表示为

$v = c_1X_1(p) + \cdots + c_hX_h(p)$, 其中 $c_i \in \mathbf{R}$. 那么 $X := c_1X_1 + \cdots + c_hX_h$ 为属于光滑分布 $L^h|_V$ 的光滑切向量场. 由定理 4.1 （P147）可知, 存在 p 的一个局部坐标系 $(W; y^i)$, $W \subset V$, 使得 $X|_W = \frac{\partial}{\partial y^1}$. 那么 X 有积分曲线 $y = (y^1, 0, \cdots, 0)$, 满足 $y(0) = p$, $y'(0) = v$. 因为 $X \in L^h|_V$, 这也是 L^h 的一条积分曲线.

30. 假设 $\{\varphi_t\}$ 与 $\{\psi_s\}$ 在 M 上的作用可交换. 对任意的 t, 因为 $\varphi_t \colon M \to M$ 为局部光滑同胚, 所以根据推论 3.4（P144）, 有 $(\varphi_t)_*Y = Y$. 进而在局部上, 根据定理 3.5（P145）, 成立

$$[X, Y] = \lim_{t\to 0} \frac{Y - (\varphi_t)_*Y}{t} = \lim_{t\to 0} 0 = 0.$$

于是在 M 上, $[X, Y] = 0$ 恒成立.

假设 $[X, Y] \equiv 0$. 根据推论 3.4, 只要证明 $(\varphi_t)_*Y = Y$ 对所有 t 成立. 由于 $t = 0$ 时命题成立, 我们希望证明 $(\varphi_t)_*Y$ 关于 t 不变. 对任意的 $g \in C^\infty(M)$ 和 $t_0 \in \mathbf{R}$, 根据定理 3.5, 我们有

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\,t}\bigg|_{t=t_0} [(\varphi_t)_*Y(g)] &= \frac{\mathrm{d}}{\mathrm{d}\,t}\bigg|_{t=t_0} [Y(g \circ \varphi_t)] \\
&= \lim_{s\to 0} \frac{Y(g \circ \varphi_{t_0+s}) - Y(g \circ \varphi_{t_0})}{s} \\
&= \lim_{s\to 0} \frac{(\varphi_s)_*Y(g \circ \varphi_{t_0}) - Y(g \circ \varphi_{t_0})}{s} \\
&= \lim_{s\to 0} \frac{(\varphi_s)_*Y - Y}{s}(g \circ \varphi_{t_0}) \\
&= [X, Y](g \circ \varphi_{t_0}) = 0.
\end{aligned}$$

所以对任意的 t, 成立 $(\varphi_t)_*Y(g) = (\varphi_0)_*Y(g) = Y(g)$, 也即 $(\varphi_t)_*Y = Y$.

31. 每个 X_i 在 p 的某个邻域 V_i 上生成了一个局部单参数变换群, 记为 $\{\varphi_t^{(i)}\}$. 令 $V = \cap V_i$, 则 V 仍是 p 的邻域. 任取 V (或更小的邻域) 上的一个坐标系 $(V; x^i)$, 其中 $1 \le i \le n := \dim M$. 在 V 上, 由线性无关性, $\mathrm{span}\{X_1, \cdots, X_h\}$ 是一个 h 维的线性空间. 所以可以假设

$$X_1, \cdots, X_h, \frac{\partial}{\partial x^{h+1}}, \cdots, \frac{\partial}{\partial x^n}$$

线性无关, 进而构成了 $\mathscr{X}(V)$ 上的一组基. 由上一题的结论, 因为 $[X_i, X_j] = 0$, $\varphi_t^{(i)}$ 与 $\varphi_s^{(j)}$ 可交换. 定义

$$\varphi(y^1, \cdots, y^n) := \varphi_{y^1}^{(1)} \circ \cdots \circ \varphi_{y^h}^{(h)}(0, \cdots, 0, y^{h+1}, \cdots y^n),$$

则 φ^{-1} 给出了一个 V 上的坐标系, 且满足 $X_i = \frac{\partial}{\partial y^i}$, $\forall 1 \le i \le h$.

32. (1) 在每个分量 $\frac{\partial}{\partial x^{i_\alpha}}$ 或 $\mathrm{d}\,x^{j_\beta}$ 上, 分别有 $(\varphi_{-(t+s)})_* = (\varphi_{-t})_* \circ (\varphi_{-s})_*$, 以及 $(\varphi_{-(t+s)})^* = (\varphi_{-t})^* \circ (\varphi_{-s})^*$. 根据 (5.14) 的定义, 就有 $\Phi_{t+s} = \Phi_t \circ \Phi_s$. 同理, $\Phi_{t+s} = \Phi_s \circ \Phi_t$.

(2) 我们首先有

$$\frac{\mathrm{d}}{\mathrm{d}\,t}\bigg|_{t=s} \Phi_t\tau = \lim_{t\to 0} \frac{\Phi_{t+s}\tau - \Phi_s\tau}{t}.$$

利用上一小问的两个等式和定义 5.2（P163）, 我们分别得到

$$\frac{\mathrm{d}}{\mathrm{d}\,t}\bigg|_{t=s} \Phi_t\tau = \lim_{t\to 0} \frac{\Phi_t(\Phi_s\tau) - \Phi_s\tau}{t} = \mathscr{L}_X(\Phi_s\tau),$$

以及

$$\frac{\mathrm{d}}{\mathrm{d}\,t}\bigg|_{t=s} \Phi_t\tau = \lim_{t\to 0} \frac{\Phi_s(\Phi_t\tau) - \Phi_s\tau}{t}$$

$$= \Phi_s \left(\lim_{t\to 0} \frac{\Phi_t \tau - \tau}{t} \right)$$
$$= \Phi_s \mathscr{L}_X(\tau).$$

33. 我们用 t_0 表示确定的实数 t, 以示区别. 边界条件显然成立. 为了凑 $X^i(x)$ 这一项, 考虑 $f_{t_0} := g \circ \varphi_{t_0} \circ \varphi$, 有 $f(t+t_0, x) \equiv f_{t_0}(t, x)$. 所以

$$\frac{\partial f}{\partial t}(t_0, x) = \frac{\partial f_{t_0}}{\partial t}(0, x) = X_x(g \circ \varphi_{t_0})$$
$$= \sum_{i=1}^{n} X^i(x) \frac{\partial (g \circ \varphi_{t_0})}{\partial x^i}(x).$$

而 $g \circ \varphi_{t_0}(x) = f(t_0, x)$, 故

$$\frac{\partial f}{\partial t}(t_0, x) = \sum_{i=1}^{n} X^i(x) \frac{\partial f}{\partial x^i}(t_0, x).$$

34. 这里 $X = (X_1, X_2) = (y, -x)$，它生成的局部单参数变换群为

$$\varphi_t(r\sin\theta, r\cos\theta) = (r\sin(\theta + t), r\cos(\theta + t)).$$

由第 33 题，由 $g(x, y) = y \sin x$，

$$f(t, \mathbf{x}) = g(\varphi(t, \mathbf{x})) = r\cos(\theta + t)\sin(r\sin(\theta + t)).$$

35. 考虑另一个局部坐标系 (V, y^i), 设 $\frac{\partial}{\partial x^i} = a_i^j \frac{\partial}{\partial y^j}$, 则 $dy^j = a_i^j dx^i$.

$$\begin{aligned}
(\Phi_t(\tau))(\mathbf{x}) &= \tau^{i_1\cdots i_p}_{j_1\cdots j_q}(\gamma_{\mathbf{x}}(t))(\varphi_{-t})_*\left(\frac{\partial}{\partial x^{i_1}}|_{\gamma_{\mathbf{x}}(t)}\right) \otimes \cdots \otimes (\varphi_t)^*(dx^{j_q}|_{\gamma_{\mathbf{x}}(t)}) \\
&= \tau^{i_1\cdots i_p}_{j_1\cdots j_q}(\gamma_{\mathbf{x}}(t))(\varphi_{-t})_*\left(a_{i_1}^{s_1}\frac{\partial}{\partial y^{s_1}}|_{\gamma_{\mathbf{x}}(t)}\right) \otimes \cdots \otimes (\varphi_t)^*(a_{l_q}^{j_q} dy^{l_q}|_{\gamma_{\mathbf{x}}(t)}) \\
&= \tau^{i_1\cdots i_p}_{j_1\cdots j_q}(\gamma_{\mathbf{x}}(t)) a_{i_1}^{s_1}\cdots a_{i_p}^{s_p} \cdot a_{l_1}^{j_1}\cdots a_{l_q}^{j_q}(\varphi_{-t})_*\left(\frac{\partial}{\partial y^{s_1}}|_{\gamma_{\mathbf{x}}(t)}\right) \otimes \cdots \otimes (\varphi_t)^*(dy^{l_q}|_{\gamma_{\mathbf{x}}(t)}) \\
&= \tau^{s_1\cdots s_p}_{l_1\cdots l_q}(\gamma_{\mathbf{x}}(t))(\varphi_{-t})_*\left(\frac{\partial}{\partial y^{s_1}}|_{\gamma_{\mathbf{x}}(t)}\right) \otimes \cdots \otimes (\varphi_t)^*(dy^{l_q}|_{\gamma_{\mathbf{x}}(t)}).
\end{aligned}$$

36. 若函数 $f_1, f_2, \cdots, f_r$ 可以作为点 $p \in U$ 的某个邻域内的坐标函数 F 的组成部分，即存在函数 $g_{r+1}, \cdots, g_n$ 使得 $F = \left(f_1, \cdots, f_r, g_{r+1}, \cdots, g_n\right)^t$ 是坐标函数.
则有 $dF = \left(df_1, \cdots, df_r, dg_{r+1}, \cdots, dg_n\right)^t$ 满秩, 因此有 $df_1, \cdots, df_r$ 线性无关.
另一方面，若 $df_1, \cdots, df_r$ 线性无关, 我们用自然坐标扩充该线性无关组，得到坐标函数 $F = \left(f_1, \cdots, f_r, x^{i_{r+1}}, \cdots, x^{i_n}\right)^t$, 则函数 $f_1, f_2, \cdots, f_r$ 可以作为点 $p \in U$ 的邻域 U 内的这个坐标函数 F 的组成部分.

37. 若关于 a, b 的方程 $0 = a\omega^1 + b\omega^2 = (ax^1 + bx^2)dx^1 + (ax^2 + bx^1)dx^2$ 只有零解，即 $\begin{pmatrix} x^1 & x^2 \\ x^2 & x^1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = 0$ 只有零解 $\iff \begin{vmatrix} x^1 & x^2 \\ x^2 & x^1 \end{vmatrix} \neq 0$, 即 $(x^1)^2 \neq (x^2)^2$, 由此得到区域

$$U = \mathbf{R}^2 \setminus (\{y = x\} \cup \{y = -x\}).$$

$$\begin{cases} \omega^1 = x^1 dx^1 + x^2 dx^2 \\ \omega^2 = x^2 dx^1 + x^1 dx^2 \end{cases} \Rightarrow \begin{cases} \frac{\partial}{\partial x^1} = x^1\delta_1 + x^2\delta_2 \\ \frac{\partial}{\partial x^2} = x^2\delta_1 + x^1\delta_2 \end{cases} \Rightarrow \begin{cases} \delta_1 = \frac{1}{(x^1)^2 - (x^2)^2}\left(x^1\frac{\partial}{\partial x^1} - x^2\frac{\partial}{\partial x^2}\right) \\ \delta_2 = \frac{1}{(x^1)^2 - (x^2)^2}\left(-x^2\frac{\partial}{\partial x^1} + x^1\frac{\partial}{\partial x^2}\right) \end{cases}.$$

38. 设有 2 重 $C^\infty(M)$- 线性映射 Φ: $\mathscr{X}(M)\times\mathscr{X}(M)\to C^\infty(M)$, 我们要在 M 上定义一个光滑的 $(0,2)$ 型张量场 $\tilde{\Phi}$, 使得在每一点 $p\in M$, 有

$$\tilde{\Phi}(p)(X(p)\otimes Y(p))=\Phi(X,Y)(p),\ \ \forall X,\ Y\in(M).$$

(1) 首先注意到, 映射 Φ: $\mathscr{X}(M)\times\mathscr{X}(M)\to C^\infty(M)$ 有局部性.
若 $(X,Y),(X_1,Y_1)\in\mathscr{X}(M)\times\mathscr{X}(M)$, 且在开子集 $U\subset M$ 上有 $(X,Y)|_U=(X_1,Y_1)|_U$, 我们证明有 $\Phi(X,Y)|_U=\Phi(X_1,Y_1)|_U\in C^\infty(U)$.
事实上，$\forall p\in U$，存在 p 的邻域 V，使得 $\overline{V}$ 紧致, 且 $p\in V\subset\overline{V}\subset U$. 且存在函数 $f\in C^\infty(M)$ 使得 $f|_V\equiv 1, f|_{M\setminus U}\equiv 0$. 于是 $f\cdot(X-X_1)$ 是 M 上的零向量场.
而 $\Phi(0,Y)=\Phi(0,Y)+\Phi(0,Y)=2\Phi(0,Y)\Rightarrow\Phi(0,Y)=0$, 故有

$$0=\Phi(f\cdot(X-X_1),Y)=f\Phi(X-X_1,Y)=f(\Phi(X,Y)-\Phi(X_1,Y)).$$

特别地，在 p 点，有 $\Phi(X,Y)(p)=\Phi(X_1,Y)(p)$, 故有 $\Phi(X,Y)|_U=\Phi(X_1,Y)|_U$.
同理可证 $\Phi(X,Y)|_U=\Phi(X,Y_1)|_U$, 则由 2 重 $C^\infty(M)$- 线性知，$\Phi(X,Y)|_U=\Phi(X_1,Y_1)|_U$.

(2) 由 Φ 的局部性，对于 M 上的任意开子集 U, $\Phi|_U$ 是 2 重 $C^\infty(M)$ - 线性映射: $\mathscr{X}(U)\times\mathscr{X}(U)\to C^\infty(U)$.

(3) 设 (U,x^i) 是 p 点的任意一个局部坐标系，又设 $\mathbf{w},\mathbf{v}\in\mathscr{X}(U)$，且有局部表示 $\mathbf{w}=w^i\frac{\partial}{\partial x^i}$, $\mathbf{v}=v^j\frac{\partial}{\partial x^j}$.
特别地，我们有：

$$\Phi(\mathbf{w},\mathbf{v})(p)=w^iv^j\Phi(\frac{\partial}{\partial x^i},\frac{\partial}{\partial x^j})(p),$$

$\Phi(\mathbf{w},\mathbf{v})(p)$ 线性地依赖于切向量场 $\mathbf{w},\mathbf{v}$ 在 p 点处的分量，与 $\mathbf{w},\mathbf{v}$ 在 p 以外的值无关.
现在可以在每一个点 $p\in M$ 定义 $(0,2)$ 型张量场 $\tilde{\Phi}\in T_p^*M\otimes T_p^*M$ 如下: 任取 $\mathbf{w}_0$, $\mathbf{v}_0\in T_pM$, 作 $\mathbf{w}_0,\mathbf{v}_0$ 在 p 点邻域 U 内的任意光滑扩张 $\mathbf{w},\mathbf{v}\in\mathscr{X}(U)$, 使得 $(\mathbf{w},\mathbf{v})(p)=(\mathbf{w}_0,\mathbf{v}_0)(p)$.
令 $(\tilde{\Phi}(p))(\mathbf{w}_0,\mathbf{v}_0)=\Phi(\mathbf{w},\mathbf{v})(p)$, 由上式知此式有意义，即 $(\tilde{\Phi}(p))(\mathbf{w}_0,\mathbf{v}_0)$ 的值与光滑扩张 $\mathbf{w},\mathbf{v}\in\mathscr{X}(U)$ 的选取无关.
这样，我们便得到了光滑流形 M 上的 $(0,2)$ 型张量场 $\tilde{\Phi}$（光滑性是显然的）.

39. 定义 $\tilde{\varphi}(X,\alpha)=\alpha(\varphi(X))$, 容易验证这是一个 2 重线性映射，且对每个自变量都是 $C^\infty(M)$-线性的，由定理 5.1（P159）知命题成立.

40. $\forall f,g\in C^\infty(M),Y,Z\in\mathscr{X}(M)$，

$$\begin{aligned}\mathscr{L}_X\varphi(fY+gZ)&=[X,\varphi(fY+gZ)]-\varphi([X,fX+gZ])\\&=[X,f\varphi(Y)+g\varphi(Z)]-\varphi(f[X,Y]+g[X,Z])\\&=f[X,\varphi(Y)]+g[X,\varphi(Z)]-f\varphi([X,Y])-g\varphi([X,Z])\\&=f\mathscr{L}_X\varphi(Y)+g\mathscr{L}_X\varphi(Z).\end{aligned}$$

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2. $\varphi_1\circ\varphi_2^{-1}(y_1,\cdots,y_n)=\left(\frac{y_1}{\sum y_i^2},\cdots,\frac{y_n}{\sum y_i^2}\right)$ 为光滑函数，类似可证 $\varphi_2\circ\varphi_1^{-1},\varphi_\alpha^{\pm}\circ\varphi_2,\varphi_2\circ(\varphi_\alpha^{\pm})^{-1}$ 光滑，因此 $U_i,i=1,2$ 与 $\varphi_\alpha^{\pm}$ 光滑相容，因此给出了同一个光滑结构.
5. $f(x,y,z)=(\sqrt{x^2+y^2}-R)^2+z^2-R^2$ 为 $\mathbf{R}^3$ 到 $\mathbf{R}$ 的光滑映射，$\nabla f=\left(\frac{x}{\sqrt{x^2+y^2}},\frac{y}{\sqrt{x^2+y^2}},2z\right)\neq 0$ 说明它的秩处处为 1，由定理 5.4（P103），$f^{-1}(0)$ 是一个 2 维光滑流形。
7. 设 $p,f(p)$ 处的局部坐标卡为 $(U,\psi_1),(V,\psi_2)$，取 $v:h\mapsto\frac{\mathrm{d}}{\mathrm{d}t}(h\circ\gamma(t))$，其中 $\gamma\subset U$ 为一条道路.

   由 $f_*=0$ 可知 $\frac{\mathrm{d}}{\mathrm{d}t}(\psi_2^i\circ f\circ\gamma(t))=0$（其中 $\psi_2=(\psi_2^1,\psi_2^2)$），由 U 道路连通可知，f 在 U 上为常值，由 M 的连通性可知 f 在 M 上为常值.
9. f^* 在自然基底下的矩阵就是（见 P78-80）

$$J_f=\begin{pmatrix}\frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2}\\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2}\end{pmatrix}=\begin{pmatrix}e^{x_2} & x_1e^{x_2}+1\\ e^{x_2} & x_1e^{x_2}-1\end{pmatrix}.$$

   f_* 在自然基底下的矩阵是其转置矩阵 J_f^T.

   显然有 f 光滑，由 $\det J=2e^{x_2}\neq 0$ 得 f 是光滑同胚.

   Remark: 经典错误是认为 f^* 和 f_* 的表示矩阵都是 J_f, 这种错误出现的原因大概是看了陈维桓的课本中的相关表述而没有理解. 实际上由对偶性, 显然它们互为转置.
11. (1) $\forall v\in T_p(X)$，

$$(g\circ f)_*(v)=v(\cdot\circ g\circ f)=f_*(v(\cdot\circ g))=g_*\circ f_*(v).$$

    (2) $\forall\alpha\in T^*_{g\circ f(p)}(X)$，

$$(g\circ f)^*(\alpha)=\alpha((g\circ f)_*\circ\cdot)\overset{(1)}{=}\alpha(f_*\circ g_*\circ\cdot)=f^*(\alpha(g_*\circ\cdot))=f^*\circ g^*(\alpha).$$

12. 假设 f 的秩（即 f_* 的秩）处处为 n，则 $f(M)$ 是开集，又由 M 是紧集且 f 连续得 $f(M)$ 是紧集，矛盾。
13. σ 在每个平面 $\{x_3=a\}$ 上是一个旋转，因此 $\sigma|_{S^2}$ 给出 S^2 到它自身的双射。

    从 $\sigma|_{S^2}$ 与 $\sigma|_{S^2}^{-1}$ 的表达式容易看出它们都是光滑映射，故 $\sigma|_{S^2}$ 是光滑同胚。
14. 设 $p:\mathbb{R}^2\to T^2$ 为投影映射，则对于小邻域 U，$p|_U$ 为同胚，因此 $p|_U^{-1}\circ\varphi\circ\mathrm{id}_{\mathbb{R}}(t)=(\sqrt{2}t,t)+(a,b)$, 可见 φ 光滑.

    由 Dirichlet 逼近定理可知，$\sqrt{2}\mathbb{Z}$ 在 $[0,1)$ 中稠密，于是对于 $\overline{(x,y)}\in T^2$，$\forall\varepsilon>0$，存在 $k\in\mathbb{Z}$ 使得 $|\sqrt{2}(y+k)-x|\bmod 1<\varepsilon$, 因此 $d((x,y),\varphi(y+k))<\varepsilon$，于是 $\varphi(\mathbb{R})$ 稠密.
15. 由连续函数的开集的逆像为开集，由 M 移植的拓扑细于 N 诱导的拓扑。
16. 参考 §4 例 2,3（P83,84），取 M 为

$$G:\mathbf{R}\to\mathbf{R}^2,t\mapsto(2\cos(2\arctan t-\frac{\pi}{2}),\sin(2(2\arctan t-\frac{\pi}{2})))$$

    的像（附带微分结构），N 为

$$F:\mathbf{R}\to\mathbf{R}^2,t\mapsto(2\cos(t-\frac{\pi}{2}),\sin(2(t-\frac{\pi}{2})))$$

    的像（附带微分结构），曲线 γ 为 F 限制在 $(\pi-1,\pi+1)$ 上。

    $t=\pi$ 时 $\gamma(\pi)=(0,0)$，$\gamma'(0)=(-2,-2)$，而 $T_{\gamma(0)}M$ 为直线 $y=-x$，故 $\gamma'(0)\notin T_{\gamma(0)}M$。

17. 由于 φ 是同胚，不妨设 φ 为包含映射 i，$M=\varphi(M)\subseteq N$，否则取 $f\circ\varphi^{-1}$ 即可。
对任意一点 $p\in N$，由定理 2.3，存在 N 在 p 的邻域 U_p 以及 $g_p\in C^\infty(N)$ 使得 $g_p|_{U_p\cap M}=g|_{U_p\cap M}$。由于 $\{U_p\mid p\in M\}$ 再加上 $N\backslash M$ 是 N 的开覆盖，由单位分解定理，存在 N 上局部有限项非零的光滑函数 $\{f_i\}_{i=0}^\infty$ 使得 $0\le f_i\le 1$，$\sum\limits_{i=0}^{\infty} f_i=1$，且对 $i\ge 1$，$\operatorname{supp} f_i\subset U_{p_i}$，以及 $\operatorname{supp} f_0\subset N\backslash M$，故在 M 上 $\sum\limits_{i=1}^{\infty} f_i=1$。
令 $f=\sum\limits_{i=1}^{\infty} f_i g_{p_i}\in C^\infty(N)$，由 $\operatorname{supp} f_i\cap M\subset U_{p_i}\cap M\subset\{q\in M\mid g_{p_i}=g\}$ 得到实际上 $f_ig_{p_i}=f_ig$，所以在 M 上
$$f=\sum_{i=1}^{\infty} f_ig_{p_i}=\sum_{i=1}^{\infty} f_ig=g.$$
18. 由 f_1,f_2 光滑得 f 也光滑，由切映射 f_{1*},f_{2*} 非退化得 $f_*=(f_{1*},f_{2*})$ 非退化，即 f 也是浸入，由 f_1,f_2 是同胚得 f 也是同胚，所以 $(f,M_1\times M_2)$ 是嵌入子流形。
19. 沿用 §1 例 4（P59）的记号。π 显然是满射，且对任意 $1\le\alpha\le n+1$，$(\psi_\alpha\circ\pi)(x^1,\cdots,x^{n+1})=\left(\frac{x^1}{x^\alpha},\cdots,\frac{x^{\alpha-1}}{x^\alpha},\frac{x^{\alpha+1}}{x^\alpha},\cdots,\frac{x^{n+1}}{x^\alpha}\right)$ 是光滑的，其 Jacobi 矩阵有子方阵
$$\left(\frac{\partial(\psi_\alpha\circ\pi)^i}{\partial x^j}\right)_{\substack{1\le i\le n,\\ 1\le j\le n+1,j\ne\alpha}}=\frac{1}{x_\alpha}I_n,$$
故 $\operatorname{rk}(\psi_\alpha\circ\pi)=n$，所以 π 是淹没。
20. 设 U 是 M 中的开集，由于对任意 $p\in U$，由定理 4.1（P81）得存在其邻域 V 使得 $f(V)$ 是 $f(p)$ 的邻域，得到 $f(U)$ 是开集，即 f 是开映射。
21. 假设存在这样的 f，由于 f 是 $1-1$ 的，任取一点 $p\in M$，有 $f(M\backslash\{p\})=\mathbf{R}\backslash\{f(p)\}$，但是 $\dim M\ge 2$ 表明 $M\backslash\{p\}$ 是连通的，它在连续映射 f 下的像也是连通的，矛盾。
22. 笛卡尔叶形线 $f_1(x,y)=x^3+xy+y^3=0$，在 $(0,0)$ 点自交，在该点附近任何嵌入映射关于叶形线（作为流形）的拓扑和 $\mathbf{R}^2$ 的子空间拓扑不同胚，所以不是嵌入子流形。
对 $f_2(x,y)=x^3+xy+y^3-f(\frac13,\frac13)=0$，不难验证 $\frac{\partial f_2}{\partial x}=3x^2+y\ne 0$，由隐函数定理，存在可逆的连续递增函数 g 使得 $y=g(x)$，所以 $(i,f^{-1}(f((\frac13,\frac13))))$ 是嵌入子流形。
注意到 $f_3(x,y)=x^3+xy+y^3-f(-\frac13,-\frac13)=0$ 能因式分解为 $(x+y+\frac13)\big((x+\frac13)^2-(x+\frac13)(y+\frac13)+(y+\frac13)^2\big)=0$，有孤立点 $(-\frac13,-\frac13)$，不是流形，更不是嵌入子流形。
23. 对于 $x_0=(x_0^1,\dots,x_0^n)\in g^{-1}(c)$，不妨设 $\dfrac{\partial g}{\partial x^n}(x_0^1,\dots,x_0^n)\ne 0$，由隐函数定理，存在 $(x_0^1,\dots,x_0^{n-1})$ 的邻域 V，在其上有连续可微的函数 $\varphi(x^1,\dots,x^{n-1})$ 使得 $g(x^1,\dots,x^{n-1},\varphi_0(x^1,\dots,x^{n-1}))=c$. 故对于 $(x_0^1,\dots,x_0^n)\in g^{-1}(c)$，其在 $g^{-1}(c)$ 上有邻域
$$\tilde V=\{(x^1,\dots,x^{n-1},\varphi(x^1,\dots,x^{n-1}))\mid(x^1,\dots,x^{n-1})\in V\}$$
有 $\tilde V$ 与 V 同胚，从而 $g^{-1}(0)$ 为 $n-1$ 维子流形. 由于
$$\frac{\partial(x^1,\dots,x^{n-1},\varphi(x^1,\dots,x^{n-1}))}{\partial(x^1,\dots,x^{n-1})}$$
$$\left(\frac{\partial(x^i,\varphi)}{\partial x^j}\right)_{1\le i,j\le n-1}$$

秩为 $n-1$，$g^{-1}(c)$ 为 U 的浸入. $g^{-1}(c)\hookrightarrow U$ 的单一性自然成立，且对于 $(x_0^1,\dots,x_0^n)\in U$，取局部坐标

$$\begin{aligned} y^i &= x^i - x_0^i, \qquad 1\le i\le n-1,\\ y^n &= x^n-\varphi(x^1,\dots,x^n), \end{aligned}$$

得到 $g^{-1}(c)$ 为嵌入.

24. 将 W 与 $i(W)$ 视作同一流形. 设 $\dim M=m,\dim N=n,\dim W=w$. 对于 $p\in M$，有 N 中局部坐标系 $V,\{y^i\}$ 使得

$$y^i(f(p))=0,\forall\, 1\le i\le n,$$

$$W\cap V=\{s\in V\mid y^i(s)=0,\forall\, w+1\le i\le n\},$$

f 作为 $M\to N$ 光滑，即 f^i 对 $1\le i\le n$ 光滑，从而自然对 $1\le i\le w$ 光滑. 故 f 作为 $M\to W$ 光滑.

25. 任取一点 $p\in S^n$，不妨设 $p=(x_0^1,\dots,x_0^{n+1}),x_0^{n+1}>0$. 分别取 p 在 S^n 中局部坐标 (U,φ) 和 $f(p)$ 在 $\mathbf{RP}^n$ 中局部坐标 (V,ψ) 使得 $f(U)\subset V$ 并且

$$\begin{aligned} \varphi(x^1,\dots,x^{n+1}) &= (x^1,\dots,x^n),\\ \psi([(x^1\cdots x^{n+1})]) &= \Big(\frac{x^1}{x^{n+1}}\cdots\frac{x^n}{x^{n+1}}\Big), \end{aligned}$$

从而得证. 而其秩处处为 n 可由直接计算得知.

26. 对于 $x,y\in\mathbf{R}$，对 $\tilde a<a<x<y<b<\tilde b$，存在 $\varphi\in C^\infty(\mathbf{R})$ 使得

$$\varphi|_{[a,b]}=1,\varphi|_{\mathbf{R}\setminus(\tilde a,\tilde b)}=0.$$

取 $\lambda<1$ 足够小，使得对 $\varphi_\lambda(t):=\varphi(\lambda(t-\frac{a+b}{2})+\frac{a+b}{2})$ 有 $|\varphi_\lambda'(t)|<\frac{1}{y-x}$. 取

$$f(t)=t+(y-x)\varphi_\lambda(t),$$

由

$$f'(t)\ge 1-(y-x)|\varphi_\lambda'(t)|>0$$

得到 f 是同胚.

而对于 $x,y\in\mathbf{R}^n$，令 $f^i:\mathbf{R}\to\mathbf{R}$ 为使得 $f^i(x^i)=y^i$ 的函数. 令

$$f(t)=(f^1(t^1)\cdots f^n(t^n)),$$

那么光滑同胚 $f:\mathbf{R}^n\to\mathbf{R}^n$ 满足 $f(x)=y$，并且保持一个有界开集之外的点不动.

27. 若 x,y 处于同一个局部坐标内，由上一题，可构造光滑同胚 $f:M\to M$ 使得 $f(x)=y$.

对一般的情况，由 M 的连通性，存在道路 $\gamma:[0,1]\to M,\gamma(0)=x,\gamma(1)=y$，取 γ 的有限开覆盖 $\{U_i\}_{i=1}^N$ 使得 $x\in U_1,y\in U_N$，且 $U_i\cap U_{i+1}\neq\varnothing$.

取一系列点 $\{p_i\}_{i=1}^{N-1}$ 使得 $p_i\in U_i\cap U_{i+1}$，并且令 $p_0=x,p_N=y$. 取光滑同胚 $f_i:M\to M,f_i(p_i)=p_{i+1}$，$0\le i\le N-1$，再令 $f=f_{N-1}\circ f_{N-2}\circ\cdots\circ f_0$ 即可.

28. C^1- 浸入子流形。

    不是子流形。

    C^∞- 浸入子流形。

    Remark:(3):Need reconsideration.

29. 对任意点 $p \in X$, 存在 n 条曲线 $\gamma_1(t), \cdots, \gamma_n(t)$: $(-\varepsilon, \varepsilon) \to X$ 使得 $\{\gamma_1'(0), \cdots, \gamma_n'(0)\}$ 是 T_pX 的一组基.

    因此 $\eta_1(t) = \varphi(\gamma_1(t)), \cdots, \eta_n(t) = \varphi(\gamma_n(t))$ 是 Y 中的 n 条曲线, 且 $\{\eta_1'(0), \cdots, \eta_n'(0)\} = \{d\varphi_p(\gamma_1'(0)), \cdots, d\varphi_p(\gamma_n'(0)\}$ 是 $T_{\varphi(p)}Y$ 中的一组线性无关向量.

    注意到 $\dim X = \dim Y$, 因此 $\{\eta_1'(0), \cdots, \eta_n'(0)\}$ 是 $T_{\varphi(p)}Y$ 中的一组基.

    赋予 X 一个 Riemann 结构, 自然地, φ 在 Y 上诱导了一个 Riemann 结构. 因此, $\exp_p$ 和 $\exp_{\varphi(p)}$ 良定.

    因为 $\exp_p$ 是从 T_pX 到 X 的局部微分同胚, 所以存在 0 附近的邻域 U 使得 $\exp_p|_U$ 是一个微分同胚.

    而 $d\varphi_p$ 是从 T_pX 到 $T_{\varphi(p)}Y$ 的线性变换, 因此 $d\varphi_p(U)$ 是开的, 且存在开子集 $V \subset d\varphi_p(U)$ 使得 $\exp_{\varphi(p)}|_V$ 是微分同胚.

    如此, 我们得到从 $\exp_p(d\varphi_p^{-1}(V))$ 到 $\exp_{\varphi(p)}(V)$ 的微分同胚

    $$g = \exp_{\varphi(p)} \circ d\varphi_p \circ \exp_p^{-1}.$$

    所以, 对每个点 p, 存在一个开邻域 $U_p \subset X$ 使得 $\varphi(U_p)$ 在 Y 中是开集.

    因此 $\varphi(X) = \bigcup_{p \in X} \varphi(U_p)$ 是 Y 中的开子集.

30. 设 X 为 k 维流形，(U_α, ϕ_α) 为其局部坐标，则有如下图表，

    $$\begin{array}{ccc} U_\alpha & \xrightarrow{\varphi} & \mathbf{R}^n \\ \downarrow{\scriptstyle \phi_\alpha} & & \downarrow{\scriptstyle \pi} \\ \mathbf{R}^k & \longrightarrow & \mathbf{R}^m \end{array}$$

    $\pi \circ \varphi \circ \phi_\alpha^{-1}$: $\mathbf{R}^k \to \mathbf{R}^m$, $(x_1, x_2, \cdots, x_k) \mapsto (y_1, y_2, \cdots, y_m)$,

    $$\mathrm{rk}(\frac{\partial y_j}{\partial x_i})_{1 \le i \le k}^{1 \le j \le m} = k.$$

    所以

    $$\mathrm{rk}(\frac{\partial y_j}{\partial x_i})_{1 \le i \le k}^{1 \le j \le n} = k,$$

    所以 X 为浸入子流形。

31. 错误示范: 由于任意一条有向直线由其到原点距离以及其方向确定，故这个集合可看作 $S^1 \times R$, 其中 S^1 表示方向，R 表示其到原点的距离（带方向，所以可正可负），显然这是一个 2 维可定向光滑流形。

    Remark: 经典错误, 默认所有向量丛都是平凡丛. 不能排除 Mobius 丛. 应当具体构造坐标覆盖来说明.

32. 错误示范: 设 $(v_1, v_2 \cdots, v_m)$ 为 $\mathbf{R}^n$ 中 m 个线性无关向量，则 $\mathrm{span}(v_1, v_2 \cdots, v_m)$ 为 m 维平面，设 $(u_1, u_2 \cdots, u_m)$ 为另一组基，其过渡矩阵为 $A_{m \times m}$。在 $F(m, n)$ 中构造等价关系 $u = v \times A$, 若 $\det A > 0$, 则 uRv。令 $L(m, n) = F(m, n)/R$, 由书中对 Grassman 流形的讨论可知这是一个 $A2$，$T2$ 空间，以下给出其光滑结构，令 $U_{i_1, i_2 \cdots, i_m} = \{v \in F(m, n)$，其由

$i_1, i_2, \cdots, i_m$ 行构成的主子式非 0}，则 $U_{i_1,i_2\cdots,i_m}$ 为其开覆盖，由 (5.7)（P93）类似可得其转换矩阵光滑，$L(m,n)$ 为 $m(n-m)$ 维流形。

Remark: 显然是错误的, 和 31 题的结论都不符合. 问题在于默认所有平面过原点. 正确的维数应该是 $(m+1)(n-m)$.

33. (1) $n=1$ 时，S^1 为可定向流形，$n>1$ 时，S^n 可由两个开集覆盖，且 $U_\alpha \cap U_\beta$ 连通（其中 $U_\alpha = S^n - (0,0\cdots,0,1), U_\beta = S^n - (0,0\cdots,0,-1)$，故 S^n 为可定向流形。

(2) 考虑将 S^n 嵌入 $\mathbf{R}^{n+1}$, 给定 S^n 的定向等价于给出 $\mathbf{R}^{n+1}$ 中 n 个线性无关的单位向量 $v_1, v_2, \cdots, v_n$，则必存在一个法向量 v，使得这 $n+1$ 个向量在 $\mathbf{R}^{n+1}$ 标准坐标下行列式 >0,n 为奇数时，$n+1$ 为偶数, 此时 $p\circ v_1, p\circ v_2\cdots, p\circ v$ 行列式仍大于 0，而 n 为奇数时, 行列式小于 0，改变了定向。

34. $\forall \alpha \in M$，由于 M 连通故道路连通，所以可以通过 $[0,1]: p\to\alpha$ 的连续延拓给出 α 的定向，故以下仅需说明 α 处的定向与道路的选取无关即可。设 t_1, t_2 为两条不同的道路，且在 α 出有不同的定向，则逆转 t_2 在 α 出的定向记作 $-t_2$，考虑则 $-t_2^{-1}\circ t_1$ 为经过 p 的闭道，且逆转了 p 点的定向，与条件矛盾，故 α 处的定向与道路选取无关，故 M 可定向（由 P112 的说明可得）。

35. 当 n 为奇数时，S^n 的对径映射是保定向的，所以由 S^n 到 RP^n 的商映射自然诱导了一个 RP^n 的定向。当 n 为偶数时，对径映射反转定向，从而在 RP^n 上通过继承 S^n 结构得到的定向不能自洽，所以不可定向.

Remark 1：由 Hatcher, *Algebraic Topology*, P238 Corollary 3.28，一个 n 维紧致无边流形 M 是可定向的当且仅当 $H_{n-1}(M)$ 的扭子群（torsion subgroup）是平凡的, 又由 Hatcher P144 Example 2.42 有 $H_{n-1}(\mathbf{RP}^n)$ 的扭子群平凡当且仅当 $n-1$ 为偶数。

Remark 2：或使用这一结论：对 n 维连通闭流形 M，$H_n(M) = \begin{cases} \mathbb{Z}, M \text{ 可定向} \\ 0, M \text{ 不可定向} \end{cases}$（见 Hatcher P236 Theorem 3.26）。

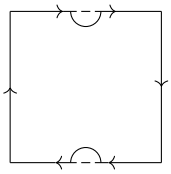
Remark 3：由 Poincaré 对偶定理（见 Hatcher P241 Theorem 3.30），容易说明当 n 为偶数时 RP^n 不可定向.

Remark 4：还可以使用 n 次微分形式（等价地，n 次 de Rham 上同调群）分析最高次微分形式在对径变换下的良定义性, 类似可得结论.

36. 沿 S^2 的水平的大圆（作为一条给定方向的闭路径）取标架场 $\{p; e_1, e_2\}$，使得 e_1 是该路径的切向量，e_2 竖直向上，那么粘合对径点后，在同一点两个 e_1 方向相反而两个 e_2 方向相同，也就是说沿该路径传播得到了相反的定向，由推论 6.2（P113），$\mathbf{RP}^2$ 是不可定向的。

37. 参看 §5 例 2（P98）的图，沿右图中的路径取标架场 $\{p; e_1, e_2\}$，使得 e_1 是该路径的切向量，e_2 对应到左下图中是竖直向上的，即从 A 指向 D，从 B 指向 C，那么粘合成 Klein 瓶后，在 $A(=B=C=D)$ 点 e_2 翻转了方向而 e_1 的方向良定，沿该路径传播得到了相反的定向，由推论 6.2，Klein 瓶是不可定向的。

注：一个更简单的方法是，假设 $\mathbf{RP}^2$ 或 Klein 瓶可定向，如下图在 $\mathbf{RP}^2$ 上剪一个洞，或沿图中路径剪开 Klein 瓶，就得到了 Möbius 带以及限制于其上的定向，但例 1 已证明 Möbius 带不可定向，矛盾。



38. 只需验证全纯函数作为复变量的实部与虚部的函数的 Jacobi 行列式大于 0。
设 $f(x+iy) \triangleq F(x,y) = u(x,y) + iv(x,y)$ 是全纯函数，由 Cauchy-Riemann 方程，

$$J_F = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ -\frac{\partial u}{\partial y} & \frac{\partial u}{\partial x} \end{vmatrix} = \Big(\frac{\partial u}{\partial x}\Big)^2 + \Big(\frac{\partial u}{\partial y}\Big)^2 > 0.$$

39. (1) 对于使 $g(q) > c$ 的 $q \in \mathbf{R}^n$，由 g 的连续性得存在 q 的邻域 U 使得 $g|_U > c$，即 $U \subset \operatorname{Int} M$；
对于 $q \in g^{-1}(c)$，由 $\nabla g(q) \neq 0$，可取合适的光滑函数 $y^1 = y^1(x), \cdots, y^{n-1} = y^{n-1}(x)$ 使得对 $y^n = g(x) - c$ 有 $\det\big(\frac{\partial y^i}{\partial x^j}\big)|_q \neq 0$，由隐函数定理，存在 $\mathbf{R}^n$ 在 q 的邻域 U 使得 (U, y^i) 是与原坐标系 (M, x^j) C^∞- 相关的坐标卡。在这一坐标卡下，$p \in M$ 等价于 $y_n(p) = g(p) - c \geq 0$，即 p 的坐标属于 H^n，所以 M 在 q 的邻域 $U \cap M$ 光滑同胚于 H^n 的一个开子集，且 $q \in \partial M$。
因此，M 是一个 n 维带边流形。
(2) 由推论 5.5（P103）得到 $g^{-1}(c)$ 是 $n-1$ 维无边流形；
由（1）知 $g^{-1}(c) = \partial M$，所以 M 上的定向（可由 $\mathbf{R}^n$ 的定向限制得到）诱导出 ∂M 上的定向。

40. (1) 设 $\{(U_\alpha, \varphi_\alpha)\}$ 是 M 的一个定向，$\{(V_\beta, \phi_\beta)\}$ 是 N 的一个定向，那么 $\{(U_\alpha \times V_\beta, (\varphi_\alpha, \phi_\beta))\}$（包含于的定向相符的极大容许坐标卡集）就是 $M \times N$ 的一个定向，这是因为当 $(U_\alpha \times V_\beta) \cap (U_{\alpha'} \times V_{\beta'}) \neq \varnothing$ 时，

$$\det\Big(\frac{\partial((\varphi_{\alpha'}, \phi_{\beta'}) \circ ((\varphi_\alpha, \phi_\beta)^{-1})^i}{\partial(x_\alpha^j, y_\beta^k)}\Big) = \det\Big(\frac{\partial(\varphi_{\alpha'} \circ \varphi_\alpha^{-1})^i}{\partial x_\alpha^j}\Big) \det\Big(\frac{\partial(\phi_{\beta'} \circ \phi_\beta^{-1})^i}{\partial y_\beta^k}\Big) > 0.$$

(2) 即证明若 $M \times N$ 是可定向的，则 M, N 都是可定向的。
设 $\{(W_\gamma, \psi_\gamma)\}$ 是 $M \times N$ 的一个定向，$\{(U_\alpha \times V_\beta, (\varphi_\alpha, \phi_\beta))\}$ 为其中全体这种形式的元素，显然有 $M = \bigcup\limits_\alpha U_\alpha, N = \bigcup\limits_\beta V_\beta$。注意到当 $U_\alpha \cap U_{\alpha'} \neq \varnothing$ 时，任取一个 β，

$$\det\Big(\frac{\partial(\varphi_{\alpha'} \circ \varphi_\alpha^{-1})^i}{\partial x_\alpha^j}\Big) = \det\Big(\frac{\partial((\varphi_{\alpha'}, \phi_\beta) \circ (\varphi_\alpha, \phi_\beta)^{-1})^i}{\partial(x_\alpha^j, y_\beta^k)}\Big) > 0,$$

所以 $\{(U_\alpha, \varphi_\alpha)\}$ 是 M 的一个定向，同理 $\{(V_\beta, \phi_\beta)\}$ 是 N 的一个定向。
(3) 设 $\{(U_\alpha, \varphi_\alpha)\}$ 是带边流形 M 的微分结构，$\{(V_\beta, \phi_\beta)\}$ 是流形 N 的微分结构，那么 $\{(U_\alpha \times V_\beta, (\varphi_\alpha, \phi_\beta))\}$ 是带边流形 $M \times N$ 的微分结构，这是因为 $(\varphi_\alpha, \phi_\beta)$ 是 $U_\alpha \times V_\beta$ 到 $H^m \times \mathbf{R}^n = H^{m+n}$ 的开子集的同胚（这里 $m = \dim M, n = \dim N$，$H^{m+n} = \{x \in \mathbf{R}^{m+n} \mid x_m \geq 0\}$），且 $(p, q) \in \partial(M \times N)$ 当且仅当存在坐标卡使 $(\varphi_\alpha(p), \phi_\beta(q)) \in \partial H^{m+n}$，当且仅当 $\varphi_\alpha(p) \in \partial H^m$，当且仅当 $p \in \partial M$，所以 $\partial(M \times N) = \partial M \times N$。

41. 称为光滑 Urysohn 定理，是单位分解定理（定理 2.7，P66）的一个推论。
$\{M \backslash A, M \backslash B\}$ 是 M 的一个开覆盖，根据单位分解定理，存在局部有限的加细开覆盖 $\{V_i\}_{i=1}^\infty$ 以及光滑函数 $\{f_i\}_{i=1}^\infty$ 使得 $0 \leq f_i \leq 1$，$\sum\limits_{i=1}^\infty f_i = 1$，且 $\operatorname{supp} f_i \subset V_i \subset M \backslash A$ 或 $M \backslash B$，也就是说 $f_i|_A \equiv 0$ 或 $f_i|_B \equiv 0$。

令 $\psi=\sum\limits_{f_i|_B\equiv 0} f_i$，则 $0\le\psi\le 1$，$\psi|_B\equiv 0$，且由于若 $f_i|_B\neq 0$ 就有 $f_i|_A\equiv 0$，得

$$\psi|_A=\Big(\sum_i f_i-\sum_{f_i|_B\neq 0} f_i\Big)\Big|_A\equiv 1,$$

而且 $\{V_i\}_{i=1}^{\infty}$ 局部有限表明 $\psi=\sum\limits_{f_i|_B\equiv 0} f_i$ 局部为有限个非零 f_i 之和，所以 ψ 也光滑。

1.13 习题: 陈维桓微分流形/第四章外微分式

原网页

2. 在 S^n 上有标准 Riemann 度量, 该度量在对径映射下不变，$\mathbf{RP}^n$ 作为其在对径映射下的商空间，可自然诱导出 $\mathbf{RP}^n$ 上的一个黎曼度量，下面给出该度量的局部坐标：
   只给出在 $U=\{[x_1,\dots,x_{n+1}]\mid x_{n+1}>0,\sum(x_i)^2=1\},\phi([x_1,\dots,x_{n+1}])=[x_1,\dots,x_n]$ 的局部坐标下的黎曼度量，其他一个局部坐标可类似给出。
   局部上 U 的黎曼度量与 S^n 相同，我们将 S^n 视作 $\mathbb{R}^{n+1}$ 的子流形. 任意 U 上的点 p, 该点处有切空间的一组基 $\{\frac{\partial}{\partial x_i}\mid i=1,\dots,n\}$, 且有 $\frac{\partial}{\partial x_k}=(0,\cdots,1,\cdots,0,\frac{-x_k}{\sqrt{1-\sum x_i^2}})$, 故我们定义 $g_{kk}=1+\frac{x_k^2}{1-\sum x_i^2}$, 对于 $k=1,2,\cdots,n.$ $g_{ij}=0$, 对于 $i\neq j.$, 这样就给出了 U 上的黎曼度量，进而可给出全空间上的黎曼度量。
   体积元素即流形上大范围定义的正的 n 次外微分式（不一定要用黎曼度量来定义），这等价于流形是可定向的（定理 1.4，P178），故当且仅当 n 为奇数时 $\mathbf{RP}^n$ 上能定义体积元素。
3. (1) 由 φ 是反对称的 r 重线性映射得 $i(X)\varphi$ 是反对称的 $r-1$ 重线性映射，且对每个 X_i 是 $C^\infty(M)$ - 线性的，故 $i(X)\varphi\in A^{r-1}(M)$。
   (2) 同第一章第 44 题 (1)。
   (3) 同第一章第 44 题 (2)。
4. (1)

$$\begin{aligned}[(i(X)\circ d)(\omega)](Y_1,\dots Y_r)=&[i(X)(d\omega)](Y_1,\dots Y_r)\\=&d\omega(X,Y_1,\cdots Y_r)\\=&X(\omega(Y_1,\cdots,Y_r))+\sum_{\lambda=1}^{r}(-1)^r Y_\lambda(\omega(X,Y_1,\cdots,\hat{Y_\lambda},\cdots,Y_r))\\&+\sum_{\lambda<\mu}(-1)^{\lambda+\mu+1}\omega(X,[Y_\lambda,Y_\mu],Y_1,\dots,\hat{Y_\lambda},\cdots,\hat{Y_\mu},\cdots,\ Y_r)\\&+\sum_{\lambda=1}^{\lambda=r}(-1)^\lambda\omega([X,Y_\lambda],Y_1,\cdots,\hat{Y_\lambda},\dots,Y_r).\end{aligned}$$

   记 $i(X)(\omega)=\phi\in A^{r-1}(M)$，

$$\begin{aligned}[(d\circ i(X))(\omega)](Y_1,\dots Y_r)=&d\phi(Y_1,\dots Y_r)\\=&\sum_{\lambda=1}^{r}(-1)^{r+1}Y_\lambda(\phi(Y_1,\dots,\hat{Y_\lambda},\dots,Y_r))+\sum_{\lambda<\mu}(-1)^{\lambda+\mu}\phi([Y_\lambda,Y_\mu],Y_1,\dots,\hat{Y_\lambda},\end{aligned}$$

$$=\sum_{\lambda=1}^{r}(-1)^{r+1}Y_\lambda(\omega(X,Y_1,\ldots,\hat{Y_\lambda},\ldots,Y_r))+\sum_{\lambda<\mu}(-1)^{\lambda+\mu}\omega(X,[Y_\lambda,Y_\mu],Y_1,..$$

$$[(i(X)\circ d+d\circ i(X))(\omega)](Y_1,\cdots Y_r)=X(\omega(Y_1,\cdots,Y_r))+\sum_{\lambda=1}^{r}(-1)^{\lambda}\omega([X,Y_\lambda],Y_1,\cdots,\hat{Y_\lambda},\cdots,Y_r)$$

$$(\mathscr{L}_X(\omega))=\lim_{t\to 0}\frac{(\Phi_t)^*(\omega)-\omega}{t}.$$

仿照定理 3.5（P146）即可证明。

(2) $\mathscr{L}_X\circ d=(i(X)\circ d+d\circ i(X))\circ d=d\circ i(X)\circ d=d\circ(i(X)\circ d+d\circ i(X))=d\circ\mathscr{L}_X$。

(3) $\mathscr{L}_X(\varphi\wedge\psi)=(i(X)\circ d+d\circ i(X))(\varphi\wedge\psi)$。

已知：

$$d(\varphi\wedge\psi)=(d\varphi)\wedge\psi+(-1)^r\varphi\wedge(d\psi),$$

$$i(X)(\varphi\wedge\psi)=(i(X)\varphi)\wedge\psi+(-1)^r\varphi\wedge(i(X)\psi),$$

$$\begin{aligned}(i(X)\circ d)(\varphi\wedge\psi)&=(i(X))((d\varphi)\wedge\psi+(-1)^r\varphi\wedge(d\psi))\\&=[(i(X)\circ d)\varphi]\wedge\psi+(-1)^r(d\varphi)\wedge(i(X)\psi)+(-1)^r(i(X)\varphi)\wedge d\psi+\varphi\wedge[(i(X)\circ d)\end{aligned}$$

$$\begin{aligned}(d\circ i(X))(\varphi\wedge\psi)&=d\circ((i(X)\varphi)\wedge\psi+(-1)^r\varphi\wedge(i(X)\psi))\\&=[(d\circ i(X))\varphi]\wedge\psi+(-1)^r(i(X)\varphi)\wedge d\psi+(-1)^r(d\varphi)\wedge(i(X)(\psi))+\varphi\wedge[(d\circ i(X)\end{aligned}$$

$$\begin{aligned}&\mathscr{L}_X(\varphi\wedge\psi)\\=&[(i(X)\circ d)\varphi]\wedge\psi+\varphi\wedge[(i(X)\circ d)\psi]+[(d\circ i(X))\varphi]\wedge\psi+\varphi\wedge[(d\circ i(X))\psi]\\=&\mathscr{L}_X(\varphi)\wedge\psi+\varphi\wedge\mathscr{L}_X(\psi)\end{aligned}$$

5. (1) $xdy\wedge dx-(1+z)dy\wedge dz$.

(2) $-2yzdz\wedge dy-2dx\wedge dz$.

(3) $(2x^2-x-xz-2xy^2z+2z)dx\wedge dy\wedge dz$.

(4) $f^*\omega=(6u^2+2uv+u^3v^2-9u^3-3u^2v)du+(u^4v-3u^3-u^2v)dv$,
$f^*(d\omega)=(2u^3v-2u-6u^2-2uv)du\wedge dv$.

6. $\omega=x_{r+1}dx_1\wedge dx_2\cdots\wedge dx_r$ 在 $x_{r+1}=0$ 上某点为 0, 但该点 $d\omega\neq 0$。

7. 注意到 $\omega=d\big(\frac{1}{2}\ln(x^2+y^2)\big)$，即 ω 是恰当微分式，于是也是闭微分式。

8. (1) $d\omega=\sum_{i=1}^{n}\frac{\partial f_i}{\partial x^i}dx_1\wedge dx_2\cdots\wedge dx_n=\frac{n-m}{(\sum_{i=1}^{n}(x^i)^2)^{\frac{m}{2}}}dx_1\wedge dx_2\cdots\wedge dx_n$。

(2) 由 (1)，$m=n$。

(3) 假设$\omega = df$,则在平面$x^3 = x^4 = \cdots = x^n = 0$上,$\omega = \dfrac{x^1dx^2 - x^2dx^1}{((x^1)^2 + (x^2)^2)^{\frac{n}{2}}} \wedge dx^3 \wedge \cdots \wedge dx^n$. 在无边子流形$U = \{(\cos t, \sin t, x^3, \cdots, x^n) \mid 0 \leq t \leq 2\pi, 0 \leq x^k \leq 1, k = 3, 4, \cdots, n\}$上对$\omega$积分，计算知：

$$\int_\gamma \omega = 2\pi \neq 0,$$

由 Stokes 公式，知与ω是恰当微分式矛盾。

9. $\forall p$，取$p \in V \subseteq \overline{V} \subseteq W \subseteq \overline{W} \subseteq U$，其中$\overline{V}$是紧致的，由第二章引理 2.2（P63），存在函数$\varphi \in C^\infty(M) : \varphi|_V = 1, \varphi|_{M\setminus\overline{W}} = 0$。令$\tilde{\omega} = \omega \cdot \varphi(x)$便得到了想要的微分形式。

10. (1) 不难验证$|v| = 1$。又$\forall w \in T_pM$ $(p = f(u^1, \ldots, u^n))$, $w = \dfrac{d}{dt} f(u^1(t), \ldots, u^n(t))$, 内积$|\langle v, w\rangle| = |\sum(-1)^{i+1} A^i \dfrac{d}{dt} f^i(u^1(t), \ldots, u^n(t))| = |d\dfrac{1}{A} \det(M)| = 0$。其中$M$是$n+1$阶方阵，后$n$行为$f$关于$u_i$的 Jacobi 矩阵的转置，第一行是$\sum_{i=2}^{n+1}$第 i 行$\cdot \dfrac{\partial u_{i-1}}{\partial t}$

(2) $\mathbf{R}^{n+1}$上的黎曼度量是$\sum dx^i \otimes dx^i$，其在M上诱导度量为：

$$g = \sum_\alpha \frac{\partial f^\alpha}{\partial u^i} \frac{\partial f^\alpha}{\partial u^j} du^i \otimes du^j.$$

(3) $g_{ij} = \sum\limits_\alpha \dfrac{\partial f^\alpha}{\partial u^i} \dfrac{\partial f^\alpha}{\partial u^j} du^i \otimes du^j$，所以矩阵$(g_{ij})_{n\times n} = (a_{ik})_{n\times(n+1)} (a_{ik})^t_{(n+1)\times n}$, 其中$a_{ik} = \dfrac{\partial f^k}{\partial u^i}$，由 Guass—Binet 公式计算知$M$的单位元素为

$$A du^1 \wedge du^2 \wedge \cdots \wedge du^n.$$

现在用第 3 题 (3)] 来计算$i(v)(dx^1 \wedge \cdots \wedge dx^{n+1})$，事实上原式

$$\begin{aligned}
&= \sum_\alpha (-1)^{\alpha+1} dx^1 \wedge \ldots i(v)(dx^\alpha) \wedge \cdots \wedge dx^{n+1} \\
&= \sum_\alpha (-1)^{\alpha+1} dx^\alpha(v) dx^1 \wedge \cdots \wedge dx^{n+1} \\
&= \sum_\alpha (-1)^{\alpha+1} v^\alpha A^\alpha du^1 \wedge \ldots \hat{du^\alpha} \wedge \cdots \wedge du^{n+1} \\
&= A du^1 \wedge du^2 \wedge \cdots \wedge du^n.
\end{aligned}$$

11. 对于$\mathbf{R}^{n+1}$中$n+1$个线性无关向量$v_1, \ldots, v_{n+1}$，其中v_{n+1}为其余n个向量法向量，若它们在标准基下表示矩阵行列式> 0，称v_{n+1}为前n个向量的正法向量，否则称为负法向量。因此：

(f, M)上存在光滑处处非零法向量v，

$\Leftrightarrow (f, M)$存在局部坐标系使得存在法向量v，相对于各点切空间的标准基嵌入$\mathscr{R}$后得到的n个向量处处为正法向量，

$\Leftrightarrow (f, M)$存在局部坐标系使得转移函数 Jacobi 矩阵恒正，（$\Rightarrow$: 是显然的；$\Leftarrow$: 由单位法向量只有两个，分别相对同一组向量为正、负法向量，v定义为该点相对局部坐标的正单位法向量；可看出其良定性，光滑性则由法向量是曲面方程的梯度得知）

$\Leftrightarrow M$可定向。

13. $\mathbf{RP}^3 = S^3/_{x\sim -x}$，且与 S^3 为局部光滑同胚，故可以在 S^3 上构造后通过商映射诱导出 $\mathbf{RP}^3$ 上的 1 次微分，并证明其良定性。

    $\S^3 = \left\{(x_1,x_2,x_3,x_4) \mid \sum (x_i)^2 = 1\right\}$，$\omega(x_1,x_2,x_3,x_4) \triangleq x_2dx_1 - x_1dx_2 + x_4dx_3 - x_3dx_4$，只需验证 $\omega(x) = \omega(-x)$ 即得诱导微分形式的良定性，诱导得到的 ϖ 符合题意。

14. 见 §2 例 2（P194）。
    然而在 M 上存在函数 $f = -\arctan\frac{x}{y}$ 使 $df = \omega$，故 ω 是恰当的。

15. (1) 设 α 是 r 次外微分式，由 $d\alpha = d\beta = 0$，得 $d(\alpha\wedge\beta) = d\alpha\wedge\beta + (-1)^r\alpha\wedge d\beta = 0$。
    (2) 设 α 是 r 次外微分式，$\beta = d\gamma$，由 $d\alpha = 0$，得 $d((-1)^r\alpha\wedge\gamma) = \alpha\wedge d\gamma = \alpha\wedge\beta$。

16. I 的线性从 ω 继承。I 是单射，因为若 $I(\zeta) = 0$，即 $\forall\eta\in T_xM$，$\omega(\zeta,\eta) = (I(\zeta))(\eta) = 0$，由 ω 非退化得 $\zeta = 0$。又 $\dim T_xM = \dim T_x^*M$，故 I 是同构。

17. 容易计算 $d\omega = 0$。在 $\mathbf{R}^{2n}$ 自然坐标系诱导的切空间的基底下，$\omega = \sum\limits_i dp^i\wedge dq^i$ 的矩阵为 $\bigoplus\limits_i I_i$，I_i 具有以下形式：

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

    容易验证这个矩阵是非退化的。

18. 考虑 $\wedge^n\omega\in\mathscr{A}^{2n}$, 由 ω 的非退化性得到 $\wedge^n\omega\neq 0$ 点点成立。

19. (1) $(i(X_f)\omega)(Y) = \omega(X_f,Y) = df(Y)$，即 $i(X_f)\omega = df$。
    (2) $\frac{d}{dt}(f\circ\gamma_x(t)) = \gamma_x(f) = X_f(x)(f) = X_f(f)(x) = df(X_f)(x) = \omega(X_f,X_f) = 0$。
    (3) $\mathscr{L}_{X_f}(\omega) = i(X_f)(d\omega) + d(i(X_f)(\omega)) = i(X_f)(d\omega) + d\circ df = i(X_f)(d\omega)$。
    而 $[i(X_f)(d\omega)](Y,Z) = d\omega(X_f,Y,Z) = 0$，因此 $[\mathscr{L}_{X_f}(\omega)](Y,Z) = 0$，$\forall Y,Z\in\mathscr{X}(M)$，即 $\mathscr{L}_{X_f}(\omega) = 0$。

20. (1) 由于 $\{f,g\} = \omega(X_f,X_g) = -i(X_f)i(X_g)\omega = -i(X_f)dg = \mathscr{L}_{X_g}f = -\mathscr{L}_{X_f}g$.
    (2) 由于 $\{g\circ\varphi_t\} = \mathscr{L}_{X_f}(g\circ\varphi_t) = X_f(g\circ\varphi_t) = \dfrac{d}{dt}(g\circ\varphi_t)$.
    (3) 首先，$C^\infty(M)$ 关于 Poisson 括号的封闭性显然。线性与反交换性是简单的，只验证 Jacobi 恒等式。由第 19 题 (3) 知

$$\begin{aligned}\{f,\{g,h\}\} &= X_f(\{g,h\}) = \mathscr{L}_{X_f}\{g,h\} = \mathscr{L}_{X_f}(\omega(X_g,X_H))\\ &= (\mathscr{L}_{X_f}\omega)(X_g,X_h) + \omega(\mathscr{L}_{X_f}X_g,X_h) + \omega(X_g,\mathscr{L}_{X_f}X_h)\\ &= 0 - \omega(X_h,[X_f,X_g]) + \omega(X_g,[X_f,X_h])\\ &= \{g,\{f,h\}\} - \{h,\{f,g\}\}.\end{aligned}$$

    (4) 只证封闭性：令 $i_X\omega = df$，$i_Y\omega = dg$，$i_Z\omega = dh$，则有

$$i_{X+cY}\omega = df + cdg = d(f+cg),$$

    以及由第 19 题 (3) 知

$$\begin{aligned}i_{[X,Y]}\omega &= \mathscr{L}_X(i_Y\omega) - i_Y(\mathscr{L}_X\omega) = \mathscr{L}_X(dg) - i_Y(0)\\ &= d(\mathscr{L}_Xg) = d(i_X(dg)) = d(i_X(i_Y\omega)) = d(\omega(Y,X)).\end{aligned}$$

    而线性，反交换律以及 Jacobi 恒等式直接继承向量场的李括号。

21. 由 Gauss 公式直接有

$$\int_{S^2(r_0)}\omega=\int_{r_0B^3}d\omega=\frac{1}{r_0^3}\int_{r_0B^3}3dx\wedge dy\wedge dz=\frac{1}{r_0^3}\cdot 3\cdot\frac{4}{3}\pi r_0^3=4\pi.$$

22. 直接计算得 $f^*\omega(\frac{\partial}{\partial u},\frac{\partial}{\partial v})=v\cdot(-2u)+u(u^2+v^2+1)\cdot 2v=2uv(u^2+v^2)$。那么

$$\int_D f^*\omega=\int_D 2uv(u^2+v^2)dudv=\frac{1}{2}.$$

23. 由于 $(-1)^k\omega\wedge d\eta+d\omega\wedge\eta=d(\omega\wedge\eta)$ 及 M 无边知

$$\int_M(-1)^k\omega\wedge d\eta+d\omega\wedge\eta=\int_M d(\omega\wedge\eta)=\int_{\partial M}\omega\wedge\eta=0,$$

而此即

$$\int_M d\omega\wedge\eta=(-1)^{k+1}\int_M\omega\wedge d\eta.$$

24. 简便起见，定义 $\tilde{A}=\int_0^1 tA(tx,ty,tz)dt$ 以及 $\tilde{A}_x=\int_0^1 t^2\frac{\partial A}{\partial x}(tx,ty,tz)$，类似定义 $\tilde{A}_y$，$\tilde{B}_x$ 等。由

$$\begin{aligned}d(A(ydz-zdy))&=(y\tilde{A}_xdx+(y\tilde{A}_y+\tilde{A})dy)\wedge dz-(z\tilde{A}_xdx+(z\tilde{A}_z+\tilde{A})dz)\wedge dy\\&=-z\tilde{A}_xdx\wedge dy+(y\tilde{A}_y+z\tilde{A}_z+2A)dy\wedge dz-y\tilde{A}_xdz\wedge dx\end{aligned}$$

可得

$$\begin{aligned}d\alpha=&(x\tilde{C}_x+y\tilde{C}_y+2\tilde{C}-z\tilde{A}_x-z\tilde{B}_y)dx\wedge dy+\\&(y\tilde{A}_y+z\tilde{A}_z+2\tilde{A}-z\tilde{B}_y-x\tilde{C}_z)dy\wedge dz+\\&(x\tilde{B}_x+z\tilde{B}_z+2\tilde{B}-y\tilde{A}_x-y\tilde{C}_z)dz\wedge dx.\end{aligned}$$

逐一计算，如

$$\begin{aligned}&y\tilde{A}_y+z\tilde{A}_z+2\tilde{A}-z\tilde{B}_y-x\tilde{C}_z\\=&\int_0^1 t^2y\frac{\partial A}{\partial y}dt+\int_0^1 t^2z\frac{\partial A}{\partial z}dt+\int_0^1 2tAdt-\int_0^1 t^2x\frac{\partial B}{\partial y}dt-\int_0^1 t^2x\frac{\partial C}{\partial z}dt,\end{aligned}$$

由 ω 闭知 $\frac{\partial A}{\partial x}+\frac{\partial B}{\partial y}+\frac{\partial C}{\partial z}=0$，代入得上式即

$$\int_0^1 t^2(x\frac{\partial A}{\partial x}+y\frac{\partial A}{\partial y}+z\frac{\partial A}{\partial z})dt+\int_0^1 2tAdt=\int_0^1\frac{\partial(t^2A)}{\partial t}dt=A,$$

同理得另两项，合并则有 $d\alpha=Adx\wedge dy+Bdy\wedge dz+Cdz\wedge dx=\omega$。

注 1.13.1. 这是书上定理 *2.5* 计算过程的特例。

1.14 试卷: 微分流形

原网页

2025–2026 学年期末 (陈曦)

一、 (40 分)

1. 叙述光滑流形的定义，并验证

$$S^2 := \left\{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = 1\right\}$$

是光滑流形.

2. 叙述正则子流形的定义。设 $g : \mathbb{R}^n \to \mathbb{R}$ 光滑，设

$$f : \mathbb{R}^{n+1} \to \mathbb{R}; (x_1,\ldots,x_n,x_{n+1}) \mapsto x_{n+1} - g(x_1,\ldots,x_n).$$

证明 $f^{-1}(0)$ 是 $\mathbb{R}^{n+1}$ 的正则子流形.

3. 对 $F : N \to M$, 叙述推前映射 $F_{*,p} : T_pN \to T_{F(p)}M$ 的定义, 并用一组基表示它。

4. 构造外微分 d。对 N 上的 k-形式以及 $F : N \to M$, 叙述拉回 F^* 的定义.

5. 叙述 Stokes 定理。设 D 是平面中的闭圆盘，

$$\omega = (x^2-y)dx + (y^2+x)dy,$$

使用 Stokes 定理计算 $\int_{\partial D}\omega$.

二、 (60 分)

1. 考虑 $\mathbb{R}^4 = \{(x,y,z,t)\}$, 记

$$E = E_1dx + E_2dy + E_3dz,$$

$$B = B_1dy\wedge dz + B_2dz\wedge dx + B_3dx\wedge dy,$$

$$F = E\wedge dt + B.$$

(1) 计算 dF.

(2) 证明 $dF = 0$ 等价于：

$$\nabla \times E = -\frac{dB}{dt}, \quad \operatorname{div} B = 0.$$

2. 考虑导子 $D : C^\infty(M) \to C^\infty(M)$ 满足 Leibniz 律

$$D(fg) = D(f)g + fD(g).$$

(1) 证明：若 $f|_U \equiv 0$, 则 $Df|_U \equiv 0$.

(2) 证明：存在唯一的向量场 X, 使得对任意 f 成立 $Df = Xf$. 给出 $\{x^1,\ldots,x^n\}$ 下 X 的表达式.

3. 考虑

$$SO(2) := \left\{A \in GL(2,\mathbb{R}) \mid A^TA = I, \det A = 1\right\}.$$

(1) 验证 $SO(2)$ 是一维李群并给出其李代数 $\mathfrak{so}(2) = T_ISO(2)$.

(2) 证明 $SO(2)$ 中元素都形如

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

(3) 证明：

$$\Phi : S^1 \to SO(2); \cos\theta + i\sin\theta \mapsto \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

是 Lie 群的同构，并给出 $\Phi_{*,1}$ 的表达式.

4. 考虑 Liouville 1-形式 λ. (试卷上有完整的定义)
   (1) 叙述余切簇 T^*M 的定义. 设其上的坐标卡为 $(U, x^1, \ldots, x^n, p_1, \ldots, p_n)$, 证明：$\lambda = \sum_{i=1}^n p_i dx^i$.
   (2) 计算 $d\lambda$, 证明 $(d\lambda)^n$ 是一个在 $\pi^{-1}U$ 上非退化的 $2n$-形式.
   (3) 证明 $(d\lambda)^n$ 非退化与坐标卡的选取无关，由此说明它在 M 上非退化.

2025–2026 学年期末 (王志张)

1 简答题（每题 7 分）

1. 证明：不存在紧致的极小曲面。
2. 证明：无边界紧致曲面有局部凸点。
3. 证明：$SL(n)$ 是流形。
4. 给定流形 M，给出函数芽的定义。如果记 F_m 是所有在 m 点处消失的函数芽构成的理想，那么 F_m/F_m^2 作为线性空间的维数是什么？为什么？
5. 记 $\mathcal{X}(M)$ 是 M 上光滑切向量场构成的集合。如果 ω 是一个 k 形式，证明

$$\omega : \mathcal{X}(M) \times \mathcal{X}(M) \times \cdots \times \mathcal{X}(M) \to C^\infty(M)$$

给出了一个交错的 $C^\infty(M)$-多重线性函数。反过来对么？为什么？

6. 线性空间的定向是什么？光滑流形的定向是什么？给出可定向的另外两个等价定义？Lie 群总是可定向的么？为什么？
7. 在分布 $\mathscr{D}$ 上消失的 q 形式的定义是什么？什么是微分理想？证明：$\mathscr{D}$ 是对合的当且仅当 $\mathscr{I}(\mathscr{D})$ 是微分理想。

2 作业题（每题 8 分）

1. GTM 94 第四章第 18 题。
2. GTM 94 第四章第 19 题。（无提示）

3 定理默写题（15 分）

给出奇异 p 链的定义，再给出奇异同调空间的定义，定义形式 ω 在 p 链上的积分，叙述并证明 p 链形式的 Stokes 定理。

4 自己挑战（20 分）

同 22–23 年最后一题。

2022–2023 学年期末 (王志张)

本次考试延长了 30 分钟。

1.（每小题 7 分，共 49 分）

(1) 什么是正则曲面的定向？能否用外法向量定义定向？证明你的结论。

(2) 证明：$\mathbb{R}^3$ 中的紧致无边曲面一定有严格凸点。

(3) 什么是微分流形？证明矩阵群 $SL(n)$ 是微分流形。

(4) 什么是全纯向量丛？回忆复切丛的构造过程，并证明复切丛是全纯向量丛。

(5) 定义 R- 模 M, N 的张量积。定义 M 的张量代数、外代数、k 次外积。

(6) 什么是 $\Omega^k(M)$？写出 $v_1 \wedge v_2 \in \Omega^2(M)$ 在 T^2M^* 中的嵌入. 已知 $\omega \in \Omega^1(M)$, 证明：$2!\langle \mathrm{d}\omega, X, Y\rangle = X\langle \omega, Y\rangle - Y\langle \omega, X\rangle - \langle \omega, [X, Y]\rangle$, 其中 $[\cdot,\cdot]$ 代表 Lie 括号。

(7) 证明：de Rham 上同调是微分同胚不变量。

2.（8 分）设 M, N 都是连通的光滑流形. 证明:

(1) 若 M, N 都是可定向的, 则 $M \times N$ 也是可定向的.

(2) 若 M, N 之中有一个是不可定向的, 则 $M \times N$ 是不可定向的.

3.（8 分）

定义映射 $f : D \to \mathbb{R}^3$ 为

$$f(u, v) = (u, v, u^2 + v^2 + 1) \tag{1.1}$$

其中 $D = [0, 1] \times [0, 1]$, 设

$$\omega = y\mathrm{d}y \wedge \mathrm{d}z + xz\mathrm{d}x \wedge \mathrm{d}z \tag{1.2}$$

是 $\mathbb{R}^3$ 上的 2 次微分式. 求积分 $\int\limits_D f^*\omega$.

4.（15 分）

(1) 什么是投射模?

(2) 证明：若 B 是紧致无边光滑流形，E 是 B 上的光滑向量丛，则 $E(B)$ 是投射模.(Swan 定理)

(3) 证明：若 B 是紧致无边光滑流形，M 是投射 $C^\infty(B)$- 模，则存在 B 上的光滑向量丛 E 使得 $E(B) = M$.

5.（20 分）研究环面上的 Hodge 算子. 将环面 $\mathbb{S}^1 \times \mathbb{S}^1$ 与 $D = [0, 1] \times [0, 1]$ 商掉对边同向对应顶点后等同起来. 因而此时对环面上的光滑函数 $f(x, y)$, 有 $f(0, y) = f(1, y), f(x, 0) = f(x, 1)$ (且任意阶导数也满足该相等条件). 定义 Hodge 算子 $\delta := - * \mathrm{d} *$ 是从 $\Omega^k(D)$ 到 $\Omega^{k-1}(D)$ 的线性映射，其中 $*$ 满足：

$$*1 = \mathrm{d}x \wedge \mathrm{d}y, \quad *\mathrm{d}x = \mathrm{d}y, \quad *\mathrm{d}y = -\mathrm{d}x, \quad *\mathrm{d}x \wedge \mathrm{d}y = 1。$$

(1) 对 $\varphi, \psi \in \Omega^k(D)$，定义 $\langle \varphi, \psi\rangle = \int\limits_D \varphi \wedge *\psi$。证明 $\langle \cdot, \cdot\rangle$ 是 $\mathbb{R}$ 线性空间 $\Omega^k(D)$ 上的内积。

(2) 证明：在 (1) 的内积下，δ 是 d 的对偶算子. (只需写 φ 为 0-形式，ψ 为 1-形式的情形下，成立 $d\varphi$ 和 ψ 的内积等于 φ 和 $\delta\psi$ 的内积即可)

(3) 定义 $\Omega^k(D)$ 到自身的 Laplace 算子 $\Delta\omega = \mathrm{d}\delta\omega + \delta\mathrm{d}\omega$。证明：$\Delta\omega = 0$ 的充要条件是 $\mathrm{d}\omega = 0$，$\delta\omega = 0$。

(4) 若 $\omega = f\mathrm{d}x + g\mathrm{d}y$，证明：$-\Delta\omega = \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}\right)\mathrm{d}x + \left(\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2}\right)\mathrm{d}y$。

(5) 定义 1-调和形式 $H^1(M) = \{\omega \in \Omega^1(M) : \Delta\omega = 0\}$. 已知以下事实: 定义 $Z^1(M) = \{\omega \in \Omega^1(M) : d\omega = 0\}$, 我们有 $Z^1(M) = H^1(M) \oplus d(C^\infty(M))$. 根据该事实证明本题中所定义的该环面上的 1-调和形式所构成的空间同构于该环面的 1 维 de Rham 上同调群.

(6) 利用 (4) (5) 以及 Liouville 定理，计算该环面的 1 维 de Rham 上同调群。

注: 本题内容可参考陈维桓第五章黎曼几何关于 Hodge 分解的相关内容, 或查阅其他黎曼几何教材.

2021–2022 学年期末 (王志张)

本次考试延长了 15 分钟。

1.（每小题 7 分，共 49 分）

(1) 什么是正则曲面？正则曲面上是否存在微分结构？

(2) 叙述曲面上奇点指标的定义，并证明它不依赖于局部共形映射的选取。

(3) 给出簇的光滑点的定义。证明由 $\mathbb{R}^n$ 上的光滑函数 $f(x) = 0$ 定义的簇是开的光滑流形。

(4) 写出复射影空间 $\mathbb{CP}^n$ 的定义，并给出 $\mathbb{CP}^n$ 上的一个复结构。

(5) 写出向量丛的 cocyle 条件，简要说明 cocyle 条件与向量丛的存在性的联系。

(6) 给出局部坐标系下外微分 $d : A^r(M) \to A^{r+1}(M)$ 的定义。写出外微分所需满足的四条性质，并验证 d 的定义满足这四条性质。

(7) 什么是 de Rham 上同调？说明平面 $\mathbb{R}^2$ 去掉一个闭圆盘的 1 阶 de Rham 上同调群非平凡。

2.（10 分）

(1) 给出 $\mathbb{R}^{n+1}$ 的单位球面 S^n 的标准微分结构。

(2) 定义映射 $f : S^n \to RP^n$ 如下：

$$f(x_1, \cdots, x_{n+1}) = [(x_1, \cdots, x_{n+1})].$$

证明：f 是光滑映射，并且它的秩处处是 n。

3.（10 分）

设

$$\omega = Ady \wedge dz + Bdz \wedge dx + Cdx \wedge dy$$

是 $\mathbb{R}^3$ 上的 2 次外微分式，且 $d\omega = 0$。令

$$\begin{aligned}\alpha = &\int_0^1 tA(tx, ty, tz)dt(ydz - zdy)\\ &+ \int_0^1 tB(tx, ty, tz)dt(zdx - xdz)\\ &+ \int_0^1 tC(tx, ty, tz)dt(xdy - ydx),\end{aligned}$$

验证 $d\alpha = \omega$。

4.（15 分）

(1) 给出 $C^\infty(U)$ 上的导算子的定义。证明 $C^\infty(U)$ 上的导算子全体 $\mathcal{F}(U)$ 是 $C^\infty(U)$- 模。

(2) 切丛 TM 在 U 上的截面 $TM(U)$ 是 $C^\infty(U)$- 模。

(3) 证明 $TM(U)$ 与 $\mathcal{F}(U)$ 模同构。

5.（16 分）

定义层 $\mathcal{S}$ 是 X 的所有开集 U 到 Abel 群的映射。对于开集 $V \subset U$，定义层上的限制映射 r_{VU}，满足以下条件：* $r_{UU} = id_U$，* 对于 $U \subset V \subset W$，$r_{WU} = r_{WV} \circ r_{VU}$，* 对于开集 $U = \bigcup\limits_{\alpha \in I} U_\alpha$ 与 $f_\alpha \in \mathcal{S}(U_\alpha)$，其中 I 是指标集，若满足 $r_{WU_\alpha}(f_\alpha) = r_{WU_\beta}(f_\beta)$，其中 $W = U_\alpha \cap U_\beta$，则存在 $f \in \mathcal{S}(U)$ 使得 $r_{U_\alpha U}(f) = f_\alpha$。

(1) 对照向量丛之间的态射的概念，试给出层之间的映射的定义。进一步地，试给出层同构的定义。

(2) 设 E 是 X 上的向量丛，记 $E(U)$ 为 X 的开集 U 上的截面。试证明 $E(U)$ 给出了 X 上连续函数环的模层。

(3) 定义 X 的所有开集 U 到 Abel 群 G 的摩天大楼层为

$$i_p(G)(U) = \begin{cases} G & p \in U, \\ 0 & p \notin U. \end{cases}$$

试说明它是层。

(4) 对于拓扑空间 X，证明其上的任何向量丛不与摩天大楼层同构。